MOR Number and Ver /MOR 编号和版本号: MOR-TJ414002101.04

Production Site Address/生产厂区地址:

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)

凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）

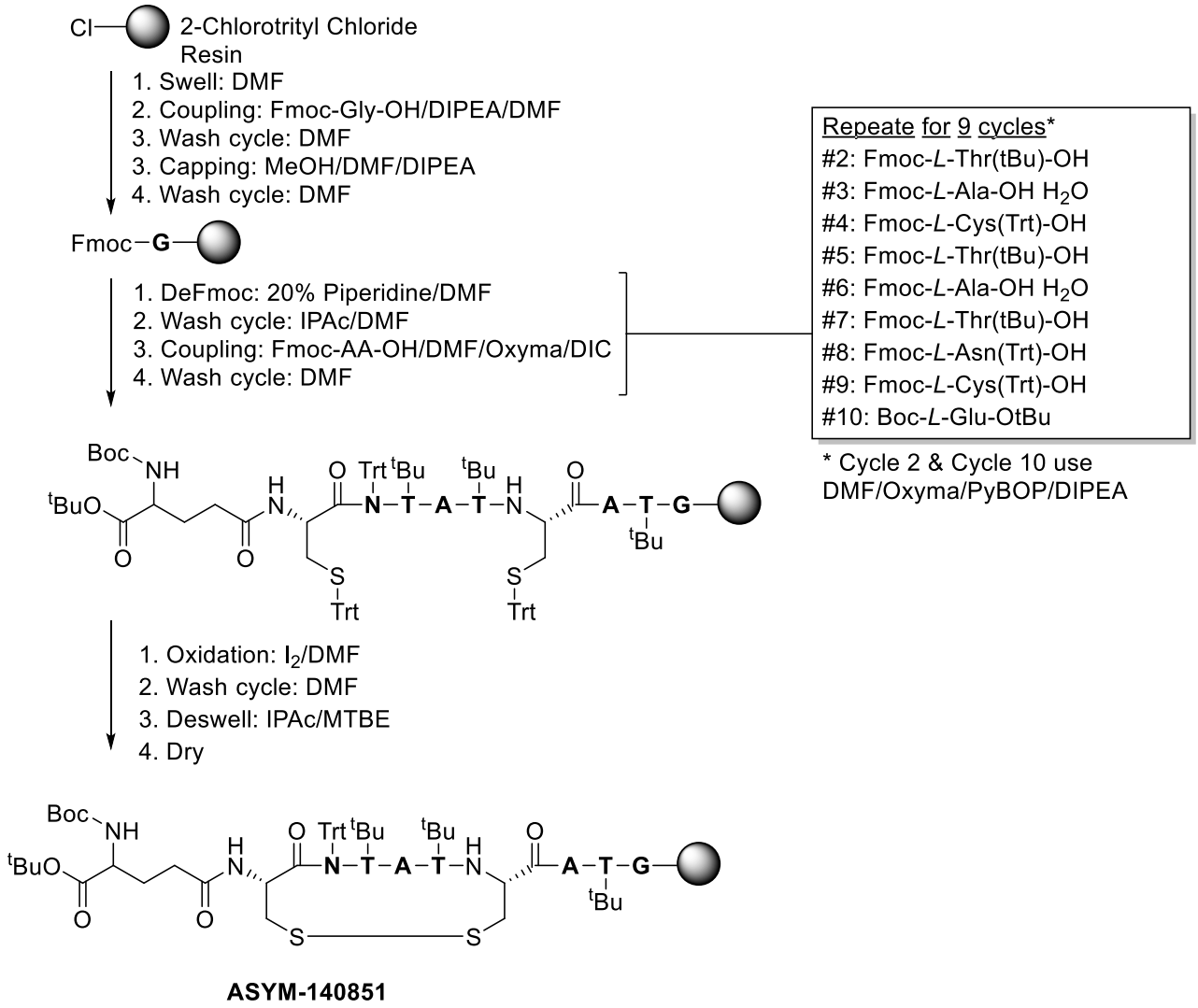
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Master Operation Record For

Boc-L3-SS peptidyl resin/140851

Synthesis Route

合成路线



ASYM-140851

Batch size (output range): 157.69~292.80 kg

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Code list for reagents, critical RM, RSM and intermediate/product

试剂，关键原料，主原料和中间体/产品的代码表

Material Name /物料名称	Asymchem Code/凯莱英代码	Client Code/客户代码	Control Number/内部控制号
2-Chlorotrityl chloride resin/2-氯三苯甲基氯树脂	N/A	N/A	YL003422
OxymaPure/ 2-肟氰乙酸乙酯	N/A	N/A	YL005180
PyBOP/1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	N/A	N/A	YL002855
DIPEA/N,N-二异丙基乙胺	N/A	N/A	YL000555
Methanol/甲醇	N/A	N/A	YL000031
DIC/N,N'-二异丙基碳二亚胺	N/A	N/A	YL003564
Fmoc-Gly-OH/芴甲氧羰基-甘氨酸	ASYM-138200	N/A	YL004846
Fmoc-L-Thr(tBu)-OH/芴甲氧羰基-叔丁基-L-苏氨酸	ASYM-138389	N/A	YL003563
Fmoc-L-Ala-OH•H_2O/芴甲氧羰基-L-丙氨酸一水合物	ASYM-138204	N/A	YL006160
Fmoc-L-Cys(Trt)-OH /Fmoc-S-三苯甲基-L-半胱氨酸	ASYM-138397	N/A	YL003565
Fmoc-L-Asn(Trt)-OH/Fmoc-N-三苯甲基-L-天冬酰胺	ASYM-138519	N/A	YL004843
Boc-L-Glu-OtBu/Boc-L-谷氨酸-1-叔丁酯	ASYM-140413	N/A	YL008618
Iodine/碘	N/A	N/A	YL003092
DMF/N,N-二甲基甲酰胺	N/A	N/A	YL000336
Isopropyl acetate/乙酸异丙酯	N/A	N/A	YL000932
MTBE/甲基叔丁基醚	N/A	N/A	YL000557
In-process product/中间体 ASYM-140851	ASYM-140851	N/A	ZJ007316

Batch size (output range): 157.69~292.80 kg

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Operation Record for 140851

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1.20. (10) Dispense Boc-L-Glu-OtBu/分装 Boc-L-谷氨酸-1-叔丁酯 93

1.21. Dispense PyBOP/分装 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐 95

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Batch size (output range): 157.69~292.80 kg

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Master Operations Record (MOR) Approvals /MOR 批准

Authored by/编写者:

Author& Date/撰写人及日期: Signed by [刘亚超 ALS11087, Yachao Liu], at '10/29/2026 7:26:22 PM'

Approved By/批准:

Project Director & Date /项目主管及日期: Signed by [张宇 ALS12718, Yu Zhang], at '10/29/2026 7:30:29 PM'

Production Director & Date /生产部主管及日期: Signed by [张忠尧, Zhongyao Zhang], at '10/29/2026 7:35:28 PM'

QA project Manager & Date /QA 项目经理及日期: _____

Client Approval & Date/客户审批及日期: _____

(The last signature date on this page is the effective date. /本页最后一个签字日期即为生效日期)

Batch size (output range): 157.69~292.80 kg

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Operations Record (OR) Issue /OR 签发

Batch Number/生产批号: _____

Approved by QA/QA 批准人: _____

Date/日期: _____

Batch size (output range): 157.69~292.80 kg

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Intermediate Information/中间体信息

Expected yield/预期收率：80~130%

Calculation of theoretical output/理论产量计算:

Theoretical peptidyl resin output/理论肽树脂产出重量= resin weight 树脂重量+resin weight 树脂重量× degree of substitution 取代度× net MW gain 中间体分子量增量=_____kg +_____kg ×_____mol/kg× 1498.89 g/mol ÷ 1000 = ___ kg

Calculation of expected gain output weight/预期产量计算:

Expected output =Theoretical output × Expected yield

预期产量=理论肽树脂重量×预期收率=_____* 80% ~_____* 130% _=_____~_____kg

Actual output weight /实际产量:

Physical output of peptidyl resin weight/实际肽树脂产量: _____Kg

Calculation of weight yield /实际收率计算方法:

Weight yield 增重收率= (Actual output 实际产量/Theoretical output 理论产量)×100%

=(_____kg/_____kg) ×100% = _____%

According to the operation record, the Lot# of the finished product is assigned as/

根据操作记录，该批最终产品的批号为: _____

Batch size (output range): 157.69~292.80 kg

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Deviation & Change Log/生产偏差及变更记录

The corresponding deviations and changes related to this batch should be recorded in the following Table:
与本次相关的偏差和变更应该在下表中记录:

No. / 序号	Operation Instruction. No./ 操作指令编号	Deviation report No. / 偏差报告编号	No. / 序号	Operation Instruction No. / 操作指令编号	Change file No. / 变更文件编号
1			1		
2			2		
3			3		
4			4		
5			5		
6			6		
7			7		
8			8		
9			9		

Batch size (output range): 157.69~292.80 kg

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Incident Log/事件调查登记

The corresponding incident related to this batch should be recorded in the following Table: /与本次相关的异常事件应该在下表中记录:

No. /序号	Operation Instruction. No./操作指令编号	Incident No. /事件编号	Incident brief description/事件简述	Filled by/Date 填写人/日期
1				

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

No. /序号	Operation Instruction. No./操作指令编号	Incident No. /事件编号	Incident brief description/事件简述	Filled by/Date 填写人/日期
2				

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

No. /序号	Operation Instruction. No./操作指令编号	Incident No. /事件编号	Incident brief description/事件简述	Filled by/Date 填写人/日期
3				

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Parameter change log in recipe/配方参数修改登记表

No./ 序号	Instruction./ 操作号	Current parameter/ 当前参数	Updated parameter/ 更新后的参数	Revision justification and reference document (if any)/ 修改原因以及文件索引（如有）	Performed by/Date 执行人/日期	Checked by/Date 核实人/日期

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

No./ 序号	Instruction./ 操作号	Current parameter/ 当前参数	Updated parameter/ 更新后的参数	Revision justification and reference document (if any)/ 修改原因以及文件索引（如有）	Performed by/Date 执行人/日期	Checked by/Date 核实人/日期

Batch size (output range): 157.69~292.80 kg

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Equipment Information List/设备使用清单

Pre-batch Equipment Confirmation Checklist//每次生产前设备检查确认表

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12, Module 13/ 12 号楼 13 车间	2，3，4	Peptide Solid Phase Synthesizer / 多肽固相合成仪 Equipment # 设备编号: TJ4S-1213-Sy16 Size /设备大小:3000 L MOC :Hastelloy/材质：哈氏合金	Peptide Solid/ 固相合成& Dry/烘料	☐ Level I ☐Level II Cleaning record number/清洁记录编号: _____		
Building 12, Module 13/ 12 号楼 13 车间	2，3，4	Activation tank//激活罐 Equipment # 设备编号: TJ4S-1213-Sy16-R01 Size /设备大小:1500 L MOC :Stainless Steel/材质：不锈钢	Activation/活化& Dissolve/溶解	☐ Level I ☐Level II Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12, Module 13/ 12 号楼 13 车间	3	Agitator tank/搅拌罐 Equipment # 设备编号: TJ4S-1200-AT01 Size /设备大小: 500L MOC : Hastelloy/材质：哈氏合金	Prepare DIC/DMF solution/ 配 DIC/DM 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
Building 12, Module 13/ 12 号楼 13 车间	3	Agitator tank /搅拌罐 Equipment # 设备编号: TJ4S-1213-AT09 Size /设备大小: 500L MOC : Glass lined /材质：搪玻璃	Prepare I_2/DMF solution/ 配碘/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
Building 12, Module 13/ 12 号楼 13 车间	3	Agitator tank/搅拌罐 Equipment # 设备编号: TJ4S-1213-SPS01-AT01 Size /设备大小: 3000L MOC : Stainless Steel /材质：不锈钢	Prepare 20% Piperidine/DMF solution/配 20% 哌啶/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12, Module 13/ 12 号楼 13 车间	3	Agitator tank/搅拌罐 Equipment # 设备编号: TJ4S-1213-SPS01-AT02 Size /设备大小: 3000L MOC : Stainless Steel /材质: 不锈钢	Prepare 20% Piperidine/DMF solution/配 20% 哌啶/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	1	Glove box/手套箱 Equipment # 设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Subpackage/分装	Level I		
Building 12/ 12 号楼	1	Glove box/手套箱 Equipment # 设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Subpackage/分装	Level I		
Building 12/ 12 号楼	1	Glove box/手套箱 Equipment # 设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Subpackage/分装	Level I		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	1	Glove box/手套箱 Equipment # 设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Subpackage/分装	Level I		
Building 12/ 12 号楼	1	Glove box/手套箱 (TJ4S-1200-GB04/GB06) Equipment # 设备编号: _____ MOC : Halar /材质: 衬哈	Iodine subpackage/分装	Level I		
Building 12/ 12 号楼	3.2.1	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P04 MOC : Stainless Steel /材质: 不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	Level I		
		Hose/管路 Equipment #设备编号:_____ MOC : Stainless Steel /材质: 不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	Level I		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	3.4.2	Peristaltic Pump/蠕动泵 Equipment #设备编号: _____	Add Methanol/甲醇加料	Level 0		
		Single use flexible hose/一次性加料管路 Equipment #设备编号:_____	Add Methanol/甲醇加料	Released for single use/一次性使用放行		
Building 12/ 12 号楼	3.4.3	Peristaltic Pump/蠕动泵 Equipment #设备编号: _____	Add DIPEA/ N,N-二异丙基乙胺加料	Level 0		
		Single use flexible hose/一次性加料管路 Equipment #设备编号:_____	Add DIPEA/ N,N-二异丙基乙胺加料	Released for single use/一次性使用放行		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	3.6.7	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P04 MOC : Stainless Steel /材质：不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	Level 0		
		Hose/管路 Equipment #设备编号:_____ MOC : Stainless Steel /材质：不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	Level 0		
Building 12/ 12 号楼	3.9.6	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P03 MOC : Stainless Steel /材质：不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	Level I		
		Hose/管路 Equipment #设备编号:_____ MOC : Stainless Steel /材质：不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	Level I		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	3.13.3	Diaphragm Pump/隔膜泵 Equipment #设备编号: _____	DMF / N,N-二甲基甲酰胺	Level I		
		Flexible hose/软管 Equipment #设备编号:_____ MOC : Stainless Steel /材质: 不锈钢	DMF / N,N-二甲基甲酰胺	Level I		
Building 12/ 12 号楼	3.13.3	Peristaltic Pump/蠕动泵 Equipment #设备编号: _____	Add DIC /N,N-二异丙基碳二亚胺加料	Level 0		
		Single use flexible hose/一次性加料管路/ Equipment # #设备编号:_____	Add DIC /N,N-二异丙基碳二亚胺加料	Released for single use/一次性使用放行		
Building 12/ 12 号楼	3.13.7	Metering pump/计量泵 Equipment #设备编号: TJ4S-1200-AT01-P01 MOC : Stainless Steel /材质: 不锈钢	Add DIC/DMF Solution /加 DIC/DMF 溶液	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
		Pipeline/管路 Equipment #设备编号: 140021-step01-P01 MOC : Stainless Steel /材质: 不锈钢	Add DIC/DMF solution /加 DIC/DMF 溶液	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.15.6	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P03 MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
		Pipeline/管路 Equipment #设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.18.6	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P03 MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
		Pipeline/管路 Equipment #设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.21.6	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P03 MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
		Pipeline/管路 Equipment #设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.24.6	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P03 MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
		Pipeline/管路 Equipment #设备编号: _____ MOC : Stainless Steel /材质: 不锈钢	Add DIC /N,N-二异丙基碳二亚胺加料	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.28.3	Diaphragm Pump/隔膜泵 Equipment #设备编号: _____	DMF / N,N-二甲基甲酰胺	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
		Flexible hose/软管 Equipment #设备编号:_____	DMF / N,N-二甲基甲酰胺	☐Level 0 ☐Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.28.3	Peristaltic Pump/蠕动泵 Equipment #设备编号: _____	Add DIC /N,N-二异丙基碳二亚胺加料	Level 0		
		Single use flexible hose /一次性加料管路 Equipment # #设备编号:_____	Add DIC /N,N-二异丙基碳二亚胺加料	Released for single use/一次性使用放行		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	3.28.7	Metering pump/计量泵 Equipment #设备编号: TJ4S-1200-AT01-P01 MOC : Stainless Steel /材质：不锈钢	Add DIC/DMF Solution /加 DIC/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Pipeline/管路 Equipment #设备编号: 140021-step01-P01 MOC : Stainless Steel /材质：不锈钢	Add DIC/DMF solution /加 DIC/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
Building 12/ 12 号楼	3.30.7	Metering pump/计量泵 Equipment #设备编号: TJ4S-1213-Sy16-P04 MOC : Stainless Steel /材质：不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Hose/管路 Equipment #设备编号:_____ MOC : Stainless Steel /材质：不锈钢	Add DIPEA/ N,N-二异丙基乙胺加料	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	3.32.1	Diaphragm Pump/隔膜泵 Equipment #设备编号: _____	DMF / N,N-二甲基甲酰胺	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Flexible hoses/软管 Equipment #设备编号:_____	DMF / N,N-二甲基甲酰胺	Level I		
Building 12/ 12 号楼	3.32.8	Metering pump/计量泵 Equipment #设备编号: TJ4S-1107-MPu003 MOC : Halar /材质: 衬哈 Meter MOC 流量计材质: Tantalum/钽	Add Iodine/DMF solution /加碘/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Pipeline/管路 Equipment #设备编号: 140021-step01-P02 MOC : PTFE/Halar /材质: 衬四氟/衬哈	Add Iodine/DMF solution /加碘/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期
Building 12/ 12 号楼	☐3.33.1	Diaphragm Pump/隔膜泵 Equipment #设备编号: _____	DMF / N,N-二甲基甲酰胺	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Flexible hoses/软管 Equipment #设备编号:_____	DMF / N,N-二甲基甲酰胺	Level I		
Building 12/ 12 号楼	☐3.33.8	Metering pump/计量泵 Equipment #设备编号: TJ4S-1107-MPu003 MOC : Halar /材质：衬哈 NOR Meter MOC 流量计材质：Tantalum/钽	Add Iodine/DMF solution /加碘/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		
		Pipeline/管路 Equipment #设备编号: 140021-step01-P02 MOC : PTFE/Halar /材质：衬四氟/衬哈	Add Iodine/DMF solution /加碘/DMF 溶液	☐Level 0 ☐ Level I Cleaning record number/清洁记录编号: _____		

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Title: 140021 Step: 1 Preparation of: 140851

Facility and Module/ 厂房和车间	Operation Instruction No.操作指令编号	Equipment/ 设备名称	Function/ 用途	Level of cleaning check/清洁检测核查级别	Performed by & Date/ 执行人及日期	Checked By&Date 核实人及日期

Note: The operator should fill this table before the equipment is used/

备注：此表格应该由操作者在使用设备前进行填写。

Batch size (output range): 157.69~292.80 kg

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Raw Material Summary /原料信息

Raw Material Summary Table/原料使用汇总表

Limiting material:/基准物料*:

Operation Instruction. No. 操作指令编号	Raw Material Name 原料名称	Average resin Substitution/取代度 (mmol/g)**	Standard Input Qty/标准投料量（kg）	Total loading moles/投料总摩尔量（mol）**	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围（kg）	Actual quantity charged/实际投料量（kg）
1.1/3.1.2	**2-Chlorotrityl chloride resin/2-氯三苯甲基氯树脂**		**65.00**		**±0.2%**	**64.87~65.13kg**	
			Entered by & Date/填写人及日期:				
			Checked by& Date/核实人及日期:				

* The limiting material is the basis for the other material charging input in the process. For example, if the limiting material is deemed as 1.0 eq/1.0 W/1.0 Vol, the other materials quantities are caculated based on their target molar/weight/volume ratios against the limiting material accordingly. /该物料是其他物料投料的基础，也就是说，如果该物料为1.00eq，则其他物料的数量以该物料为基础进行计算。

The Average resin substitution and Total loading moles are calculated by using the QA approved validated spreadsheet in batch record attachment.

Limiting material mole (1.00 eq) = Total loading moles (mol)/ 限量物料摩尔数（1.00 eq）= 总上载摩尔数（mol）= _____ (mol)

Limiting material weight (W) = Actual quantity charged (kg)/ 限量物料重量（W）= 实际投料量（kg）= _____ (kg)

Limiting material volume (Vol) = Actual quantity charged (L)/ 限量物料体积（Vol）= 实际投料量（L）= _____ (L)

The acceptance criteria for the substitution degree of 2-CTC resin is 1.4 to 1.6 mmol/g/ 2 -氯三苯甲基氯树脂的取代度合格标准为：1.4–1.6 mmol/g。

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Non-Limiting material:/非基准物料

For below table. /对于下表:

* Standard Batch Qty 标准投料量=Limiting material quantity/基准物料数量×Qty Ratio to limiting material/与基准物料质量比;

Acceptable weighing range 可接受称量范围=Standard Batch Qty 标准投料量×（1±Weighing accuracy 称量精确度）

* For materials that need to be converted with assay, need to indicate in parentheses after the Standard Batch Qty. value that the value means the quantity converted with assay. /对于需要进行含量折算的物料，需要在标准投料量数值后加括号注明该数值是指折算含量后的投料量。

** The meaning of the value should be the same as the value of 'Standard Batch Qty.'. For materials that need to be converted with assay, need to indicate in parentheses after the 'Acceptable Weighing Range' value that the value means the quantity converted with assay. /数值的含义应该与"标准投料量"一列中数值的含义相同，对于需要进行含量折算的物料，需要在"可接受范围"数值后，加括号注明该数值是指折算含量后的投料量。

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量*（kg）	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围（kg）
3.0	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±6%	347.66~392.04
☐3.0	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±6%	347.66~392.04
3.1.3	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.1.6	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.2/3.1.7	Fmoc-Gly-OH 芴甲氧羰基-甘氨酸	-	297.31	-	-	2.25		±3%	

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Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.1.8	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.1.9	DMF N,N-二甲基甲酰胺	0.948	73.10	5.0	4.74	-	308.10	±6%	289.61~326.59
3.2.1	DIPEA N,N-二异丙基乙胺	0.742	129.24	-	-	6.00		±3%	
3.3.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.3.6	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26
3.4.1	DMF N,N-二甲基甲酰胺	0.948	73.10	9.0	8.53	-	554.45	±6%	521.18~587.72
3.4.2	Methanol 甲醇	0.792	32.04	-	-	10.00		±3%	
3.4.3	DIPEA N,N-二异丙基乙胺	0.742	129.24	-	-	5.00		±3%	
3.4.6	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.5.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.5.4	IPAc 乙酸异丙酯	0.873	102.13	8.0	6.98	-	453.70 (each wash/ 单次)	±6%	426.48~480.92
3.5.5	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.5.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.6.1	DMF N,N-二甲基甲酰胺	0.948	73.10	4.0	3.79	-	246.35	±5%	234.03~258.67
1.3/3.6.2	Fmoc-L-Thr(tBu)-OH 芴甲氧羰基-叔丁基-L-苏氨酸	-	397.47	-	-	2.00		±3%	
1.4/3.6.4	PyBOP 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	-	520.39	-	-	2.00		±3%	
1.5/3.6.5	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.6.6	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.6.7	DIPEA N,N-二异丙基乙胺	0.742	129.24	-	-	3.00		±3%	
3.7.1	DMF N,N-二甲基甲酰胺	0.948	73.10	3.0	2.84	-	184.60	±6%	173.52~195.68
3.7.3	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.7.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26
3.8.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.8.4	IPAc 乙酸异丙酯	0.873	102.13	8.0	6.98	-	453.70 (each wash/ 单次)	±6%	426.48~480.92
3.8.5	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.8.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.9.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.6/3.9.2	Fmoc-L-Ala-OH•H_2O 芴甲氧羰基-L-丙氨酸一水合物	-	329.35	-	-	2.00		±3%	
1.7/3.9.4	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	
3.9.5	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.9.6	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	2.20		±3%	
3.10.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.10.6	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26
3.11.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.11.4	IPAc 乙酸异丙酯	0.873	102.13	8.0	6.98	-	453.70 (each wash/ 单次)	±6%	426.48~480.92

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.11.5	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.11.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.12.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.8/3.12.2	Fmoc-L-Cys(Trt)-OH Fmoc-S-三苯甲基-L-半胱氨酸	-	585.71	-	-	2.00		±3%	
1.9/3.12.3	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	3.00		±3%	
3.12.4	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.13.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.13.3	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.13.3	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	3.30		±3%	

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.13.10	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26
3.14.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.14.4	IPAc 乙酸异丙酯	0.873	102.13	8.0	6.98	-	453.70 (each wash/ 单次)	±6%	426.48~480.92
3.14.5	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.14.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.15.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.10/3.15.2	Fmoc-L-Thr(tBu)-OH 芴甲氧羰基-叔丁基-L-苏氨酸	-	397.47	-	-	2.00		±3%	
1.11/3.15.4	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.15.5	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.15.6	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	2.20		±3%	
3.16.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.16.6	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70 (each wash/ 单次)	±6%	463.14~522.26
3.17.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.17.4	IPAc 乙酸异丙酯	0.873	102.13	8.0	6.98	-	453.70 (each wash/ 单次)	±6%	426.48~480.92
3.17.5	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.17.7	DMF N,N-二甲基甲酰胺	0.948	73.10	8.0	7.58	-	492.70	±6%	463.14~522.26
3.18.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
1.12/3.18.2	Fmoc-L-Ala-OH•H2O 芴甲氧羰基-L-丙氨酸一水合物	-	329.35	-	-	2.00		±3%	
1.13/3.18.4	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	
3.18.5	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.18.6	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	2.20		±3%	
3.19.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.19.6	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17
3.20.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.20.4	IPAc 乙酸异丙酯	0.873	102.13	10.0	8.73	-	567.45 (each wash/ 单次)	±6%	533.40~601.50

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.20.5	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.20.7	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.21.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.14/3.21.2	Fmoc-L-Thr(tBu)-OH 芴甲氧羰基-叔丁基-L-苏氨酸	-	397.47	-	-	2.00		±3%	
1.15/3.21.4	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	
3.21.5	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.21.6	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	2.20		±3%	
3.22.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.22.6	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.23.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.23.4	IPAc 乙酸异丙酯	0.873	102.13	10.0	8.73	-	567.45 (each wash/ 单次)	±6%	533.40~601.50
3.23.5	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.23.7	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.24.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.16/3.24.2	Fmoc-L-Asn(Trt)-OH Fmoc-N-三苯甲基-L-天冬酰胺	-	596.68	-	-	2.00		±3%	
1.17/3.24.4	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	
3.24.5	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.24.6	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	2.20		±3%	

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.25.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.25.6	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17
3.26.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.26.4	IPAc 乙酸异丙酯	0.873	102.13	10.0	8.73	-	567.45 (each wash/ 单次)	±6%	533.40~601.50
3.26.5	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.26.7	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.27.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.18/3.27.2	Fmoc-L-Cys(Trt)-OH Fmoc-S-三苯甲基-L-半胱氨酸	-	585.71	-	-	2.00		±3%	
1.19/3.27.3	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	3.00		±3%	

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.27.4	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.28.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.28.3	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.28.3	DIC N,N'-二异丙基碳二亚胺	0.815	126.20	-	-	3.30		±3%	
3.28.10	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17
3.29.1	20% Piperidine solution 20%哌啶/N,N-二甲基甲酰胺溶液	0.931	-	8.0	7.44	-	483.60	±6%	454.58~512.62
3.29.4	IPAc 乙酸异丙酯	0.873	102.13	10.0	8.73	-	567.45(each wash/单次)	±6%	533.40~601.50
3.29.5	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.29.7	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	44 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
3.30.1	DMF N,N-二甲基甲酰胺	0.948	73.10	6.0	5.69	-	369.85	±5%	351.36~388.34
1.20/3.30.2	Boc-L-Glu-OtBu Boc-L-谷氨酸-1-叔丁酯	-	303.35	-	-	2.00		±3%	
1.21/3.30.4	PyBOP 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	-	520.39	-	-	2.00		±3%	
1.22/3.30.5	Oxyma Pure 2-肟氰乙酸乙酯	-	142.11	-	-	2.00		±3%	
3.30.6	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.30.7	DIPEA N,N-二异丙基乙胺	0.742	129.24	-	-	3.00		±3%	
3.31.2	DMF N,N-二甲基甲酰胺	0.948	73.10	1.0	0.95	-	61.75	±6%	58.05~65.46
3.31.6	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17
3.32.1	DMF N,N-二甲基甲酰胺	0.948	73.10	4.0	3.79	-	246.35	±6%	231.57~261.13

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
1.23/3.32.3	Iodine / 单质碘	-	253.81	-	-	5.00		±3%	
3.32.5	DMF N,N-二甲基甲酰胺	0.948	73.10	12.0	11.37	-	739.05	±6%	694.71~783.39
3.32.9	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.32.11	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
3.32.13	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20 (each wash/ 单次)	±6%	579.23~653.17
☐3.33.1	DMF N,N-二甲基甲酰胺	0.948	73.10	4.0	3.79	-	246.35	±6%	231.57~261.13
1.24/ ☐3.33.3	Iodine / 单质碘	-	253.81	-	-	5.00		±3%	
☐3.33.5	DMF N,N-二甲基甲酰胺	0.948	73.10	12.0	11.37	-	739.05	±6%	694.71~783.39
☐3.33.9	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)
☐3.33.11	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20	±6%	579.23~653.17
☐3.33.13	DMF N,N-二甲基甲酰胺	0.948	73.10	10.0	9.48	-	616.20(each wash/单次)	±6%	579.23~653.17
3.34.1	IPAc 乙酸异丙酯	0.873	102.13	10.0	8.73	-	567.45 (each wash/ 单次)	±6%	533.40~601.50
3.34.2	MTBE 甲基叔丁基醚	0.740	88.15	10.0	7.41	-	481.65 (each wash/ 单次)	±6%	452.75~510.55

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operation Instruction No. 操作指令编号	Raw Material Name 原料名称	Density/ 密度(g/cm^3)	Mol. Wt. of material/ 原料分子量	Volume/Weight Ratio to limiting material/与基准原料质量体积比	Qty Ratio to limiting material/ 与基准原料质量比	Mol Ratio to limiting material/与基准原料摩尔比	Standard Batch Qty/标准投料量* (kg)	Weighing accuracy/称量精确度	Acceptable weighing range/可接受称量范围 (kg)

Recorded by& Date/记录人及日期(Chemist or engineer/相关技术人员): _____

Checked by& Date/核实人及日期(Chemist or engineer/相关技术人员): _____

Checked by QA & Date/QA 核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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IPC Sampling and Testing Plan /过程控制取样检测计划

✧ Sampling and testing list of intermediate/product /中间体/产品取样检测列表

Instruction No./ 指令编号	Sample Type/ 样品检测类型			Sample source description 样品来源描述	Analytical test/ 检测项目	Criterion of acceptability/ 合格标准	Analytical method# / 分析方法编号
	IPC test	FIO test	Release testing				
3.0.	X	-	-	DMF rinse solution/DMF 淋洗液	KF/含水	≤500ppm	AM-YL000336-004
3.7.6; 3.10.5; 3.13.9; 3.16.5; 3.19.5; 3.22.5; 3.25.5; 3.28.9; 3.31.5	X		-	Coupling-stage resin/ 偶联阶段树脂	Resin color/树脂颜色	Negative(the resin is yellow to orange)/ 阴性(树脂黄色至橙色)	AM-140021-043
3.32.13/3.33.13	X	-	-	Post-oxidation Resins/ 氧化后肽树脂	ASYM-140850	≤0.5%	AM-140021-011
3.35.1	X	-	-	Resins under drying/ 烘料固体	LOD/干燥失重	≤8%	AM-140021-007

Analytical report numbering method: project No. + step No. + batch No. + Ins.No. + Sampling No. in this Instruction, as CPo1234-01-01-01-09-01

所有分析报告的编号方式为项目号 + 步数 + 批次 + 操作指令数 + 指令中的第几次取样，如 CPo1234-01-01-01-09-01

Confirm in MOR the decimal places of the IPC test acceptance criteria is consistent with the analytical method.

确认 MOR 中的 IPC 检测可接受标准的小数位数和分析方法的小数位数一致。

Analytical Laboratory Manager & Date / 分析实验室经理签字及日期

Signed by [杜雪峰, Xuefeng Du], at '10/29/2026 7:33:04 PM'

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Recipe Code against the Unit Operation/基于单元操作的配方命名

Instruction No./指令编号	Recipe usage/配方用途	Recipe name/配方命名	Batch name/批次命名
3	Synthesis/合成	14002101M01V02	CPo140021_____01_____

Recipe name: project number " CCl#+Step No.+M+serial No. +V+version No/ 配方命名：CCl#+步骤号+M ++序列号+V+2 位版本号）

Batch name: project number " + step No.+ batch No. / 批次命名：项目号+步骤号+批次号

Batch size (output range): 157.69~292.80 kg

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Operation Instructions and Execution/操作指令和执行记录

Operations Instruction
操作指令

Instructions/说明:

1. Key operation/关键操作 (refer to DS-QA-001 参考 DS-QA-001):
   The CPP (if any), CIPCs (if any) will be indicated with ★in the MOR./对于关键工艺参数（如有），关键过程控制参数（如有）将在 MOR 中★用进行标记。
2. In each operation instruction, the process parameter should be defined in the format of target with allowed operation range. /在操作指令处参数实例处一律以操作目标值加上允许操作范围的方式体现。
3. Chinese version will prevail in this document. Where applicable, English translation is added to aid comprehension by reviewer whose native language is not Chinese. /本记录以中文为准，英文翻译是用于帮助非中文的审阅者理解的。
4. In this document, some non-process operations, such as check as equipment readiness, hose connection, starting material, auxiliary solution preparation, may be performed in advance when appropriate. /在本记录中，非工艺流程操作，例如设备准备，管道连接，物料准备，辅助溶液配制可以在合适的时候提前操作。
5. Various confirmations/各类确认
   5.1. Before material charging and dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current charging. /加物料或分装物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
   5.2. Before solid material charging, confirm/加固体物料前，确认:
   - No non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
   - The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

   5.3. Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
   5.4. After solid material charging, confirm/固体物料投料之后，确认:
   - The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
   - The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
   - The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

   5.5. Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。
6. The agitating speed control refers to the control in a normal steady process status exclude the operation-switch stage or interruption for some observation/sampling operation et.al. /搅拌速度控制指正常工艺稳定状态下的搅拌速度，不包括操作间的切换阶段或者为观察或者取样等操作为目的的扰动。
7. The temperature control has priority to the material charging rate control from safety consideration，the material charging rate can further slow-down or stop to an temperature-controllable status. /从安全角度考虑，相比加料速度的控制，温度控制具有优先权，物料加料速度可以进一步放慢或者停止使温度处于可控状态。
8. 8-10L purified water should be discharged at the point of use before use. /纯化水每次使用前，应先从用水点放出约 8-10L 水。
9. Check the cleanness of glass sample bottle prior to use, and rinse the sample bottle with reaction/workup solvent or mixture. /使用玻璃取样瓶前需要检查是否清洁，同时使用反应/后处理溶剂或体系润洗取样瓶。
10. In the Master Operation Record (MOR), the responsibility of roles are defined as follows:/在该主操作记录中，对于各类角色职责定义如下:

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

- **Performed by:** The person who executes the synthesis procedure. For manual operations (e.g., material charging, sampling), the person performing the operation by manual is the performer; For automated operations (e.g., automated peptide synthesis cycle operation), it is the person who initiates the automatic operation of the on-site monitoring program and is also the account login person during the automatic operation stage of the program.
  执行人: 为现场实际执行合成工序作业的人员。对于手动投料的操作（如投料、取样），操作人员即为执行人；对于程序自动运行的操作（如多肽自动合成循环操作等），执行人为当班现场监控程序自动运行的人员，也是程序自动运行阶段的账号登录人员。
- **Transcribed by:** The person who is responsible for transcribing electronic data automatically generated by equipment onto the paper batch operation record. Since computerized systems operate at high speed, the Transcriber can transcribe relevant information from the original electronic data such as audit trails and transcribe it into the batch operation record. Shift personnel shall complete the transcription of data required by the batch record for production operations that have been completed during their shift.
  转录人: 负责将设备自动化电子数据誊写至纸版批操作记录的人员。因计算机化系统运行速率较高，转录人可依据审计追踪等原始电子数据源，将相关信息摘抄至批操作记录中。当班人员应完成本班涉及到的已完成生产操作中，批记录中要求的数据摘抄工作。
- **Checked by:** The second person who witness the activity (such as operation, data, and record) is performed correctly through supervising on the site.
  核实人: 由第二人通过现场亲眼目击确证相应的活动（如操作，数据和记录的）被完全正确地执行的人员。

Batch size (output range): 157.69~292.80 kg

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Process Safety information /工艺安全信息

1.2-CTC resin: OEB3, resin can not swell too long time and avoid excessive drying caused by resin water absorption deactivation. / CTC 树脂 OEB3，树脂不可溶胀过长时间以及避免过量的抽干引起树脂吸水失活。

2. It is necessary to confirm that De-Fmoc of resin is completed before activation. The addition of DIPEA requires attention to the temperature increase of mixture. /需要确认肽树脂脱 Fmoc 洗涤进行完毕再开始活化氨基酸。滴加 N,N'-二异丙基乙胺过程需要注意内温会升高。

3. OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。

4. PyBOP is high energy compounds, impact sensitive, skin sensitizing. /1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐是高能化合物，撞击敏感性，皮肤致敏。

5. DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'-二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。

6.IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。

7.DMF/N,N'-二甲基甲酰胺： Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

8.Piperidine 哌啶： Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。

9.DIPEA/N,N'-二异丙基乙胺： It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状

10.MTBE/甲基叔丁基醚： Inhalation: May cause dizziness or suffocation. Contact may irritate or burn eyes or skin. May be harmful if swallowed. /吸入:可引起头晕或窒息。接触可能刺激或灼伤眼睛或皮肤。吞食可能有害。

11.MeOH/甲醇: Exposure to excessive vapor causes eye irritation, head- ache, fatigue and drowsiness. High concentrations can produce central nervous system depression and optic nerve damage. 50,000 ppm will probably cause death in 1 to 2 hrs. Can be absorbed through skin. Swallowing may cause death or eye damage. /接触过多的蒸汽会引起眼睛刺激、头痛、疲劳和困倦。高浓度可导致中枢神经系统抑制和视神经损伤。50000 ppm 可能会在 1 到 2 小时内导致死亡。可通过皮肤吸收。吞咽可能导致死亡或眼睛损伤

13. ASYM-140851： OEB3； OEL(10~100ug/m3). Exposure control level ≤10ug/m^3./暴露控制水平≤10ug/m^3。

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	53 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

Ins.1/操作 1: Dispense/分装

Wear nitrile rubber gloves and 3M 6000 series Full-face piece respirator, 6006CN cartridges, Particulate filter, White disposable chemical protective clothing. 戴丁腈橡胶手套、3M 6000 系列全面罩, 6006CN 滤毒盒, 颗粒物滤棉, 白色防化服。

1.1.Dispense 2-chlorotrityl chloride resin/分装 2-氯三苯甲基氯树脂

Before material dispensing , confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense 2-chlorotrityl chloride resin. /分装 65.00 kg2-氯-三苯甲基氯树脂（Acceptable weighing range/可接受质量范围: 64.87 kg~ 65.13 kg）。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL003422

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number. /手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.1 Dispense 2-chlorotrityl chloride resin/分装 2-氯三苯甲基氯树脂

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量					Total resin Mole/总摩尔量 (mol)*	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight /毛重	Tare weight /皮重	Net weight/净重	No. /编号	Substitution /取代度 (mmol/g)			
3.1.2					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
					___/_____				
	Total weight /合计重量				-				
	Total No. of bags/合计袋数				-				
Begin/开始		End/结束							

*Note/备注: Average resin substitution and total loading moles are calculated using the QA-approved validated spreadsheet included in the batch record attachment.. /平均树脂取代度及总上载摩尔数通过批记录附件中 QA 批准的已验证电子表格计算。

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.2. (1) Dispense Fmoc-Gly-OH/分装芴甲氧羰基-甘氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。
Before solid material dispensing, confirm/固体物料分装前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期:_____

Dispense Fmoc-Gly-OH. /分装_____kg芴甲氧羰基-甘氨酸（2.25eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL004846

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.2 (1) Dispense Fmoc-Gly-OH/分装芴甲氧羰基-甘氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight /皮重	Net weight/净重			
3.1.7					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.3.(2) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date. Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Thr(tBu)-OH. /分装_____kg芴甲氧羰基--叔丁基-L-苏氨酸（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL003563

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.3(2) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.6.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.4.Dispense PyBOP/分装 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐

Before material dispensing,, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense PyBOP. /分装_____kg 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL002855

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.4 Dispense PyBOP/分装 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.6.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.5.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing,, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.5.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.6.5					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.6. (3) Dispense Fmoc-L-Ala-OH•H_2O/分装芴甲氧羰基-L-丙氨酸一水合物

Before material dispensing,, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体分装物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Ala-OH•H_2O. /分装_____kg芴甲氧羰基-L-丙氨酸一水合物（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL006160

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.6.(3) Dispense Fmoc-L-Ala-OH•H2O/分装芴甲氧羰基-L-丙氨酸一水合物

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.9.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.7.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.7.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.9.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后（开放性操作后），确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.8.(4) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Cys(Trt)-OH. /分装【_____kg Fmoc-S-三苯甲基-L-半胱氨酸（2.00eq）】（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL003565

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.8.(4) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.12.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.8.(4) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

★Note: The quantity of Fmoc-L-Cys(Trt)-OH is a CPP./备注：Fmoc-S-三苯甲基-L-半胱氨酸总用量是关键工艺参数

★NOR limit/正常操作范围：_____kg~_____ kg

Filled by& Date/填写人及日期：_____

Checked by & Date/核实人及日期：_____

★After dispensing, the operator shall confirm that Critical Process Parameters (CPP) is within the respective Normal Operating Ranges (NOR)/ 分装结束后，操作人员须确认关键工艺参数（CPP）均处于正常操作范围（NOR）内：

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.9.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（3.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.9.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.12.3					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.10.(5) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Thr(tBu)-OH. /分装_____kg芴甲氧羰基--叔丁基-L-苏氨酸（2.00eq）（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL003563

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.10 (5) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.15.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.11.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.11.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.15.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.12.(6) Dispense Fmoc-L-Ala-OH•H_2O/分装芴甲氧羰基-L-丙氨酸一水合物

Before material dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Ala-OH•H_2O. /分装_____kg芴甲氧羰基-L-丙氨酸一水合物（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL006160

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.12.(6) Dispense Fmoc-L-Ala-OH•H_2O/分装芴甲氧羰基-L-丙氨酸一水合物

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.18.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.13.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.13.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.18.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total/合计				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.14.(7) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Thr(tBu)-OH. /分装_____kg芴甲氧羰基--叔丁基-L-苏氨酸（2.00eq）（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL003563

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.14 (7) Dispense Fmoc-L-Thr(tBu)-OH/分装芴甲氧羰基-叔丁基-L-苏氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.21.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.15.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date. Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.15.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.21.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.16.(8) Dispense Fmoc-L-Asn(Trt)-OH/分装 Fmoc-N-三苯甲基-L-天冬酰胺

Before material dispensing, confirm the material is within the retest date/ the expiry date. Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Asn(Trt)-OH. /分装_____kg Fmoc-N-三苯甲基-L-天冬酰胺（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL004843

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.16. (8) Dispense Fmoc-L-Asn(Trt)-OH/分装 Fmoc-N-三苯甲基-L-天冬酰胺

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.24.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后（开放性操作后），确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.17.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date.Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.17 Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.24.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

Document Copy: Check that version # matches current version # in QualityDocs.
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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.18.(9) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Fmoc-L-Cys(Trt)-OH. /分装_____kg Fmoc-S-三苯甲基-L-半胱氨酸（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL003565

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.18.(9) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.27.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.18.(9) Dispense Fmoc-L-Cys(Trt)-OH /分装 Fmoc-S-三苯甲基-L-半胱氨酸

★Note:The quantity of Fmoc-L-Cys(Trt)-OH is a CPP./备注：Fmoc-S-三苯甲基-L-半胱氨酸总用量是关键工艺参数

★NOR limit/正常操作范围：_____kg~ _____kg

Filled by& Date/填写人及日期：_____

Checked by & Date/核实人及日期：_____

★After dispensing, the operator shall confirm that Critical Process Parameters (CPP) is within the respective Normal Operating Ranges (NOR)/ 分装结束后，操作人员须确认关键工艺参数（CPP）均处于正常操作范围（NOR）内:

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

Document Copy: Check that version # matches current version # in QualityDocs.
Retrieved by Simon Cushen on 29 Oct 2026 at 12:35PM GMT+00:00.

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.19.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（3.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.19 Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.27.3					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.20.(10) Dispense Boc-L-Glu-OtBu/分装 Boc-L-谷氨酸-1-叔丁酯

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Boc-L-Glu-OtBu. /分装_____kg Boc-L-谷氨酸-1-叔丁酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL008618

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.20.(10) Dispense Boc-L-Glu-OtBu/分装 Boc-L-谷氨酸-1-叔丁酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.30.2					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.21.Dispense PyBOP/分装 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense PyBOP. /分装_____kg 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL002855

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.21 Dispense PyBOP/分装 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.30.4					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后（开放性操作后），确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.22.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/固体物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense OxymaPure. /分装_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494"原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL005180

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.22.Dispense OxymaPure/分装 2-肟氰乙酸乙酯

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.30.5					/		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.23.Dispense Iodine/分装碘

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Iodine. /分装_____kg碘（5.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494“原料汇总表”中对应的每个单元操作。

Control No./内部控制号: YL003092

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.23.Dispense Iodine/分装碘

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.32.3					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing, confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.24.Dispense Iodine/分装碘

Before material dispensing, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current dispensing. /物料分装前，确认物料在复检日期/有效期之内，确认重量是否满足本次分装使用。

Before solid material dispensing, confirm/物料分装前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /开放性操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before dispensing, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，分装前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Dispense Iodine. /分装_____kg碘（5.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。

For the Standard Batch Quantity, see Page 30~47 of 494 the information in "Raw Material Summary Table" corresponding to each unit operation/对于批标准投料量，参照 Page 30~47 of 494 "原料汇总表"中对应的每个单元操作。

Control No./内部控制号: YL003092

Scale ID. No./秤编号: _____

Lot number of LDPE powder feeding bag/低密度聚乙烯粉末投料袋批号: _____

Glovebox number/手套箱编号: _____

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. /砝码编号	Weight/砝码重量（kg）	Reading/示重（kg）	Less than half of verification scale interval/小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
			☐Y ☐N		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

1.24.Dispense Iodine/分装碘

Ins. No. for material use/分装物料计划用于的操作	Lot number/批号	Charging Qty/投入的质量			No. /编号	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Gross weight/毛重	Tare weight/皮重	Net weight/净重			
3.33.3					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
					____/____		
	Total weight /合计重量				-		
	Total No. of bags/合计袋数				-		
Begin/开始		**End/结束**					

After solid material dispensing , confirm/固体物料分装之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

Ins.2/操作 2: Equipment confirmation & Recipe Confirmation /设备确认和配方确认

2.1. Equipment confirmation/设备确认

Before using, confirm Peptide Solid Phase Synthesizer (referred as synthesis reactor) oxygen content≤1.0%.

使用前确认多肽固相合成仪 (以下简称合成釜)氧气含量≤1.0%。

3000 L Peptide Solid Phase Synthesizer (referred as synthesis reactor)

多肽固相合成仪 (以下简称合成釜): TJ4S-1213-Sy16

Oxygen detector ID. No. /氧气检测器编号: _____

Oxygen content/氧气含量=_____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期: _____

Before using, confirm the synthesis reactor TJ4S-1213-Sy16's calibration information is within valid date. /使用前确认合成釜 TJ4S-1213-Sy16 的仪表在校准有效期内。

Checked by & Date/核实人及日期: _____

Before using, confirm that cleaning of the synthesis reactor TJ4S-1213-Sy16 has been completed/使用前，确认合成釜 TJ4S-1213-Sy16 清洁已完成

Checked by & Date/核实人及日期: _____

Before using, confirm solid-phase peptide synthesizer activation tank (referred as activation tank) oxygen content≤1.0%.

使用前确认多肽固相合成仪激活罐(以下简称激活罐)氧气含量≤1.0%。

1500 LSolid-phase peptide synthesizer activation tank (referred as activation tank)

多肽固相合成仪激活罐(以下简称激活罐): TJ4S-1213-Sy16-R01

Oxygen detector ID. No. /氧气检测器编号: _____

Oxygen content/氧气含量=_____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

2.1. Equipment confirmation/设备确认

Before using, confirm the activation tank TJ4S-1213-Sy16-R01's calibration information is within valid date. /使用前确认激活罐 TJ4S-1213-Sy16-R01 的仪表在校准有效期内。

Checked by & Date/核实人及日期: _____

Before using, confirm that cleaning of the activation tank TJ4S-1213-Sy16- R01 has been completed/ 使用前，确认激活罐 TJ4S-1213-Sy16- R01 清洁已完成

Checked by & Date/核实人及日期: _____

2.2. Recipe confirmation/配方确认

The recipe is edited and established on the synthesis equipment according to the requirements of Ins.3. /配方是根据指令 3 的工艺操作要求在合成设备上编辑和建立的。

Confirm synthesis recipe. /确认已经建立的合成配方:

1.Double editing / 双人编辑:Two engineers shall edit the electronic recipe separately for the same project. 由两名工程师对同一项目分别进行电子配方编辑

2.Consistency comparison / 一致性比对:Compare the electronic recipes edited by the two engineers, confirm each difference between them one by one, and correct any errors. To ensure the electronic recipe to be run is reviewed and accurate. 将双人编辑后的电子配方进行比对，对两个电子配方的不同点逐一进行确认，对编辑有误的电子配方进行纠正，确保在即将运行的电子配方是经过复核且准确的。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Recipe name /配方名称: 14002101M01V02

Batch name /批次命名: CPo140021_____01_____

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

2.3.Tank confirmation/罐区确认

The 20%v/v Piperidine/DMF solution is prepared and the bottom valve of tank TJ4S-1213-SPS01-AT01、TJ4S-1213-SPS01-AT02 is opened. /

20%v/v 哌啶/DMF 溶液完成配制，储罐 TJ4S-1213-SPS01-AT01、TJ4S-1213-SPS01-AT02 底阀已开启;

Confirmed by & Date/确认人及日期: _____

Checked by & date/核实人及日期: _____

The preparation record for 20%v/v Piperidine Record details/配制 20%v/v 哌啶记录详见: _____

Confirm that the DMF supply system from tank farm is supplied. /确认罐区 N,N-二甲基甲酰胺供应系统已准备就绪;

Confirmed by & Date/确认人及日期: _____

Checked by & date/核实人及日期: _____

Confirm that the IPAc supply system from tank farm is supplied. /确认罐区乙酸异丙酯储罐供应系统已准备就绪;

Confirmed by & Date/确认人及日期: _____

Checked by & date/核实人及日期: _____

Confirm that the MTBE supply system from tank farm is prepared. /确认罐区甲基叔丁基醚储罐供应系统已准备就绪;

Confirmed by & Date/确认人及日期: _____

Checked by & date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

2.4.Run Recipe /配方运行

2.4.1 Invoke synthesis recipe. /调用已经建立的合成配方:

Recipe name /配方名称: 14002101M01V02

Batch name /批次命名: CPo140021_____01_____

Move to Modules /调用模块: 3.0

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

2.4.2Review of electronic recipe/配方复核

When reviewing an electronic recipe, the items to be checked include but are not limited to the recipe name, instructional operational logic, key process parameters, alarm parameters, and other parameters that may affect product quality. This is to ensure the electronic recipe meets the requirements for project production and GMP management./ 复核电子配方时，复核的内容包括但不限于配方名称、指令运行逻辑、关键工艺参数、报警参数及其他可能影响产品质量的参数，以确保该电子配方能够满足项目生产及 GMP 管理的要求。

Final review / 最终复核: Before the running, production personnel and process chemist shall conduct the final review of the electronic recipe. After the review is correct, they shall sign and date on the Operation Record (OR). 正式运行前，生产人员与工艺技术人员需共同对电子配方进行最终复核；复核无误后，在操作记录（OR）上签名并注明日期。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

2.4.3 Confirm the synthetic recipe is selected correctly. And the peptide solid phase synthesizer is ready for use. The instrument is ready for all work. /确认合成配方选用正确，固相合成仪各项工作准备就绪。

Before the electronic recipe is run, it is necessary to review the key information such as the name, version, and batch number of the electronic recipe to be run. 电子配方运行前，需要对即将运行的电子配方的名称、版本、运行批号等关键信息进行复核。

If the electronic recipe is locked before running, it possible that the electronic recipe cannot be run directly. After confirming that the information is correct, it can be unlocked and run. 若运行前电子配方处于锁定状态，可能无法直接运行该电子配方，可在确认信息无误后进行解锁后，正式运行该电子配方。

Click run button and start synthesis. /点击运行，开始合成。

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.3/操作 3: Synthesis/合成

3.0.Pre-treatment/预处理

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

	Times /次数
3.0(1). DMF Addition and KF Testing/检测含水: **Add 369.85kg DMF（5.69W,6.0Vol）by spray balls into activation tank TJ4S-1213-Sy16-R01.Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Stir for 10min±5min (target: 10min). Then transfer the solution into synthesis reactor TJ4S-1213-Sy16. Confirm complete transfer. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Stir for 10min±5min (target: 10min). Stop the stirring and turn off the nitrogen. Sample the draining solution for water content analysis.Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.If KF≤500ppm, move to next operation. If KF>500 ppm, repeat Module 3.0 until KF≤500 ppm.** **通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N,N-二甲基甲酰胺（5.69W,6.0Vol）(Acceptable weighing range/可接受称量范围：347.66kg~392.04kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。搅拌 10min±5min（目标时间:10min）后，转移至固相合成釜 TJ4S-1213-Sy16.中。确认转移完成。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10min±5min（目标时间:10min）后，停止搅拌，关闭氮气，取样检测体系 KF，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。若含水≤500ppm，则进行下一操作；若含水＞500ppm，则重复进行模块 3.0 直至含水≤500ppm**	**1**

Add 369.85kg DMF（5.69W,6.0Vol）by spray balls into activation tank TJ4S-1213-Sy16-R01/通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N,N-二甲基甲酰胺（5.69W,6.0Vol）

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.0 Pre-treatment/预处理

Start the agitator of activation tank to 115±5rpm (target: 115 rpm). Stir for 10min±5min (target: 10min). /开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。搅拌 10min±5min（目标时间:10min）

Pre-treatment/预处理	Time point/时间节点	Wash /洗涤
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始**	
	End/结束	
	Transcribed by & Date/转录人及日期:	

Then transfer the solution into synthesis reactor TJ4S-1213-Sy16. Confirm complete transfer./转移至固相合成釜 TJ4S-1213-Sy16，确认转移完成

Transfer/转移	**Begin/开始**	**End/结束**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Stir for 10min±5min (target: 10min). Stop the stirring and turn off the nitrogen. Sample the draining solution for water content analysis.. /开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10min±5min（目标时间:10min）停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始**	
	End/结束	
	Transcribed by & Date/转录人及日期:	
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始**	
	End/结束	
	Transcribed by & Date/转录人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.0 Pre-treatment/预处理

Standard sampling method/标准取样方法

Approximately 10~20 ml of the mixture is withdrawn from the reactor via a dedicated in-line sampler and submit for analysis testing. During sampling, should wear nitrile gloves and 3M 6000 series Full-face piece respirator, 6006CN cartridges.
通过在线取样器从反应釜内抽取约取体系约 10~20ml，直接送样检测。取样过程中，应该戴丁腈手套、3M 6000 系列全面罩, 6006CN 滤毒盒。

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Water content/ 水分	Result filled by &Date/结果填写人及日期
CPo140021-_____-01-_____-3.0-01				

Drain via diaphragm pump until there is no drip, the operator visually confirms the draining end point.If KF≤500ppm, move to next operation. If KF>500 ppm, repeat Module 3.0 until KF≤500 ppm./ 通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。若含水≤500ppm，则进行下一操作；若含水 > 500ppm，则重复进行模块 3.0 直至含水≤500ppm

Drain waste after sampling./取样后排废

Filter/过滤	**Begin/开始**	**End/结束**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期	

Confirms KF≤500ppm. /确认含水≤500ppm。

☐**Confirms KF≤500ppm, move to Modules 3.1-3.32. /确认含水≤500ppm。调用模块：3.1-3.32。**

☐**KF>500ppm, move to Module 3.0(2) /含水>500ppm，进行模块3.0(2)。**

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.0 Pre-treatment/预处理

☐**3.0(2). DMF Addition and KF Testing/检测含水:**

Add 369.85kg DMF（5.69W,6.0Vol）by spray balls into activation tank TJ4S-1213-Sy16-R01.Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Stir for 10min±5min (target: 10min). Then transfer the solution into synthesis reactor TJ4S-1213-Sy16. Confirm complete transfer. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Stir for 10min±5min (target: 10min). Stop the stirring and turn off the nitrogen. Sample the draining solution for water content analysis.Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.If KF≤500ppm, move to next operation. If KF>500 ppm, repeat Module 3.0 until KF≤500 ppm.

通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N,N-二甲基甲酰胺（5.69W,6.0Vol）（Acceptable weighing range/可接受称量范围：347.66kg~392.04kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。搅拌 10min±5min（目标时间:10min）后，转移至固相合成釜 TJ4S-1213-Sy16.中。确认转移完成。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10min±5min（目标时间:10min）后，停止搅拌，关闭氮气，取样检测体系 KF，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。若含水≤500ppm，则进行下一操作；若含水＞500ppm，则重复进行模块 3.0 直至含水≤500ppm

Add 369.85kg DMF（5.69W,6.0Vol）by spray balls into activation tank TJ4S-1213-Sy16-R01/通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N,N-二甲基甲酰胺（5.69W,6.0Vol）

☐ **DMF addition/ 加 N,N-二甲基甲酰胺**	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Start the agitator of activation tank to 115±5rpm (target: 115 rpm). Stir for 10min±5min (target: 10min). /开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。搅拌 10min±5min（目标时间:10min）

Pre-treatment/预处理	Time point/时间节点	Wash /洗涤
Stir/搅拌	**Performed by & Date/执行人及日期:**	

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.0 Pre-treatment/预处理

Pre-treatment/预处理	Time point/时间节点	Wash /洗涤
Stir /搅拌	Begin/开始	
	End/结束	
	Transcribed by & Date/转录人及日期:	

Then transfer the solution into synthesis reactor TJ4S-1213-Sy16. Confirm complete transfer./转移至固相合成釜 TJ4S-1213-Sy16.中。确认转移完成

Transfer/转移	Begin/开始	End/结束
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Stir for 10min±5min (target: 10min). Stop the stirring and turn off the nitrogen. Sample the draining solution for water content analysis.. /开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10min±5min（目标时间:10min）停止搅拌，关闭氮气

Bubble/鼓氮气	Performed by & Date/执行人及日期:	
Bubble/鼓氮气	Begin/开始	
	End/结束	
	Transcribed by & Date/转录人及日期:	
Stir/搅拌	Performed by & Date/执行人及日期:	
Stir /搅拌	Begin/开始	
	End/结束	
	Transcribed by & Date/转录人及日期:	

2nd KF IPC Standard sampling method/2nd KF IPC 标准取样方法

Approximately 10–20 ml of the mixture is withdrawn from the reactor via a dedicated in-line sampler and submit for analysis testing. During sampling, should wear nitrile gloves and 3M 6000 series Full-face piece respirator, 6006CN cartridges.
通过在线取样器从反应釜内抽取约取体系约 10~20ml，直接送样检测。取样过程中，应该戴丁腈手套、3M 6000 系列全面罩, 6006CN 滤毒盒。

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.0 Pre-treatment/预处理

2nd KF IPC **Sampling number/ 第二次 KF 检测取样编号**	**Sampling time/取样时间**	**Sampled by & Date/取样人及日期**	**Water content/ 水分**	**Result filled by &Date/结果填写人及日期**
CPo140021_____-01-_____-3.0-02				

Drain via diaphragm pump until there is no drip, the operator visually confirms the draining end point.If KF≤500ppm, move to next operation. If KF>500 ppm, repeat Module 3.0 until KF≤500 ppm./ 通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。若含水≤500ppm，则进行下一操作；若含水＞500ppm，则重复进行模块 3.0 直至含水≤500ppm :

Drain waste after sampling./取样后排废

Filter/过滤	**Begin/开始**	**End/结束**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Invoke synthesis recipe. /调用已经建立的合成配方:

Recipe name /配方名称 14002101M01V02

Batch name /批次命名：CPo140021_____01_____

☐**Move to modules 3.1~3.32/调用 3.1~3.32 模块;**

☐ **Rerun to modules 3.0/重复调用 3.0 模块;**

（From/上接: _____**; To/下转:** _____**);**

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Confirm the synthetic recipe is selected correctly and the instrument is ready for work. /确认合成配方选用正确，仪器各项工作准备就绪。

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Click run button and start synthesis. /点击运行，开始合成。

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.Resin swelling/树脂溶胀

CAUTION/注意：2-Chlorotrityl chloride resin: OEB3:
The resin shall not be allowed to swell for an excessive duration. Over-drainage shall be avoided, as excessive drying may cause the resin to absorb water and subsequently lose activity. / 2-氯三苯甲基氯树脂 OEB3）：树脂不可溶胀过长时间。应避免过度抽干，因过度干燥可能导致树脂吸水并进而失活。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

The date when this operation stating is defined as manufacture date./此操作开始的日期被定义为生产日期。

3.1.1 Adjust the synthesis reactor jacket temperature TJ4S-1213-Sy16 to 25±5℃ (target: 25℃). **将合成釜夹套调温至 25±5℃ (目标温度：25℃)**	**Times/次数**
	1

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

3.1.2Add 65.00 kg 2-Chlorotrityl chloride resin (1.00eq) with powder feeding bags after dispensing. **通过投料袋向合成釜 TJ4S-1213-Sy16 中加入 65.00kg 2-氯三苯甲基氯树脂（1.00eq）(Acceptable weighing range/可接受称量范围：64.87~65.13 kg)。**	**Times/次数**
	1

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢No non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.1.Resin swelling/树脂溶胀

Prior to charging resin, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./投树脂前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge 2-Chlorotrityl chloride resin/加 2-氯三苯甲基氯树脂	**Performed by & Date/执行人及日期:**	
Charge 2-Chlorotrityl chloride resin/加 2-氯三苯甲基氯树脂	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

2-Chlorotrityl chloride resin confirmation/2-氯三苯甲基氯树脂确认:

Control No./内部控制号: YL003422

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No. /内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.Resin swelling/树脂溶胀

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
Total/合计		**Confirm the net weight is within the input range/确认合计净重在投料范围内**	**Recorded by & Date/记录人及日期**	**Checked by & Date/ 核实人及日期**
		☐Yes/是; ☐No/否		

After solid material charging, confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.Resin swelling/树脂溶胀

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.1.3 Rinse synthesis reactor (TJ4S-1213-Sy16) wall by spray balls with 492.70 kg (7.58W, 8.0Vol) DMF. **通过淋洗球向合成釜（TJ4S-1213-Sy16）中加入 492.70 kg（7.58W,8.0Vol）N,N-二甲基甲酰胺（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg）。**	**Times /次数**
	1

DMF addition/加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.1.4 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target 5min). Then the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min (目标时间：5min)后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。	**Times /次数**
	1

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.1.5 /结束时间记录于 3.1.5
Transcribed by & Date/转录人及日期:	

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.1.5 /结束时间记录于 3.1.5
Transcribed by & Date/转录人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.1.Resin swelling/树脂溶胀

	Times /次数
3.1.5 Adjust the internal temperature to 25±5℃ (target: 25℃). Continue stirring for 45±15min (target 45min) at 25±5℃ (target: 25℃), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **将内温调温至 25±5℃(目标温度：25℃)，控内温 25±5℃(目标温度：25℃)，继续搅拌 45±15min (目标时间：45min)后停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ℃， High limit/高限： 28℃**	**1**

Adjust to 25±5℃/调温至 25±5℃	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Continue stirring for 45±15min (target 45min) at 25±5℃ (target: 25℃), stop stirring,turn off the nitrogen. /继续搅拌 45±15min (目标时间：45min)后停止搅拌，关闭氮气。		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins3.1.4 /开始时间记录于 3.1.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.1.4 /开始时间记录于 3.1.4	
Transcribed by & Date/转录人及日期:		
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.（1）Dissolution of Fmoc-Gly-OH/芴甲氧羰基-甘氨酸溶解

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Note:备注：Ins 3.1.6~Ins3.1.8 and Ins 3.1.9~Ins3.1.10 need to be performed simultaneously.操作 3.1.6~操作 3.1.8 与操作 3.1.9~操作 3.1.10 同时进行

3.1.6 Add solvent to the activation tank/激活罐加溶剂:	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 25±5ºC (target: 25ºC). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into the activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 25±5ºC（目标温度：25ºC）。向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N,N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围：351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm).**	**1**

DMF addition/加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			
Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)			
Stir/搅拌	**Performed by & Date/执行人及日期:**		
Stir /搅拌	**Begin/开始:**		
	The end time is documented in Ins 3.3.3 结束时间记录于 3.3.3		
Transcribed by & Date/转录人及日期:			

3.1.7 Add Fmoc-Gly-OH with powder feeding bags into the activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-甘氨酸（2.25eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）。**	**Times /次数**
	1

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.1.（1）Dissolution of Fmoc-Gly-OH/芴甲氧羰基-甘氨酸溶解

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-Gly-OH/ 加芴甲氧羰基-甘氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-Gly-OH/ 加芴甲氧羰基-甘氨酸	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		
Fmoc-Gly-OH confirmation/芴甲氧羰基-甘氨酸确认: Control No./内部控制号: YL004846		

Check items/检查项	**Is it correct?/是否正确**	**Performed by & Date/ 执行人及日期**	**Checked by & Date/ 核实人及日期**
Material name/物料名称	☐**Yes/是;** ☐**No/否**		
Project batch number/ 项目批次号	☐**Yes/是;** ☐**No/否**		
Operation number/ 操作号	☐**Yes/是;** ☐**No/否**		
Control No. /内部控制号	☐**Yes/是;** ☐**No/否**		
Raman Scan Confirmation/拉曼扫描确认	☐**Conforms/相符** ☐ **Does not conform/不相符**		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.（1）Dissolution of Fmoc-Gly-OH/芴甲氧羰基-甘氨酸溶解

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
Total/合计		**Confirm the net weight is within the input range/确认合计净重在投料范围内**	**Recorded by & Date/计录人及日期**	**Checked by & Date/ 核实人及日期**
		☐Yes/是; ☐No/否		

After solid material charging ,confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意: DMF/N,N-二甲基甲酰胺: Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.1.（1）Dissolution of Fmoc-Gly-OH/芴甲氧羰基-甘氨酸溶解

	Times /次数
3.1.8 Rinse the port of the activation tank TJ4S-1213-Sy16-R01 with 61.75kg (0.95W, 1.0Vol) DMF by spray balls. Adjust internal temperature of the mixture to 15±5ºC. Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min at 15±5ºC (target: 15ºC).Confirm no visible solids and later use. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W, 1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗激活罐投料口。内温调温至 15±5ºC。控内温 15±5ºC（目标温度:15ºC），维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min，确认无明显固体，待用。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：12 ºC，High limit/高限： 18ºC**	**1**

Rinse the port of the activation tank TJ4S-1213-Sy16-R01 with 61.75kg (0.95W, 1.0Vol) DMF by spray balls. 通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W, 1.0Vol）。

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Adjust internal temperature of the mixture to 15±5ºC/内温调温至 15±5oC。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min at 15±5ºC (target: 15ºC) and later use./ 控内温 15±5ºC（目标温度:15ºC），维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min，待用。

Stir/保温	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.1.Resin Post-Swelling Wash /树脂溶胀后洗涤

Note:备注：Ins 3.1.6~Ins3.1.8 and Ins 3.1.9~Ins3.1.10 need to be performed simultaneously.操作 3.1.6~操作 3.1.8 与操作 3.1.9~操作 3.1.10 同时进行

3.1.9 Post-Swelling Wash/溶胀后洗涤:	**Times/次数**
Add 308.10kg DMF (4.74W, 5.0Vol) by spray balls into synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 308.10kg N,N-二甲基甲酰胺（4.74W, 5.0Vol）。（Acceptable weighing range/可接受称量范围：289.61kg~326.59kg）。**	**1**

DMF addition/加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.1.10 Post-Swelling Wash/溶胀后洗涤:	**Times/次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h).Bubble with nitrogen for 5±2 min (target: 5min). . Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min (目标时间：5min)后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h).Bubble with nitrogen for 5±2 min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring,turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min (目标时间：5min)后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.1.Resin Post-Swelling Wash /树脂溶胀后洗涤

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.2.（1）Fmoc-Gly-OH activation /芴甲氧羰基-甘氨酸活化

CAUTION/注意：DIPEA/N,N-二异丙基乙胺：It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Checked by & Date/核实人及日期：_____

	Times/ 次数
3.2.1 Add DIPEA (6.00eq) into activation tank TJ4S-1213-Sy16-R01 through metering pump at a reference rate of 20~40 kg/h at 15±5°C (target: 15°C). The addition rate depends on temperature control. **控内温 15±5°C (目标温度：15°C)，向激活罐 TJ4S-1213-Sy16-R01 通过计量泵加入_____kg N,N-二异丙基乙胺（6.00eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 20~40kg/h，实际以控内温为主。** **Weighing of DIPEA/N,N-二异丙基乙胺称重:** **Control Number/内部控制号：YL000555 Scale ID. No./秤编号：_____** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date, confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期：_____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.2（1）Fmoc-Gly-OH activation /芴甲氧羰基-甘氨酸活化

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

DIPEA Addition/加 N,N-二异丙基乙胺 **Control Number/内部控制号: YL000555**

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.2.（1）Fmoc-Gly-OH activation /芴甲氧羰基-甘氨酸活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIPEA Addition/加 N,N-二异丙基乙胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.2（1）Fmoc-Gly-OH activation /芴甲氧羰基-甘氨酸活化

CAUTION/注意：DIPEA/N,N-二异丙基乙胺：It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Checked by & Date/核实人及日期：_____

3.2.2 At 15±5℃ (target: 15℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Stir for 5±5min (Target:5min) after addition. **控内温 15±5℃(目标温度：15℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 5±5min(目标时间：5min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：12 ℃，High limit/高限：18℃**	**Times/ 次数** **1**

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
	Stirring start/搅拌开始时间	Transfer start/转移开始时间
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

3.3.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times /次数**
	1

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.3.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W，1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

3.3.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min).Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture (DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times /次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min).Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins 3.3.5/结束时间记录于操作 3.3.5	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.3.5/结束时间记录于操作 3.3.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture (DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16./
将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.1.6/开始时间记录于 3.1.6	
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

3.3.4 Adjust temperature to 25±5°C in synthesis reactor TJ4S-1213-Sy16. **合成釜 TJ4S-1213-Sy16 内温调温至 25±5°C。**	**Times/次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

3.3.5 The mixture is allowed to react at internal temperature【25±5°C (target: 25°C, starts timing when the temperature reaches 20°C)】for 2.5h~5.0h (target:3h).Stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5°C（目标温度:25°C）】(保温开始以温度达到 20°C 开始计时)，反应 2.5h~5.0h（目标时间:3h）. 停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 °C，High limit/高限：28°C**	**Times/次数** **1**

The total reaction Fmoc-Gly-OH coupling time: NMT 480min (Sourced from the holding time study in the PAR study report.)/芴甲氧羰基-甘氨酸偶联总的反应时间不超过 480min(来源于 PAR 研究报告中稳定时间研究)。

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.3.7./偶联反应结束时间即为 Ins 3.3.7 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

Internal temperature/内部温度	Highest values/最高值:	Lowest values/最低值:

Transcribed by & Date/转录人及日期:

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins 3.3.3/开始时间记录于操作 3.3.3	
Transcribed by & Date/转录人及日期		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.3.3/开始时间记录于操作 3.3.3	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.3.6 Activation Tank Rinse/激活罐洗涤	**Times/次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 492.70kg DMF (7.58W, 8.0Vol) by spray balls.Circulate and clean the activating liquid transfer pipeline 1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 492.70kg N, N-二甲基甲酰胺（7.58W ,8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg），循环清洗激活液转移管路 1，和液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

Operation name/ 操作名称	Time point/时间节点	Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1	liquid charging port cleaning pipeline 2,/ 液体加料口清洗管路 2	Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3
		Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N,N-二甲基甲酰胺	Performed by & Date/ 执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	Performed by & Date/ 执行人及日期:			
Circulate and clean/ 循环清洗	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

3.3.7 Post-coupling wash./偶联后洗涤	Times /次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 .Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2 min (target:5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.3.（1）Fmoc-Gly-OH coupling/芴甲氧羰基-甘氨酸偶联

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Transfer/转移	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间: 10min）后，停止搅拌，关闭氮气.

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Stir /搅拌	Performed by & Date/ 执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./ 通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4.（1）Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.4.1 Post-coupling wash./偶联后洗涤	**Times /次数**
Add 554.45 kg DMF (8.53W, 9Vol) by spray balls into synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 554.45 kg N,N-二甲基甲酰胺（8.53W, 9Vol）(Acceptable weighing range/可接受称量范围: 521.18kg~587.72kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5min±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5min±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins 3.4.5/结束时间记录于 3.4.5
Transcribed by & Date/转录人及日期: _____	
Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins 3.4.5/结束时间记录于 3.4.5
Transcribed by & Date/转录人及日期:	

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.4. （1）Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

CAUTION/注意： Methanol/甲醇: Exposure to excessive vapor causes eye irritation, head- ache, fatigue and drowsiness. High concentrations can produce central nervous system depression and optic nerve damage. 50,000 ppm will probably cause death in 1 to 2 hrs. Can be absorbed through skin. Swallowing may cause death or eye damage. /接触过多的蒸汽会引起眼睛刺激、头痛、疲劳和困倦。高浓度可导致中枢神经系统抑制和视神经损伤。50000 ppm 可能会在 1 到 2 小时内导致死亡。可通过皮肤吸收。吞咽可能导致死亡或眼睛损伤。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.4.2 Add methanol (10.00eq) through peristaltic pump into synthesis reactor TJ4S-1213-Sy16. Adjust the internal temperature to 17~23ºC(Target:20ºC). **通过蠕动泵向合成釜 TJ4S-1213-Sy16 中加入_____kg 甲醇（10.00eq）(Acceptable weighing range/可接受称量范围: _____kg~_____kg)，将内温调温至 17~23ºC（目标温度 20ºC）。**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date. Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4. （1）Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Weighing of methanol/甲醇称重:

Control Number/内部控制号: YL000031

Scale ID. No./秤编号: _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.4. Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐Non-full drum /非整桶											
☐Full drum 整桶 ☐Non-full drum /非整桶											
Total/合计											

Methanol addition/加甲醇	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.4.Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DIPEA/N,N-二异丙基乙胺：It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

★Note: The temperature of charging DIPEA is a CPP./ 备注：加 N,N-二异丙基乙胺温度是关键工艺参数。

	Times /次数
3.4.3 Add DIPEA (5.00eq) (reference rate of 20~40 kg/h) into synthesis reactor TJ4S-1213-Sy16 at【17~23°C】（Target:20 °C）. The addition rate depends on temperature control. **控内温【17~23°C】(目标温度 20°C)，向合成釜 TJ4S-1213-Sy16 中加入_____kg N,N-二异丙基乙胺（5.00eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考滴加速度 20~40kg/h，以控内温为主。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限: 18°C，High limit/高限：22°C**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date.Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4.Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Weighing of DIPEA/N,N-二异丙基乙胺称重:

Control Number/内部控制号: YL000555 Scale ID. No./秤编号: _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.4.Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIPEA Addition/加 N,N-二异丙基乙胺	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	142 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.4 Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Internal temperature/内部温度	Highest values/最高值:	Lowest values/最低值:

Transcribed by & Date/转录人及日期:

★After capping is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 封端结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The temperature of charging DIPEA is a CPP./ 备注：加 N,N-二异丙基乙胺温度是关键工艺参数。

★Prior to capping (DIPEA addition), the operator shall confirm that and the reactor temperature is within the Normal Operating Range (NOR) :17.0~23.0 ºC /封端前（N,N-二异丙基乙胺投料）前，操作人员须确认反应釜温度处于正常操作范围（NOR）内:17.0~23.0 ºC

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

3.4.4 Adjust synthesis reactor TJ4S-1213-Sy16 internal temperature to 25±5 ℃ (target: 25 ℃) after addition. 加毕，将合成釜 TJ4S-1213-Sy16 内温调温至 25±5 ℃(目标温度：25 ℃)。 Over- internal temperature warning value/内温超温预警设置数值: Low limit/低限: 22 ºC ， High limit/高限： 28ºC	Times /次数
	1

Adjust/调温	Performed by & Date/执行人及日期:	
Adjust/调温	Begin/开始:	End/结束

Transcribed by & Date/转录人及日期:

3.4.5 The mixture is allowed to react at internal temperature 25±5ºC (target: 25ºC), start timing when temperature reaches 20ºC) for 60min±30min (target: 60min), Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. 控内温 25±5ºC（目标温度 25ºC，20ºC 开始计时），反应 60min±30min（目标时间：60min），停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	Times /次数
	1

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4 Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the capping serves as the start time of Wash 1 specified in Ins 3.4.6./封端反应结束时间即为 Ins 3.4.6. 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**End/结束:**	
Stir /搅拌	The start time is documented in Ins3.4.1 /开始时间记录于 3.4.1	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.4.1 /开始时间记录于 3.4.1	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4.Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

	Times/ 次数
3.4.6 Post-capping wash/封端后洗涤: **Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After Stir for 15min±2min (target: 15min). stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min (目标时间: 15min)后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.4.Capping after Fmoc-Gly-OH coupling /芴甲氧羰基-甘氨酸偶联后封端

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N,N-二甲基甲酰胺	Performed by & Date/执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min (目标时间: 15min)				
Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Stir /搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Transfer/转移	Begin/开始			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times /次数
3.5.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). **Add 483.60kg 【20% (V/V) piperidine/DMF (7.44W, 8.0Vol)】 solution into the synthesis reactor.** **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

	Times /次数
3.5.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5 ºC (target: 25ºC). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5ºC(目标温度：25 ºC)。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins 3.5.3/结束时间记录于 3.5.3

Transcribed by & Date/转录人及日期:

Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). /开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins 3.5.3/结束时间记录于 3.5.3

Transcribed by & Date/转录人及日期:

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ºC(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.5.3 Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护	**Times /次数**
After stirring for 110~130 min (target: 2 h) at 25±5 ℃(target: 25 ℃), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温 25±5℃(目标温度：25 ℃)， 搅拌 110~130 min（目标时间：2 h）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.5.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.5.1	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.5.4/脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.5.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.5.2 /开始时间记录于 3.5.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

★After De-Fmoc is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 脱保护结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: 20%piperidine quantity is a CPP./备注：20%哌啶溶液质量是关键工艺参数

★The operator shall confirm that 20%piperidine quantity is within the Normal Operating Range (NOR) 【454.58kg~512.62kg】/操作人员须确认 20%哌啶质量处于正常操作范围（NOR）内：【454.58kg~512.62kg】

20%Piperidine quantity confirmation/20%哌啶溶液质量确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

CAUTION/注意：IPAC/乙酸异丙酯:
Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.5.4~Ins3.5.8 and Ins 3.6.1~Ins3.6.3 need to be performed simultaneously.操作 3.5.4~操作 3.5.8 与操作 3.6.1~操作 3.6.3 同时进行

	Times/次数
3.5.4 Post-deprotection wash/脱保护后洗涤 **Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min).Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.** **向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg）					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。					
Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.5.5 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg）。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.5.6 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.5.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围: 463.14kg~522.26kg)。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.5.（2）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.5.8 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2 min (target: 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2 min (target: 15 min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2 min（目标时间：15 min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**Times /次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.5.4~Ins3.5.8 and Ins 3.6.1~Ins3.6.3 need to be performed simultaneously.操作 3.5.4~操作 3.5.8 与操作 3.6.1~操作 3.6.3 同时进行

3.6.1 Add solvent to the activation tank/激活罐加溶剂:	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). Add DMF (3.79kg/kg, 4.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C(目标温度: 20°C)。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 246.35kg N, N-二甲基甲酰胺（3.79kg/kg, 4.0Vol）(Acceptable weighing range/可接受称量范围: 234.03kg~258.67kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度：20°C）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add DMF (3.79kg/kg, 4.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01.通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 246.35kg N, N-二甲基甲酰胺（3.79kg/kg, 4.0Vol）(Acceptable weighing range/可接受称量范围: 234.03kg~258.67kg)

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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操作指令

3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins 3.7.4/结束时间记录于 3.7.4

Transcribed by & Date/转录人及日期:

3.6.2 Add Fmoc-L-Thr(tBu)-OH (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-叔丁基-L-苏氨酸（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Thr(tBu)-OH/ 加芴甲氧羰基-叔丁基-L-苏氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Thr(tBu)-OH/ 加芴甲氧羰基-叔丁基-L-苏氨酸	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Fmoc-L-Thr(tBu)-OH confirmation/芴甲氧羰基-叔丁基-L-苏氨酸确认：Control No./内部控制号：YL003563

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.6.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min.Confirm no visible solids and later use.Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 5±5°C(target: 5°C). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm),继续搅拌至少 15min，确认无明显固体，激活罐温度 TJ4S-1213-Sy16-R01 调温至 5±5°C(目标温度: 5°C)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限: 2 °C , High limit/高限: 8°C**	**Times /次数**
	1

Maintain &Stir /保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain &Stir /保持搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.7.4/结束时间记录于 3.7.4	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

CAUTION/注意: PyBOP is high energy compounds, impact sensitive, skin sensitizing. /1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐是高能化合物，撞击敏感性，皮肤致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times /次数
3.6.4Add PyBOP (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 5±5℃ (target: 5ºC). **控内温 5±5℃（目标温度:5℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg~ kg）。**	**1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Charge PyBOP /加 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	**Performed by & Date/执行人及日期:**	
Charge PyBOP /加 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

PyBOP confirmation/1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐确认：Control No./内部控制号：YL002855

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Operations Instruction
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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/ 拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意: OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.6.5 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 5±5℃ (target: 5ºC). **控内温 5±5℃（目标温度:5℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）**	**Times/次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Before solid material charging, confirm/加固体物料前，确认:
➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前；目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

OxymaPure confirmation/2-肟氰乙酸乙酯确认：Control No./内部控制号：YL005180

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging, confirm/固体物料投料之后，确认:
- The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
- The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
- The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Transcribed by & Date/转录人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.6.6 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls.. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）(Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times/次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DIPEA/N,N-二异丙基乙胺：It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	163 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times /次数
3.6.7 Add DIPEA (3.00eq) through metering pump into activation tank TJ4S-1213-Sy16-R01 at a reference rate of 20~40 kg/h at5±5ºC (target: 5ºC). The addition rate depends on temperature control. **控内温 5±5ºC(目标温度：5ºC)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N-二异丙基乙胺（3.00eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 20~40kg/h，实际以控内温为主。** **Weighing of DIPEA/ N,N-二异丙基乙胺称重:** **Control Number/内部控制号：YL000555 Scale ID. No./秤编号：_____** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

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Batch size (output range): 157.69~292.80 kg

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Add DIPEA (3.00eq) through metering pump into activation tank TJ4S-1213-Sy16-R01 at 5±5ºC (target: 5ºC). The addition rate depends on temperature control.控内温 5±5℃(目标温度：5℃)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入 kg N,N-二异丙基乙胺（3.00eq）

DIPEA Addition/加 N,N-二异丙基乙胺	**Begin/开始:**	**End/结束:**
DIPEA Addition/加 N,N-二异丙基乙胺	**Performed by & Date/执行人及日期:**	

DIPEA Addition/加 N,N-二异丙基乙胺 **Control Number/内部控制号：YL000555**

Batch size (output range): 157.69~292.80 kg

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3.6（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIPEA Addition/加 N,N-二异丙基乙胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.6.（2）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times/次数
3.6.8 At 5±5℃ (target: 5℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target: 20min) after addition. **滴毕，控内温 5±5℃(目标温度：5℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 20±5min(目标时间：20min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：2 ºC ， High limit/高限： 8ºC**	**1**

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.7.2/Ins3.7.2 的转移开始时间
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.7.4 /结束时间记录于 3.7.4	
Transcribed by & Date/转录人及日期:		

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.6.1			
Ins 3.6.6			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			
★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【289.61~326.59kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内：【289.61~326.59kg】			☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

3.7.1 Solvent Charging to synthesis reactor (Pre-Coupling)固相釜加溶剂（偶联前）：	**Times/次数**
Add 184.60kg DMF(2.84W,3.0Vol) by spray balls into synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 184.60kg N,N-二甲基甲酰胺（2.84W,3.0Vol）(Acceptable weighing range/可接受称量范围：173.52kg~195.68kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。**	**1**

Add 184.60kg DMF(2.84W,3.0Vol) by spray balls into synthesis reactor TJ4S-1213-Sy16/通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 184.60kg N,N-二甲基甲酰胺（2.84W,3.0Vol）(Acceptable weighing range/可接受称量范围：173.52kg~195.68kg)

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.7.6/结束时间记录于 3.7.6

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.7.6/结束时间记录于 3.7.6

Transcribed by & Date/转录人及日期:

3.7.2 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times/次数** **1**

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.7.3 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times/次数** **1**

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

3.7.4 Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数** **1**

Batch size (output range): 157.69~292.80 kg

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.6.1/开始时间记录于 3.6.1	
Transcribed by & Date/转录人及日期:		

3.7.5 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to【25±5ºC (target: 25ºC)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5℃(目标温度：25 ℃)】。**	**Times/次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.7.6 Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联	Times/次数
The mixture is allowed to react at internal temperature【25±5 ºC (target: 25 ºC)】, start timing when the temperature reaches 20 ºC. The reaction is 3.5h~8h (target: 4h).After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black, react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start drain the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5ºC（目标温度 25ºC，20ºC 开始计时）】反应时间是 3.5h~8h(目标时间：4h)。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h (目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值：Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

The total reaction Fmoc-L-Thr(tBu)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/芴甲氧羰基-叔丁基-L-苏氨酸偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.7.8./偶联反应结束时间即为 Ins 3.7.8 洗涤 1 的开始时间	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测。

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阴性或阳性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.7.6-01			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.7.6-02			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.7.6-03			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.7.6-04			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.7.6-05			□Negative /阴性 □Positive /阳性		

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阴性或阳性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.7.6-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.7.6-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.7.6-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min 的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Stop the stirrer and turn off the nitrogen. Start drain the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./ 停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束:**
	The start time is documented in Ins3.7.1 /开始时间记录于 3.7.1

Transcribed by & Date/转录人及日期:

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**End/结束:**
	The start time is documented in Ins3.7.1 /开始时间记录于 3.7.1

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.6.1			
Ins 3.6.6			
Ins 3.7.1			
Ins3.7.3			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			

★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【521.18~587.72kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内: 【521.18~587.72kg】	☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	174 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

3.7.7 Activation Tank Rinse/激活罐洗涤	**Times/次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 492.70kg DMF (7.58W ,8.0Vol) by spray balls.Circulate and clean the activating liquid transfer pipeline 1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 492.70kg N, N-二甲基甲酰胺（7.58W ,8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg），循环清洗激活液转移管路 1，和液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Operation name/操作名称	**Time point/时间节点**	**Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1**	**liquid charging port cleaning pipeline 2,/液体加料口清洗管路 2**	**Solid charging port cleaning pipeline3 /固体加料口清洗管路 3**
		Wash /洗涤 1	**Wash /洗涤 2**	**Wash /洗涤 3**
DMF addition/加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**			
DMF addition/加 N,N-二甲基甲酰胺	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/转录人及日期:			
Circulate and clean/循环清洗	**Performed by & Date/执行人及日期:**			
Circulate and clean/循环清洗	**Begin/开始**			
	End/结束			
	Transcribed by & Date/转录人及日期:			

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.7.8 Post-coupling wash./偶联后洗涤	Times /次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h).Bubble with nitrogen for 5±2min (target:5min), start the agitator to 20±10 rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Transfer/转移	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Transfer/转移	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

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3.7.（2）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir /搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

CAUTION/注意: Piperidine 哌啶: Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.8.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to【25±5°C (target: 25°C)】. Add 483.60kg【 20% (V/V) piperidine/DMF (7.44W, 8.0Vol)】 solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至【25±5°C（目标温度：25°C）】。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	1

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认: _____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认: _____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.8.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25 ℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5 ℃(目标温度：25 ℃)。**	**Times/次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.8.3 /结束时间记录于 3.8.3

Transcribed by & Date/转录人及日期:

Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). 开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.8.3 /结束时间记录于 3.8.3

Transcribed by & Date/转录人及日期:

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.8.3 Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护	**Times/次数**
After stirring for 110~130 min (target: 2 h) at 25±5 ℃ (target: 25 ℃), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温 25±5℃(目标温度：25 ℃)， 搅拌 110~130 min（目标时间：2 h）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC ， High limit/高限： 28ºC**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.8.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.8.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.8.4/脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.8.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.8.2/开始时间记录于 3.8.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期	
	Checked by & Date/核实人及日期:	

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	180 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
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Note:备注：Ins 3.8.4~Ins3.8.8 and Ins 3.9.1~Ins3.9.3 need to be performed simultaneously.操作 3.8.4~操作 3.8.8 与操作 3.9.1~操作 3.9.3 同时进行

3.8.4 Post-deprotection wash/脱保护后洗涤	**Times /次数**
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/操作名称	**Time point/时间节点**	**Wash /洗涤 1**	**Wash /洗涤 2**	**Wash /洗涤 3**	**Wash /洗涤 4**
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg）					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					

Batch size (output range): 157.69~292.80 kg

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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始**				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.8.5 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.8.6 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point.开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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3.8.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.8.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.8.（3）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.9.（3）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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Note:备注：Ins 3.8.4~Ins3.8.8 and Ins 3.9.1~Ins3.9.3 need to be performed simultaneously.操作 3.8.4~操作 3.8.8 与操作 3.9.1~操作 3.9.3 同时进行

3.9.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度: 20℃）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度：20℃）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.10.3 /结束时间记录于 3.10.3

Transcribed by & Date/转录人及日期:

3.9.2 Add Fmoc-L-Ala-OH·H_2O (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-L-丙氨酸一水合物（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前；目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Ala-OH·H_2O/ 加芴甲氧羰基-L-丙氨酸一水合物	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Ala-OH·H_2O/ 加芴甲氧羰基-L-丙氨酸一水合物	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Ala-OH·H_2O confirmation/芴甲氧羰基-L-丙氨酸一水合物确认: Control No./内部控制号: YL006160

Batch size (output range): 157.69~292.80 kg

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3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

After solid material charging,confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.9.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain Stirring for NLT 15min. Confirm no visible solids and later use.Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 20±5℃ (Target: 20℃). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min。确认无明显固体，激活罐温度 TJ4S-1213-Sy16-R01 调温至 20±5℃ (目标温度: 20℃)。**	**Times/ 次数**
	1

Maintain &Stir /保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain &Stir /保持搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.10.3/结束时间记录于 3.10.3	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

CAUTION/注意: OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.9.4 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 20±5℃ (target: 20℃). **控内温 20±5℃（目标温度:20℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 2-肟氰乙酸乙酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg）**	**Times/ 次数**
	1

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		

OxymaPure confirmation/2-肟氰乙酸乙酯确认：Control No./内部控制号：YL005180

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/ 拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		**Confirm the net weight is within the input range/确认合计净重在投料范围内**	**Recorded by & Date/记录人及日期**	**Checked by & Date/ 核实人及日期**
		☐Yes/是; ☐No/否		

After solid material charging, confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意: DMF/N，N-二甲基甲酰胺: Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

3.9.5 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times/次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

3.9.6 Add DIC (2.20eq) through metering pump into activation tank TJ4S-1213-Sy16-R01 at a reference rate of 40~80 kg/h at 20±5°C (target: 20°C). The addition rate depends on temperature control. **控内温 20±5°C(目标温度：20°C)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 加入_____kg N,N'-二异丙基碳二亚胺（2.20eq）（Acceptable weighing range/可接受称量范围：_____kg~_____kg），参考加料速度 40~80kg/h，实际以控内温为主。** **Weighing of DIC/ N,N'-二异丙基碳二亚胺称重:** **Control Number/内部控制号：YL003564 Scale ID. No./秤编号：_____** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____**	**Times/次数**
	1

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好

2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。

3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

DIC Addition/ 加 N,N'-二异丙基碳二亚胺	**Performed by & Date/执行人及日期:**	
DIC Addition/ 加 N,N'-二异丙基碳二亚胺	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

DIC Addition/加 N,N'-二异丙基碳二亚胺 **Control Number/内部控制号: YL003564**

Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**

Transcribed by & Date/转录人及日期:

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.9（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIC Addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.9.（3）Fmoc-L-Ala-OH·H_2O activation /芴甲氧羰基-L-丙氨酸一水合物活化

	Times/次数
3.9.7 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target:20min) after addition. **滴毕，控内温 20±5℃(目标温度：20℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速：115 rpm),继续搅拌 20±5min(目标时间：20min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ºC，High limit/高限：23ºC**	**1**

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.10.1/Ins3.10.1 的转移开始时间

Transcribed by & Date/转录人及日期:

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.9.1			
Ins 3.9.5			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			

★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09kg~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内：【405.09kg~456.81kg】	**☐Yes/是** **☐No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.10.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times /次数**
	1

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。
★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

3.10.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）** **（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.10.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture (DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins 3.10.5/结束时间记录于 3.10.5	
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.10.5/结束时间记录于 3.10.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture (DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.
/激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.9.1/开始时间记录于 3.9.1	
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.10.4 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 【25±5°C (target: 25°C)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5℃(目标温度：25 ℃)】。**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

3.10.5 The mixture is allowed to react at internal temperature【25±5 °C (target: 25 °C), start timing when the temperature reaches 20 °C】. The reaction is 3.5h~8h (target: 4h).After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black, react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5°C（目标温度 25°C，20°C 开始计时）】反应时间是 3.5h~8h(目标时间：4h)。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点.** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 °C，High limit/高限：28°C 。**	**Times /次数** **1**

The total reaction Fmoc-L-Ala-OH·H_2O coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ 芴甲氧羰基-L-丙氨酸一水合物偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.10.7./偶联反应结束时间即为 Ins 3.10.7 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

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Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.10.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.10.5-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束:**
	The start time is documented in Ins 3.10.3/开始时间记录于 3.10.3

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Batch size (output range): 157.69~292.80 kg

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.10.3/开始时间记录于 3.10.3	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.9.1			
Ins 3.9.5			
Ins 3.10.2			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			

★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【463.14~522.26kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内: 【463.14~522.26kg】	☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.10.6 Activation Tank Rinse/激活罐洗涤

Times /次数

Rinse the activation tank TJ4S-1213-Sy16-R01 with 492.70kg DMF (7.58W ,8.0Vol) by spray balls.Circulate and clean the activating liquid transfer pipeline 1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min.

通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 492.70kg N, N-二甲基甲酰胺（7.58W ,8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg），循环清洗激活液转移管路 1，和液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。

3

Operation name/ 操作名称	Time point/时间节点	Clean the activating liquid transfer pipeline 1/ 循环清洗激活液转移管路 1	liquid charging port cleaning pipeline 2,/液体加料口清洗管路 2	Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3
		Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N，N 二甲基甲酰胺	Performed by & Date/ 执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	Performed by & Date/ 执行人及日期:			
Circulate and clean/ 循环清洗	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.10.7 Post-coupling wash./偶联后洗涤	**Times/次数**
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min(target:5min), start the agitator to 20±10 rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point./待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	**3**

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中

Operation name/操作名称	**Time point/时间节点**	**Wash /洗涤 1**	**Wash /洗涤 2**	**Wash /洗涤 3**
Transfer/转移	**Begin/开始**			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/核实人及日期:			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**			
Bubble/鼓氮气	**Begin/开始**			
	End/结束			
	Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.10.（3）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir/搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/核实人及日期:			

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3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.11.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5ºC (target: 25ºC). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5ºC（目标温度：25ºC）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认: _____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 ºC ~30 ºC

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认: _____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

	Times /次数
3.11.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5 ℃(目标温度：25 ℃)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.11.3 /结束时间记录于 3.11.3

Transcribed by & Date/转录人及日期:

Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). 开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.11.3 /结束时间记录于 3.11.3

Transcribed by & Date/转录人及日期:

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	206 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

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3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

	Times/次数
3.11.3 After stirring for 110~130 min (target: 2h) at 25±5°C (target: 25°C), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point. Move to next operation. **控内温 25±5°C（目标温度:25°C）搅拌 110~130 min（目标时间:2h）后，停止搅拌，关闭氮气，滤除反应液至无滴出，人员目视确认排废终点，进行下一步操作。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 °C，High limit/高限：28°C**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.11.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.11.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.11.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.11.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.11.2 /开始时间记录于 3.11.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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Note:备注：Ins 3.11.4~Ins3.11.8, Ins 3.12.1~Ins3.12.2 and Ins3.13.3~Ins3.13.4 need to be performed simultaneously.操作 3.11.4~操作 3.11.8,操作 3.12.1~操作 3.12.2，操作 3.13.3~操作 3.13.4 同时进行

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

	Times /次数
3.11.4 Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯(6.98W, 8.0Vol)(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Stir/搅拌	Performed by & Date/ 执行人及日期:				
Stir /搅拌	Begin/开始				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.11.5 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg）。**	1

DMF addition/ 加 N,N 二甲基甲酰胺	Performed by & Date/执行人及日期:		
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:	Qty/重量:
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

3.11.6 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h).Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point.开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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	QF05098	12.0	09/29/2026	210 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.11.（4）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

3.11.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。**	**1**

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.11.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.11.4~Ins3.11.8, Ins 3.12.1~Ins3.12.2 and Ins3.13.3~Ins3.13.4 need to be performed simultaneously.操作 3.11.4~操作 3.11.8,操作 3.12.1~操作 3.12.2，操作 3.13.3~操作 3.13.4 同时进行

3.12.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度: 20℃）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF dissolution total quantity is a CPP./备注：N, N-二甲基甲酰胺溶解总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度：20℃）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.13.5 /结束时间记录于 3.13.5

Transcribed by & Date/转录人及日期:

3.12.2Add Fmoc-L-Cys(Trt)-OH (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg Fmoc-S-三苯甲基-L-半胱氨酸（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times /次数** **1**

★Note:The quantity of Fmoc-L-Cys(Trt)-OH is a CPP./备注：Fmoc-S-三苯甲基-L-半胱氨酸总用量是关键工艺参数

Before material charging, confirm the material is within the retest date/ the expiry date, Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no o non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Cys(Trt)-OH / 加 Fmoc-S-三苯甲基-L-半胱氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Cys(Trt)-OH / 加 Fmoc-S-三苯甲基-L-半胱氨酸	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Cys(Trt)-OH confirmation/ Fmoc-S-三苯甲基-L-半胱氨酸确认：Control No./内部控制号：YL003565

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Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

★Note:The quantity of Fmoc-L-Cys(Trt)-OH is a CPP./备注：Fmoc-S-三苯甲基-L-半胱氨酸总用量是关键工艺参数

★NOR limit/正常操作范围: _____kg~ _____kg

Filled by& Date/填写人及日期: _____

Checked by & Date/核实人及日期: _____

★After charging, the operator shall confirm that Critical Process Parameters (CPP) is within the respective Normal Operating Ranges (NOR)/ 投料结束后，操作人员须确认关键工艺参数（CPP）均处于正常操作范围（NOR）内

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

CAUTION/注意：OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.12.3 Add Oxyma Pure (3.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 2-肟氰乙酸乙酯（3.00eq）(Acceptable weighing range/可接受称量范围: _____kg ~_____kg)。**	**Times /次数** **1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		
OxymaPure confirmation/2-肟氰乙酸乙酯确认: Control No./内部控制号: YL005180		

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

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3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		**Confirm the net weight is within the input range/确认合计净重在投料范围内**	**Recorded by & Date/记录人及日期**	**Checked by & Date/ 核实人及日期**
		☐Yes/是; ☐No/否		

After solid material charging, confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times /次数
3.12.4 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT15 min. Confirm no visible solids and later use **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min。确认无明显固体，待用。**	**1**

Rinse with DMF/N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			
Stir/搅拌	**Performed by & Date/执行人及日期:**		
Stir /搅拌	**Begin/开始:**		
	The end time is documented in Ins3.13.5 /结束时间记录于 3.13.5		
Transcribed by & Date/转录人及日期:			
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**		

	Times /次数
3.12.5 Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 5±5 ºC. Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 5±5min (Target:5min) at 5±5 ºC (target: 5ºC). **将激活罐温度 TJ4S-1213-Sy16-R01 调温至 5±5 ºC。控内温 5±5 ºC（目标温度:5 ºC）维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm),继续 5±5min(目标时间：5min)，待用。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：2 ºC，High limit/高限：8ºC**	**1**

Batch size (output range): 157.69~292.80 kg

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操作指令

3.12.（4）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Maintain &Stir/保温搅拌	**Performed by & Date/执行人及日期:**	
Maintain &Stir/保温搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

★After dissolution, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 溶解结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.12.1			
Ins 3.12.4			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			
★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内：【405.09~456.81kg】			☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	219 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

3.13.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times /次数**
	1

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.13.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall./转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。	**Times /次数**
	1

Rinse with DMF / N，N 二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

Before using, confirm hastelloy agitator tank oxygen content≤1.0%.

使用前确认哈氏合金搅拌罐氧气含量≤1.0%。

500 L Hastelloy agitator tank

哈氏合金搅拌罐: TJ4S-1200-AT01

Oxygen detector ID. No. /氧气检测器编号: _____

Oxygen content/氧气含量= _____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期: _____

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Before using, confirm the hastelloy agitator tank TJ4S-1200-AT01's calibration information is within valid date. /使用前确认哈氏合金搅拌罐 TJ4S-1200-AT01 的仪表在校准有效期内。

Checked by & Date/核实人及日期: _____

Before using, confirm that cleaning of the hastelloy agitator tank TJ4S-1200-AT01 has been completed/使用前，确认哈氏合金搅拌罐 TJ4S-1200-AT01 清洁已完成

Checked by & Date/核实人及日期: _____

CAUTION/注意: DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Note:备注: Ins 3.11.4~Ins3.11.8, Ins 3.12.1~Ins3.12.2 and Ins3.13.3~Ins3.13.4 need to be performed simultaneously.操作 3.11.4~操作 3.11.8,操作 3.12.1~操作 3.12.2，操作 3.13.3~操作 3.13.4 同时进行

3.13.3 Preparation of DIC/DMF Solution/配置 DIC/DMF 溶液	**Times /次数**
Add 61.75kg DMF (0.95W, 1.0Vol) into the hastelloy agitator tank TJ4S-1200-AT01. Start the agitator of the tank to 60±20 rpm (target: 60 rpm). Then add DIC(3.30eq) by peristaltic pump. **向哈氏合金搅拌罐 TJ4S-1200-AT01 中加入 61.75kg N,N-二甲基甲酰胺（0.95W, 1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg），开启搅拌转速至 60±20 rpm (目标转速: 60 rpm)。通过蠕动泵泵入_____kg N,N'-二异丙基碳二亚胺（3.30eq）（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）。**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好

2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。

3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Weighing of DMF / N,N-二甲基甲酰胺称重:

Control Number/内部控制号: YL000336 Scale ID. No./秤编号: _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

DMF addition/加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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操作指令

3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DMF/N，N-二甲基甲酰胺	Begin/开始:	End/结束:

Batch size (output range): 157.69~292.80 kg

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Start the agitator of the tank to 60±20 rpm (target: 60 rpm). /开启搅拌转速至 60±20 rpm (目标转速: 60 rpm)。

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.13.7 /结束时间记录于 3.13.7

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。
Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期: _____

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Weighing of DIC / N,N'-二异丙基碳二亚胺称重:

Control Number/内部控制号: YL003564 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

DIC addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Addition type 加料方式	Lot number/批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/投料前	After charging/投料后	Net weight/净重	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
☐Full drum 整桶 ☐Non-full drum /非整桶											
☐Full drum 整桶 ☐Non-full drum /非整桶											
Total/合计											

DIC / N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

3.13.4 Continue agitation of tank to 60±20 rpm (target: 60rpm). Maintain stirring for NLT 15min for later use. **维持搅拌罐搅拌转速至 60±20 rpm(目标转速：60rpm)，继续搅拌至少 15min，待用。**	**Times /次数** **1**

Maintain &Stir 保持搅拌	**Performed by & Date/执行人及日期:**
Maintain &Stir 保持搅拌	**Begin/开始:**
	The end time is documented in Ins3.13.7 /结束时间记录于 3.13.7

Transcribed by & Date/转录人及日期:

Prior to DIC/DMF solution transfer.Confirm the temperature is within 10~30℃. / DIC/DMF 溶液转移前，确认温度在 10~30℃。
Temperature/温度: _____ **□Yes/是 □No/否**

Checked by & Date/核实人及日期: _____

Solution No. /溶液编号：CPo140021-_____-01-_____-3.13.4-01

3.13.5 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times /次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.13.9 /结束时间记录于 3.13.9

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.13.9 /结束时间记录于 3.13.9	
Transcribed by & Date/转录人及日期:		
Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. /激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。		
Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.12.4/开始时间记录于 3.12.4	
Transcribed by & Date/转录人及日期:		

3.13.6 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to15~25°C（Target:20 °C）. **合成釜 TJ4S-1213-Sy16 调温至 15~25°C（目标温度:20°C）。**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

3.13.7 Transfer the DIC/DMF solution (Prepared in Ins.3.13.4) into the mixture at internal temperature 15~25°C（Target: 20°C）. **控内温 15~25°C（目标温度：20°C），转移操作 3.13.4 配制的 N,N'-二异丙基碳二亚胺/N,N-二甲基甲酰胺溶液。**	**Times /次数** **1**

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

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Upon completion of material transfer, perform nitrogen purge on the dedicated DIC/DMF transfer line: maintain nitrogen flow rate ≥ 6 m³/h for a continuous duration of ≥ 4 min to displace residual DIC/DMF solution in the line/物料转移完成后，对 DIC/DMF 专用输送管线进行氮气吹扫: 控制氮气流量≥6 m³/h，连续吹扫时长≥4 min。用以置换管线内残留的 DIC/DMF 溶液

Nitrogen Purge/氮气吹扫	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Agitator tank Stir 搅拌罐搅拌	**Performed by & Date/执行人及日期:**	
Agitator tank Stir /搅拌罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.13.3/开始时间记录于 3.13.3	
Transcribed by & Date/转录人及日期:		

3.13.8 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C). **合成釜 TJ4S-1213-Sy16 内温调温至 20±5℃(目标温度：20 ℃)。**	**Times/次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

	Times/次数
3.13.9 Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联 **The mixture is allowed to react at internal temperature【20±5 °C (target: 20-°C)】, start timing when the temperature reaches 15 °C. The reaction is 3.5h~8h (target: 4h).After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black, react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.** **控内温【20±5°C（目标温度 20°C，15°C 开始计时）】,反应时间是 3.5h~8h(目标时间：4h)。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 °C，High limit/高限：23°C**	**1**

The total reaction Fmoc-L-Cys(Trt)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ Fmoc-S-三苯甲基-L-半胱氨酸偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.13.11./偶联反应结束时间即为 Ins 3.13.11 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :15~25℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:15~25℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.
取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.13.9-01			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.13.9-02			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.13.9-03			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.13.9-04			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.13.9-05			□Negative /阴性 □Positive /阳性		
CPo140021-_____-01-_____-3.13.9-06			□Negative /阴性 □Positive /阳性		

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Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.13.9-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.13.9-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束**
	The start time is documented in Ins3.13.5 /开始时间记录于 3.13.5

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Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**End/结束**
	The start time is documented in Ins3.13.5 /开始时间记录于 3.13.5

Transcribed by & Date/转录人及日期:

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

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3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.12.1			
Ins 3.12.4			
Ins 3.13.2			
Ins3.13.3			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			

★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【521.18~587.72kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内：【521.18~587.72kg】	☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.13.10 Activation Tank Rinse/激活罐洗涤	Times /次数
Rinse the activation tank TJ4S-1213-Sy16-R01 with 492.70kg DMF (7.58W ,8.0Vol) by spray balls.Circulate and clean the activating liquid transfer pipeline 1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 492.70kg N, N-二甲基甲酰胺（7.58W ,8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)，循环清洗激活液转移管路 1，和液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

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Operation name/ 操作名称	Time point/时间节点	Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1	liquid charging port cleaning pipeline 2,/液体加料口清洗管路 2	Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3
		Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N，N 二甲基甲酰胺	Performed by & Date/ 执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	Performed by & Date/ 执行人及日期:			
Circulate and clean/ 循环清洗	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

3.13.11 Post-coupling wash./偶联后洗涤	Times /次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), start the agitator to 20±10 rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.13.（4）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中				
Transfer/转移	**Begin/开始**			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间: 10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Stir/搅拌	**Performed by & Date/ 执行人及日期:**			
Stir /搅拌	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	**Begin/开始**			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times/次数
3.14.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	1

★Note: De-Fmoc time is a CPP./备注：脱保护时间是关键工艺参数

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**

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Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.14.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5℃(目标温度：25℃)。**	**Times /次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.14.3 /结束时间记录于 3.14.3

Transcribed by & Date/转录人及日期:

Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). 开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.14.3 /结束时间记录于 3.14.3

Transcribed by & Date/转录人及日期:

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

	Times /次数
3.14.3After stirring for 【110~130 min (target: 2h)】 at 25±5℃ (target: 25℃), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point. Move to next operation. **控内温 25±5℃（目标温度:25℃）搅拌【110~130 min（目标时间:2h）】后，停止搅拌，关闭氮气，滤除反应液至无滴出，人员目视确认排废终点，进行下一步操作。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ℃，High limit/高限：28℃**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.14.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.14.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.14.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.14.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.14.2/开始时间记录于 3.14.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

★After De-Fmoc is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 脱保护结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★ Total (5) Fmoc-L-Thr(tBu)-OH De-Fmoc time is a CPP./备注：芴甲氧羰基-叔丁基-L-苏氨酸脱保护总时间是关键工艺参数

★Process chemist confirms the total reaction/ (5) Fmoc-L-Thr(tBu)-OH De-Fmoc time is within the respective Normal Operating Ranges (NOR):110min –160min

工艺技术人员确认芴甲氧羰基-叔丁基-L-苏氨酸脱保护总时间处于正常操作范围（NOR）范围内 110min –160min

➢Definition of De-Fmoc start time/脱保护（De-Fmoc）开始时间定义:

The De-Fmoc start time is defined as 20%Piperidine solution addition completion time in Ins3.14.1. /Ins3.14.1 的 20%哌啶溶液投料结束时间为脱保护反应开始时间。

➢The De-Fmoc end time is defined as start of 1st post deprotection wash in Ins3.14.4./ Ins3.14.4 的脱保护后第一次洗涤开始时间定义为脱保护结束时间。

Total De-Fmoc time/总脱保护时间 =Start of 1st post deprotection wash 脱保护第一次洗涤开始时间-End of charging 20% Piperidine solution/20%哌啶加料结束时间

	Start of 1st post deprotection wash/脱保护第一次洗涤开始时间	End of charging 20% Piperidine solution/20%哌啶加料结束时间	Total De-Fmoc time/总脱保护时间计算	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Time/时间					

★The operator shall confirm thatTotal De-Fmoc time is within the Normal Operating Range (NOR)【110min~160min】/操作人员须确认总脱保护时间处于正常操作范围（NOR）内:【110min~160min】 ☐Yes/是 ☐No/否

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
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Note:备注：Ins 3.14.4 ~Ins3.14.8 and Ins 3.15.1~3.15.3 need to be performed simultaneously.操作 3.14.4~操作 3.14.8 与操作 3.15.1~操作 3.15.3 同时进行

	Times/次数
3.14.4 Post-deprotection wash/脱保护后洗涤	
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg）					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min).-Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Stir/搅拌	Performed by & Date/执行人及日期:				
Stir /搅拌	Begin/开始				
	End/结束				
	Transcribed by & Date/转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/执行人及日期:				
	Checked by & Date/核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.14.5 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg）。**	1

DMF addition/加 N，N 二甲基甲酰胺	Performed by & Date/执行人及日期:		
DMF addition/加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:	Qty/重量:
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.14.6 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	243 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.14.（5）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.14.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）（Acceptable weighing range/可接受称量范围：463.14kg~522.26kg）。**	**1**

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.14.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min).Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期: _____		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期: _____		
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	244 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.14.4 ~Ins3.14.8 and Ins 3.15.1~3.15.3 need to be performed simultaneously.操作 3.14.4~操作 3.14.8 与操作 3.15.1~操作 3.15.3 同时进行

3.15.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5ºC(目标温度: 20ºC)。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）(Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5ºC（目标温度：20ºC）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）(Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg)

DMF addition/加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.16.3 /结束时间记录于 3.16.3
Transcribed by & Date/转录人及日期:	

3.15.2Add Fmoc-L-Thr(tBu)-OH(2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-叔丁基-L-苏氨酸（2.00eq）** **（Acceptable weighing range/可接受称量范围：_____kg~_____kg）。**	**Times/次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Thr(tBu)-OH /加芴甲氧羰基-叔丁基-L-苏氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Thr(tBu)-OH /加芴甲氧羰基-叔丁基-L-苏氨酸	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		

Fmoc-L-Thr(tBu)-OH confirmation/芴甲氧羰基-叔丁基-L-苏氨酸确认：Control No./内部控制号：YL003563

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.15.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm).Maintain stirring for NLT 15min. Confirm no visible solids. Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 20±5℃(target: 20℃). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm),继续搅拌至少 15min，确认无明显固体，激活罐 TJ4S-1213-Sy16-R01 温度调温至 20±5℃(目标温度：20℃)。**	**Times/ 次数**
	1

Maintain &Stir/保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain &Stir/保持搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.16.3/结束时间记录于 3.16.3	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

CAUTION/注意：OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.15.4 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 20±5℃ (target: 20℃). **控内温 20±5℃（目标温度:20℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）**	**Times/ 次数**
	1

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		
OxymaPure confirmation/2-肟氰乙酸乙酯确认: Control No./内部控制号: YL005180		

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		**Confirm the net weight is within the input range/确认合计净重在投料范围内**	**Recorded by & Date/记录人及日期**	**Checked by & Date/ 核实人及日期**
		☐Yes/是; ☐No/否		

After solid material charging confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

3.15.5 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF /加 N，N 二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.15.6 Add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/h at 20±5°C (target: 20°C). The addition rate depends on temperature control. **控内温 20±5°C(目标温度：20°C)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N'-二异丙基碳二亚胺（2.20eq）（Acceptable weighing range/可接受称量范围: _____kg~_____kg），参考加料速度 40~80kg/h，实际以控内温为主。** **Weighing of DIC / N,N'-二异丙基碳二亚胺称重:** **Control Number/内部控制号：YL003564 Scale ID. No./秤编号: _____** **Pump number/泵编号: _____** **Material charging pipe number/物料管路编号: _____**	**Times /次数**
	1

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好

2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。

3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期: _____

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

DIC addition/ 加 N，N 二异丙基碳二亚胺	**Performed by & Date/执行人及日期:**	
DIC Addition/ 加 N,N'-二异丙基碳二亚胺	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

DIC Addition/加 N,N'-二异丙基碳二亚胺: Control Number/内部控制号: YL003564

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

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操作指令

3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIC Addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.15.（5）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

3.15.7 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for【20±5min (Target:20min)】after addition. **滴毕，控内温 20±5℃(目标温度：20℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌【20±5min(目标时间：20min)】。**	**Times/次数** **1**

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.16.1/Ins3.16.1 的转移开始时间

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	**Weight/重量**	**Calculated by & Date/计算人及日期**	**Checked by & Date/核实人及日期**
Ins 3.15.1			
Ins 3.15.5			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			
★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09kg~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内: 【405.09kg~456.81kg】			☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.16.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。	Times /次数
	1

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

Transfer/转移	Begin/开始:	End/结束:
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.16.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. 转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。	Times /次数
	1

Rinse with DMF /加 N，N 二甲基甲酰胺淋洗	Performed by & Date/执行人及日期:		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	Begin/开始:	End/结束:	Qty/重量:
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.16.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). / 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.16.5 /结束时间记录于 3.16.5	
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.16.5 /结束时间记录于 3.16.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.
/激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.15.1/开始时间记录于 3.15.1	
Transcribed by & Date/转录人及日期:		

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3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

	Times /次数
3.16.4 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 【25±5ºC (target: 25ºC)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5℃(目标温度：25 ℃)】。**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

	Times /次数
3.16.5 The mixture is allowed to react at internal temperature【25±5 ºC (target: 25 ºC), start timing when the temperature reaches 20 ºC. The reaction is 3.5h~8h(target: 4h)】.After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black,react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5ºC（目标温度 25ºC，20ºC 开始计时）反应时间是 3.5h~8h(目标时间：4h)】.反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全; 树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC ， High limit/高限： 28ºC**	**1**

The total reaction Fmoc-L-Thr(tBu)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ 芴甲氧羰基-叔丁基-L-苏氨酸偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

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3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.16.7./偶联反应结束时间即为 Ins 3.16.7 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.
取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

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Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.16.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.16.5-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束:**
	The start time is documented in Ins3.16.3 /开始时间记录于 3.16.3
Transcribed by & Date/转录人及日期:	
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**End/结束:**
	The start time is documented in Ins3.16.3 /开始时间记录于 3.16.3
Transcribed by & Date/转录人及日期:	

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.15.1			
Ins 3.15.5			
Ins 3.16.2			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			

★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【463.14~522.26kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内: 【463.14~522.26kg】	☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.16.6 Activation Tank Rinse/激活罐洗涤	**Times /次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 492.70kg DMF (7.58W ,8.0Vol) by spray balls.Circulate and clean the activating liquid transfer pipeline 1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 492.70kg N, N-二甲基甲酰胺（7.58W ,8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)，循环清洗激活液转移管路 1，和液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Operation name/ 操作名称	**Time point/时间节点**	**Clean the activating liquid transfer pipeline 1/ 循环清洗激活液转移管路 1**	**liquid charging port cleaning pipeline 2,/液体加料口清洗管路 2**	**Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3**
		Wash /洗涤 1	**Wash /洗涤 2**	**Wash /洗涤 3**
DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/ 执行人及日期:**			
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	**Performed by & Date/ 执行人及日期:**			
Circulate and clean/ 循环清洗	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.16.7 Post-coupling wash./偶联后洗涤	Times /次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min(target:5min), start the agitator to 20±10 rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point./待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	**3**

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Transfer/转移	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/核实人及日期:			

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.16.（5）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir/搅拌	Performed by & Date/ 执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	263 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
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3.17.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认: _____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认: _____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.17.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min（目标时间：5 min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5 ℃（目标温度：25 ℃）。**	**Times /次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.17.3 /结束时间记录于 3.17.3

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Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). 开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.17.3 /结束时间记录于 3.17.3

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Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

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Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.17.3 Fmoc-L-Ala-OH·H2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护 **After stirring for 110~130 min (target: 2h) at25±5℃ (target: 25℃), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point. Move to next operation.** **控内温 25±5℃（目标温度:25℃）搅拌 110~130 min（目标时间:2h）后，停止搅拌，关闭氮气，滤除反应液至无滴出，人员目视确认排废终点，进行下一步操作。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ℃，High limit/高限：28℃**	**Times /次数** **1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.17.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.17.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.17.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.17.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.17.2 /开始时间记录于 3.17.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
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Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
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Note:备注：Ins 3.17.4~Ins3.17.8 and Ins 3.18.1~Ins3.18.3 need to be performed simultaneously.操作 3.17.4~操作 3.17.8 与操作 3.18.1~操作 3.18.3 同时进行

	Times/次数
3.17.4 Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h).Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（6.98W, 8.0Vol）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 453.70kg IPAc (6.98W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 453.70kg 乙酸异丙酯（(6.98W, 8.0Vol)）(Acceptable weighing range/可接受称量范围：426.48kg~480.92kg）					
IPAC addition/加乙酸异丙酯	**Performed by & Date/执行人及日期:**				
IPAC addition/加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
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Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。					
Stir/搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Filter/过滤	**Begin/开始**				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

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Operations Instruction
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3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

3.17.5 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。**	**1**

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.17.6 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) . Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. /开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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3.17.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 492.70kg DMF (7.58W, 8.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 492.70kg N,N-二甲基甲酰胺（7.58W, 8.0Vol）(Acceptable weighing range/可接受称量范围：463.14kg~522.26kg)。**	**1**

DMF addition/加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.17.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.17.（6）Fmoc-L-Ala-OH·H_2O De-Fmoc/芴甲氧羰基-L-丙氨酸一水合物脱保护

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.17.4~Ins3.17.8 and Ins 3.18.1~Ins3.18.3 need to be performed simultaneously.操作 3.17.4~操作 3.17.8 与操作 3.18.1~操作 3.18.3 同时进行

3.18.1 Add solvent to the activation tank/激活罐加溶剂	**Times/次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度: 20°C）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度：20°C）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.19.3 /结束时间记录于 3.19.3

Transcribed by & Date/转录人及日期:

3.18.2Add Fmoc-L-Ala-OH·H_2O (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-L-丙氨酸一水合物（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times/次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- No non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Ala-OH·H_2O / 加芴甲氧羰基-L-丙氨酸一水合物	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Ala-OH·H_2O / 加芴甲氧羰基-L-丙氨酸一水合物	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Ala-OH·H_2O confirmation/芴甲氧羰基-L-丙氨酸一水合物确认: Control No./内部控制号: YL006160

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.18.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min. Confirm no visible solids .Adjust to 20±5℃(target: 20℃). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm),继续搅拌至少 15min，确认无明显固体，调温至 20±5℃(目标温度: 20℃)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限: 17 ºC，High limit/高限: 23ºC**	**Times /次数** **1**

Maintain Stir/保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain Stir/保持搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.19.3/结束时间记录于 3.19.3	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

CAUTION/注意: OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

3.18.4 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 20±5℃ (target: 20ºC). **控内温 20±5℃（目标温度:20℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）**	**Times /次数** **1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

OxymaPure confirmation/2-肟氰乙酸乙酯确认: Control No./内部控制号: YL005180

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Internal temperature/内部温度	Highest values/最高值:	Lowest values/最低值:

Transcribed by & Date/转录人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.18.5 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）(Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/加 N，N 二甲基甲酰胺淋洗	Performed by & Date/执行人及日期:		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	Begin/开始:	End/结束:	Qty/重量:

Transcribed by & Date/转录人及日期:

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

	Times /次数
3.18.6 Add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/h at 20±5°C (target: 20°C). The addition rate depends on temperature control. **控内温 20±5°C(目标温度：20°C)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N'-二异丙基碳二亚胺（2.20eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 40~80kg/h，实际以控内温为主。** **Weighing of DIC / N,N'-二异丙基碳二亚胺称重:** **Control Number/内部控制号：YL003564 Scale ID. No./秤编号：_____** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
Confidential Information

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

DIC addition/ **加 N，N 二异丙基碳二亚胺**	**Performed by & Date/执行人及日期:**	
DIC Addition/ **加 N,N'-二异丙基碳二亚胺**	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
DIC Addition/加 N,N'-二异丙基碳二亚胺: Control Number/内部控制号: YL003564		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/ 还需加入净重目标值	Gross qty of drum that to be charged/ 待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐ Full drum 整桶 ☐ Non-full drum /非整桶											
☐ Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIC Addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.18.（6）Fmoc-L-Ala-OH·H_2O activation/芴甲氧羰基-L-丙氨酸一水合物活化

	Times/次数
3.18.7 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target: 20min) after addition. **滴毕，控内温 20±5℃(目标温度：20℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 20±5min(目标时间：20min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ºC，High limit/高限：23ºC**	**1**

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.19.1/Ins3.19.1 的转移开始时间

Transcribed by & Date/转录人及日期:

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.18.1			
Ins 3.18.5			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			
★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09kg~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内：【405.09kg~456.81kg】			☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.19.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times /次数**
	1

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

3.19.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall./转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。	**Times /次数**
	1

Rinse with DMF / N,N 二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.19.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16./开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。	**Times/次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.19.5 /结束时间记录于 3.19.5	
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.19.5 /结束时间记录于 3.19.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.
/激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.18.1/开始时间记录于 3.18.1	
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

	Times /次数
3.19.4 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 【25±5°C (target: 25°C).】 **合成釜 TJ4S-1213-Sy16 内温调温至【25±5℃(目标温度：25 ℃)】.**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

	Times /次数
3.19.5 The mixture is allowed to react at internal temperature【25±5 °C (target: 25 °C), start timing when the temperature reaches 20 °C. The minimum reaction is 3.5h~8h(target: 4h)】.After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black,react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5°C（目标温度 25°C，20°C 开始计时）反应时间是 3.5h~8h(目标时间：4h)】。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 °C，High limit/高限：28°C**	**1**

The total reaction Fmoc-L-Ala-OH·H_2O coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ 芴甲氧羰基-L-丙氨酸一水合物偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.19.7./偶联反应结束时间即为 Ins 3.19.7 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.19.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.19.5-08			☐Negative /阴性 ☐Positive /阳性		

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3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins 3.19.3/开始时间记录于 3.19.3	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.19.3/开始时间记录于 3.19.3	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期	
	Checked by & Date/核实人及日期:	
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.18.1			
Ins 3.18.5			
Ins 3.19.2			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			
★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【463.14~522.26kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内：【463.14~522.26kg】			☐Yes/是 ☐No/否
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

AUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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3.19.6 Activation Tank Rinse/激活罐洗涤	Times /次数
Rinse the activation tank TJ4S-1213-Sy16-R01 with 616.20kg DMF (9.48W,10.0Vol) by spray balls. Circulate and clean the activating liquid transfer pipeline1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐中加入 616.20kg N, N-二甲基甲酰胺（9.48W ,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，循环清洗激活液转移管路 1，液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	3

Operation name/ 操作名称	Time point/时间节点	Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1	liquid charging port cleaning pipeline 2,/ 液体加料口清洗管路 2	Solid charging port cleaning pipeline3 /固体加料口清洗管路 3
		Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N，N 二甲基甲酰胺	Performed by & Date/执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	Performed by & Date/执行人及日期:			
Circulate and clean/ 循环清洗	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	290 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

3.19.7 Post-coupling wash./偶联后洗涤

Times /次数

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16.Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.

待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

3

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中				
Transfer/转移	**Begin/开始**			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**			
Bubble/鼓氮气	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.19.（6）Fmoc-L-Ala-OH·H_2O coupling/芴甲氧羰基-L-丙氨酸一水合物偶联

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir/搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Stir/搅拌	Performed by & Date/执行人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.20.（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
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	Times/次数
3.20.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	1

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.20.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min（目标时间：5 min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5 ℃（目标温度：25 ℃）。**	**Times/次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins 3.20.3/结束时间记录于 3.20.3

Transcribed by & Date/转录人及日期:

Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). 开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins 3.20.3/结束时间记录于 3.20.3

Transcribed by & Date/转录人及日期:

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

3.20.3 After stirring for 110~130 min (target: 2h) at 25±5℃ (target: 25ºC), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point. Move to next operation. **控内温 25±5ºC（目标温度:25ºC）搅拌 110~130 min（目标时间:2h）后，停止搅拌，关闭氮气，滤除反应液至无滴出，人员目视确认排废终点，进行下一步操作。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**Times /次数** **1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.20.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.20.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.20.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.20.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.20.2 /开始时间记录于 3.20.2	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
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Note:备注：Ins 3.20.4~Ins3.20.8 and Ins 3.21.1~Ins3.21.3 need to be performed simultaneously.操作 3.20.4~操作 3.20.8 与操作 3.21.1~操作 3.21.3 同时进行

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

	Times /次数
3.20.4 Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）（Acceptable weighing range/可接受称量范围：533.40kg~601.50kg），开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）（Acceptable weighing range/可接受称量范围：533.40kg~601.50kg）					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Stir /搅拌	Performed by & Date/执行人及日期:				
Stir /搅拌	Begin/开始				
	End/结束				
	Transcribed by & Date/转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/执行人及日期:				
	Checked by & Date/核实人及日期				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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3.20.5 Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)。**	**Times /次数** **1**

DMF addition / 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition / 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

3.20.6 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.20.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times/次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	**1**

DMF addition / N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
DMF addition / N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.20.8Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**Times/次数**
	1

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.20（7）Fmoc-L-Thr(tBu)-OH De-Fmoc/芴甲氧羰基-叔丁基-L-苏氨酸脱保护

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	300 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.20.4~Ins3.20.8 and Ins 3.21.1~Ins3.21.3 need to be performed simultaneously.操作 3.20.4~操作 3.20.8 与操作 3.21.1~操作 3.21.3 同时进行

3.21.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度: 20°C）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度：20°C）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.22.3 /结束时间记录于 3.22.3

Transcribed by & Date/转录人及日期:

3.21.2 Add Fmoc-L-Thr(tBu)-OH(2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 芴甲氧羰基-叔丁基-L-苏氨酸（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date. /加物料前，确认物料在复检日期/有效期之内。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Thr(tBu)-OH /加芴甲氧羰基-叔丁基-L-苏氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Thr(tBu)-OH /加芴甲氧羰基-叔丁基-L-苏氨酸	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Thr(tBu)-OH confirmation/芴甲氧羰基-叔丁基-L-苏氨酸确认: Control No./内部控制号: YL003563

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the total weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域。
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域。
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁。

Checked by & Date/核实人及日期: _____

3.21.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min. Confirm no visible solids .Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 20±5℃ (target: 20℃). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min。确认无明显固体，激活罐温度 TJ4S-1213-Sy16-R01 调温至 20±5℃（目标温度:20℃）。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ℃，High limit/高限： 23℃**	**Times /次数**
	1

Maintain Stir/保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain Stir/保持搅拌	**Begin/开始:**	
	The end time is documented in Ins3.22.3 /结束时间记录于 3.22.3	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

CAUTION/注意：OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	304 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times /次数
3.21.4 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 20±5℃ (target: 20ºC). **控内温 20±5℃（目标温度:20℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）**	**1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date. /加物料前，确认物料在复检日期/有效期之内。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no o non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**
Transcribed by & Date /转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

OxymaPure confirmation/2-肟氰乙酸乙酯确认：Control No./内部控制号：YL005180

Check items/检查项	Is it correct?/是否正确	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/项目批次号	☐Yes/是; ☐No/否		
Operation number/操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/净重	All in/全部加入	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the total weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

After solid material charging confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域。

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域。

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁。

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.21.5 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）(Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times/次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times/次数
3.21.6 Add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/h at 20±5°C (target: 20°C). The addition rate depends on temperature control. **控内温 20±5℃(目标温度：20℃)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N'-二异丙基碳二亚胺（2.20eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 40~80kg/h，实际以控内温为主。**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date. /加物料前，确认物料在复检日期/有效期之内。Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:: _____

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Add DIC / 加 N,N'-二异丙基碳二亚胺	**Performed by & Date/执行人及日期:**		
Add DIC / 加 N,N'-二异丙基碳二亚胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Weighing of DIC/ N,N'-二异丙基碳二亚胺称重:

Control Number/内部控制号: YL003564 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/ 还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIC Addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.21.（7）Fmoc-L-Thr(tBu)-OH activation/芴甲氧羰基-叔丁基-L-苏氨酸活化

	Times/次数
3.21.7 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target:20min)after addition. **滴毕，控内温 20±5℃(目标温度：20℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 20±5min(目标时间：20min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ℃， High limit/高限： 23℃**	**1**

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.22.1/Ins3.22.1 的转移开始时间

Transcribed by & Date/转录人及日期:

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.21.1			
Ins 3.21.5			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			

★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内: 【405.09~456.81kg】	**☐Yes/是** **☐No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	311 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

	Times/次数
3.22.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**1**

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

	Times/次数
3.22.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**1**

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

	Times/次数
3.22.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture (DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**1**

Batch size (output range): 157.69~292.80 kg

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.22.5 /结束时间记录于 3.22.5	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.22.5 /结束时间记录于 3.22.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.
/激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.21.1/开始时间记录于 3.21.1	
Transcribed by & Date/转录人及日期:		

3.22.4 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 【25±5°C (target: 25°C)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5°C(目标温度：25 °C)】.**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

	Times/次数
3.22.5 The mixture is allowed to react at internal temperature【 25±5 ºC (target: 25 ºC), start timing when the temperature reaches 20 ºC. The reaction is 3.5h~8h(target: 4h)】. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black, react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5ºC（目标温度 25ºC，20ºC 开始计时）反应时间是 3.5h~8h(目标时间：4h)】。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	1

The total reaction Fmoc-L-Thr(tBu)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ 芴甲氧羰基-叔丁基-L-苏氨酸偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.22.7./偶联反应结束时间即为 Ins 3.22.7 洗涤 1 的开始时间	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.
取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

Batch size (output range): 157.69~292.80 kg

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.22.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.22.5-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束:**
	The start time is documented in Ins 3.22.3/开始时间记录于 3.22.3
Transcribed by & Date/转录人及日期:	
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**End/结束:**
	The start time is documented in Ins 3.22.3/开始时间记录于 3.22.3
Transcribed by & Date/转录人及日期:	

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.21.1			
Ins 3.21.5			
Ins 3.22.2			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			
★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【463.14~522.26kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内：【463.14~522.26kg】			☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.22.6 Activation Tank Rinse/激活罐洗涤	**Times /次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 616.20kg DMF (9.48W,10.0Vol) by spray balls. Circulate and clean the activating liquid transfer pipeline1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐中加入 616.20kg N, N-二甲基甲酰胺（9.48W ,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，循环清洗激活液转移管路 1，液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Operation name/ 操作名称	**Time point/时间节点**	**Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1**	**liquid charging port cleaning pipeline 2,/ 液体加料口清洗管路 2**	**Solid charging port cleaning pipeline3 /固体加料口清洗管路 3**
		Wash /洗涤 1	**Wash /洗涤 2**	**Wash /洗涤 3**
Add DMF/ 加 N,N-二甲基甲酰胺	**Performed by & Date/ 执行人及日期:**			
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	**Performed by & Date/ 执行人及日期:**			
Circulate and clean/ 循环清洗	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

3.22.7 Post-coupling wash./偶联后洗涤	Times /次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 .Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), start the agitator to 20±10 rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Transfer/转移	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期			

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气

Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.22.（7）Fmoc-L-Thr(tBu)-OH coupling/芴甲氧羰基-叔丁基-L-苏氨酸偶联

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir /搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	erformed by & Date/执行人及日期:			
	Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

	Times /次数
3.23.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to【 25±5ºC (target: 25ºC)】. Add 483.60kg 【20% (V/V) piperidine/DMF (7.44W, 8.0Vol)】solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至【25±5℃（目标温度：25℃）】。向合成釜中加入 483.60kg【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	1

★Note: 20%piperidine quantity is a CPP./备注：20%哌啶溶液质量是关键工艺参数

★Note:The de-Fmoc temperature is a CPP./ 备注：脱保护温度是关键工艺参数。

★Note: De-Fmoc time is a CPP./备注：脱保护时间是关键工艺参数

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：18.30%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 ºC ~30 ºC

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Prior to deprotection (piperidine addition), the operator shall confirm the jacket temperature are within the 25±5ºC (target: 25ºC) /脱保护（哌啶投料）前，操作人员须确认反应釜夹套温度均处于正常操作范围（NOR）内 25±5ºC（目标温度:25ºC）。

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

	Times /次数
3.23.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 【25±5℃ (target: 25℃) 】. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至【25±5 ℃(目标温度：25 ℃)】。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.23.3 /结束时间记录于 3.23.3	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.23.3 /结束时间记录于 3.23.3	
Transcribed by & Date/转录人及日期:		
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

3.23.3 After stirring for【110~130 min (target: 2h) at 25±5℃ (target: 25ºC)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5ºC（目标温度:25ºC）搅拌 110~130 min（目标时间:2h）】后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**Times /次数** **1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.23.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.23.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.23.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.23.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins 3.23.2/开始时间记录于 3.23.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After De-Fmoc is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 脱保护结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the total reaction/ （8）Fmoc-L-Asn(Trt)-OH De-Fmoc time is within the respective Normal Operating Ranges (NOR):110min –160min

工艺技术人员确认 Fmoc-N-三苯甲基-L-天冬酰胺脱保护总时间处于正常操作范围（NOR）范围内 110min –160min

- **Definition of De-Fmoc start time/脱保护（De-Fmoc）开始时间定义:**
  The De-Fmoc start time is defined as 20%Piperidine solution addition completion time in Ins3.23.1./ Ins 3.23.1 的 20%哌啶溶液投料结束时间为脱保护反应开始时间。
- **The De-Fmoc end time is defined as start of 1st post deprotection wash in Ins3.23.4./ Ins3.23.4 的脱保护后第一次洗涤开始时间定义为脱保护结束时间。**

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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Total De-Fmoc time/总脱保护时间 =Start of 1st post deprotection wash 脱保护第一次洗涤开始时间-End of charging 20% Piperidine solution/20%哌啶加料结束时间

Time/时间	Start of 1st post deprotection wash/脱保护第一次洗涤开始时间	End of charging 20% Piperidine solution/20%哌啶加料结束时间	Total De-Fmoc time/脱保护总时间计算	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★The operator shall confirm thatTotal De-Fmoc time is within the Normal Operating Range (NOR) 【110min~160min】/操作人员须确认总脱保护时间处于正常操作范围（NOR）内:【110min~160min】	**☐Yes/是 ☐No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**

Transcribed by & Date/转录人及日期:

★Note:The de-Fmoc temperature is a CPP./ 备注：脱保护温度是关键工艺参数。

★The operator shall confirm that the de-Fmoc internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认脱保护内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

★The operator shall confirm that 20%piperidine quantity is within the Normal Operating Range (NOR) 【454.58kg~512.62kg】/操作人员须确认 20%哌啶质量处于正常操作范围（NOR）内: 【454.58kg~512.62kg】

20%Piperidine quantity confirmation/20%哌啶溶液质量确认: _____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Note:备注：Ins 3.23.4~Ins3.23.8 and Ins 3.24.1~Ins3.24.3 need to be performed simultaneously.操作 3.23.4~操作 3.23.8 与操作 3.24.1~操作 3.24.3 同时进行

3.23.4 Post-deprotection wash/脱保护后洗涤	**Times /次数**
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。					
Filter/过滤	**Begin/开始**				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

Batch size (output range): 157.69~292.80 kg

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3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.23.5 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.23.6 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	328 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.23.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.23.（8）Fmoc-L-Asn(Trt)-OH De-Fmoc/Fmoc-N-三苯甲基-L-天冬酰胺脱保护

3.23.8 Resin wash reswelling 2/洗涤后溶胀 2	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间： 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.23.4~Ins3.23.8 and Ins 3.24.1~Ins3.24.3 need to be performed simultaneously.操作 3.23.4~操作 3.23.8 与操作 3.24.1~操作 3.24.3 同时进行

3.24.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度: 20℃）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5ºC (target: 20ºC)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5℃（目标温度：20℃）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins 3.25.3/结束时间记录于 3.25.3

Transcribed by & Date/转录人及日期:

3.24.2 Add Fmoc-L-Asn(Trt)-OH(2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg Fmoc-N-三苯甲基-L-天冬酰胺（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times/次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Asn(Trt)-OH / 加 Fmoc-N-三苯甲基-L-天冬酰胺	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Asn(Trt)-OH / 加 Fmoc-N-三苯甲基-L-天冬酰胺	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Asn(Trt)-OH confirmation/Fmoc-N-三苯甲基-L-天冬酰胺确认: Control No./内部控制号: YL004843

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

	Times /次数
3.24.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min. Confirm no visible solids. Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 20±5℃(target: 20℃). **维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min，确认无明显固体，激活罐温度 TJ4S-1213-Sy16-R01 调温至 20±5℃(目标温度：20℃)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ºC，High limit/高限：23ºC**	**1**

Maintain Stir/保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain Stir/保持搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.25.3/结束时间记录于 3.25.3	
Transcribed by & Date/转录人及日期: _____		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期: _____		

CAUTION/注意：OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

	Times /次数
3.24.4Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 20±5℃ (target: 20ºC). **控内温 20±5℃（目标温度:20℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）**	**1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

OxymaPure confirmation/2-肟氰乙酸乙酯确认：Control No./内部控制号：YL005180

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

After solid material charging, confirm/固体物料投料之后，确认:

- The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
- The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
- The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.24.5Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）(Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.24.6 Add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/h at 20±5°C (target: 20°C). The addition rate depends on temperature control. **控内温 20±5°C(目标温度：20°C)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N'-二异丙基碳二亚胺（2.20eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 40~80kg/h，实际以控内温为主。**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。
Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

DIC Addition/ 加 N,N'-二异丙基碳二亚胺	**Performed by & Date/执行人及日期:**	
DIC Addition/ 加 N,N'-二异丙基碳二亚胺	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Weighing of DIC/ N,N'-二异丙基碳二亚胺称重:

Control Number/内部控制号: YL003564 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

Addition type 加料方式	Lot number/批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/投料前	After charging/投料后	Net weight/净重	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIC Addition/加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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3.24.（8）Fmoc-L-Asn(Trt)-OH activation/Fmoc-N-三苯甲基-L-天冬酰胺活化

	Times/次数
3.24.7 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target:20min)after addition. **滴毕，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，控内温 20±5℃(目标温度：20℃)，继续搅拌 20±5min(目标时间：20min)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ºC，High limit/高限：23ºC**	**1**

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins3.25.1/Ins3.25.1 的转移开始时间

Transcribed by & Date/转录人及日期:

★After activation, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 活化结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF activation total quantity is a CPP./备注：N, N-二甲基甲酰胺活化总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.24.1			
Ins 3.24.5			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			
★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内：【405.09~456.81kg】			☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

3.25.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。	**Times /次数**
	1

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

Transfer/转移	**Performed by & Date/执行人及日期:**	
	Checked by & Date/核实人及日期:	
Transfer/转移	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.25.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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操作指令

3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

3.25.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). / 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins 3.25.5/结束时间记录于 3.25.5	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins 3.25.5/结束时间记录于 3.25.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.
/激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.24.1/开始时间记录于 3.24.1	
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3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

3.25.4 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 【25±5°C (target: 25°C)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5°C(目标温度：25 °C)】.**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期: _____

3.25.5 The mixture is allowed to react at internal temperature 25±5 °C (target: 25 °C), start timing when the temperature reaches 20 °C. The reaction is 3.5h~8h(target: 4h).After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black,react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温 25±5°C（目标温度 25°C，20°C 开始计时）反应时间是 3.5h~8h(目标时间：4h)。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全; 树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 °C，High limit/高限：28°C**	**Times /次数** **1**

The total reaction Fmoc-L-Asn(Trt)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ Fmoc-N-三苯甲基-L-天冬酰胺偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

Batch size (output range): 157.69~292.80 kg

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React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.25.7./偶联反应结束时间即为 Ins 3.25.7 洗涤 1 的开始时间。	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

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Title: 140021 Step: 1 Preparation of: 140851

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3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.25.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.25.5-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins3.25.3 /开始时间记录于 3.25.3	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.25.3 /开始时间记录于 3.25.3	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.24.1			
Ins 3.24.5			
Ins 3.25.2			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			
★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【463.14~522.26kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内：【463.14~522.26kg】			☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:			
Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.25.6 Activation Tank Rinse/激活罐洗涤	**Times/次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 616.20kg DMF (9.48W,10.0Vol) by spray balls. Circulate and clean the activating liquid transfer pipeline1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐中加入 616.20kg N, N-二甲基甲酰胺（9.48W ,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，循环清洗激活液转移管路 1，液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Operation name/操作名称	**Time point/时间节点**	**Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1**	**liquid charging port cleaning pipeline 2,/液体加料口清洗管路 2**	**Solid charging port cleaning pipeline3 /固体加料口清洗管路 3**
		Wash /洗涤 1	**Wash /洗涤 2**	**Wash /洗涤 3**
DMF addition/加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**			
DMF addition/加 N,N-二甲基甲酰胺	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/转录人及日期:			
Circulate and clean/循环清洗	**Performed by & Date/执行人及日期:**			
Circulate and clean/循环清洗	**Begin/开始**			
	End/结束			
	Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

	Times /次数
3.25.7 After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min (target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. . Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 ./ 待合成釜排废完毕后将激活罐中体系转移至合成釜中				
Transfer/转移	**Begin/开始**			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**			
Bubble/鼓氮气	**Begin/开始**			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Stir /搅拌	**Performed by & Date/ 执行人及日期:**			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.25.（8）Fmoc-L-Asn(Trt)-OH coupling/ Fmoc-N-三苯甲基-L-天冬酰胺偶联

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times /次数
3.26.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20%Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内：17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

	Times /次数
3.26.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).. Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min（目标时间：5 min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5 ℃(目标温度：25 ℃)。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min)开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.26.3 /结束时间记录于 3.26.3	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.26.3 /结束时间记录于 3.26.3	
Transcribed by & Date/转录人及日期:		

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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Operations Instruction
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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

	Times /次数
3.26.3 After stirring for 110~130 min (target: 2h) at 25±5℃ (target: 25ºC), stop stirring, turn off the nitrogen. Drain until there is no drip, the operator visually confirms the discharge end point. **控内温 25±5ºC（目标温度:25ºC）搅拌 110~130 min（目标时间:2h）后，停止搅拌，关闭氮气，滤除反应液至无滴出，人员目视确认排废终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.26.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.26.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.26.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.26.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.26.2 /开始时间记录于 3.26.2	
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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

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Note:备注： Ins 3.26.4~Ins3.26.8 ,Ins 3.27.1~Ins3.27.2 and Ins3.28.3~Ins3.28.4 need to be performed simultaneously.操作 3.26.4~Ins3.26.8 与操作 3.27.1~操作 3.27.2 和操作 3.28.3~操作 3.28.4 同时进行

3.26.4 Post-deprotection wash/脱保护后洗涤	**Times /次数**
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) . Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min).-Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Stir /搅拌	Performed by & Date/执行人及日期:				
Stir /搅拌	Begin/开始				
	End/结束				
	Transcribed by & Date/转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/执行人及日期:				
	Checked by & Date/核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.26.5 Resin wash reswelling 1/洗涤后溶胀 1	Times/次数
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	1

DMF addition/加 N,N-二甲基甲酰胺	Performed by & Date/执行人及日期:		
DMF addition/加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:	Qty/重量:
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

3.26.6 Resin wash reswelling 1/洗涤后溶胀 1	Times /次数
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

3.26.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.26.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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3.26.（9）Fmoc-L-Cys(Trt)-OH De-Fmoc/Fmoc-S-三苯甲基-L-半胱氨酸脱保护

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.26.4~Ins3.26.8 ,Ins 3.27.1~Ins3.27.2 and Ins3.28.3~Ins3.28.4 need to be performed simultaneously.操作 3.26.4~Ins3.26.8 与操作 3.27.1~操作 3.27.2 和操作 3.28.3~操作 3.28.4 同时进行

3.27.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度: 20°C）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

★Note: DMF dissolution total quantity is a CPP./备注：N, N-二甲基甲酰胺溶解总用量是关键工艺参数

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度：20°C）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.28.6 /结束时间记录于 3.28.6

Transcribed by & Date/转录人及日期:

3.27.2 Add Fmoc-L-Cys(Trt)-OH (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg Fmoc-S-三苯甲基-L-半胱氨酸（2.00eq）** **（Acceptable weighing range/可接受称量范围: _____kg~_____kg）。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Fmoc-L-Cys(Trt)-OH / 加 Fmoc-S-三苯甲基-L-半胱氨酸	**Performed by & Date/执行人及日期:**	
Charge Fmoc-L-Cys(Trt)-OH / 加 Fmoc-S-三苯甲基-L-半胱氨酸	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Fmoc-L-Cys(Trt)-OH confirmation/ Fmoc-S-三苯甲基-L-半胱氨酸确认: Control No./内部控制号: YL003565

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/ 拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

After solid material charging confirm/固体物料投料之后，确认:

➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域

➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域

➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

★Note:The quantity of Fmoc-L-Cys(Trt)-OH is a CPP./备注：Fmoc-S-三苯甲基-L-半胱氨酸总用量是关键工艺参数

★NOR limit/正常操作范围: _____kg~ _____kg

Filled by& Date/填写人及日期: _____

Checked by & Date/核实人及日期: _____

★After charging, the operator shall confirm that Critical Process Parameters (CPP) is within the respective Normal Operating Ranges (NOR)/ 投料结束后，操作人员须确认关键工艺参数（CPP）均处于正常操作范围（NOR）内

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

CAUTION/注意：OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.27.3 Add Oxyma Pure (3.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 加入_____kg 2-肟氰乙酸乙酯（3.00eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期：_____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / **加 2-肟氰乙酸乙酯**	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / **加 2-肟氰乙酸乙酯**	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
OxymaPure confirmation/2-肟氰乙酸乙酯确认：Control No./内部控制号：YL005180		

Batch size (output range): 157.69~292.80 kg

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3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	365 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.27.4 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min. Confirm no visible solids and later use **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）(Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min。确认无明显固体，待用。**	**Times /次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**
Maintain &Stir /保持搅拌	**Performed by & Date/执行人及日期:**
Maintain &Stir /保持搅拌	**Begin/开始:**
	The end time is documented in Ins3.28.6 /结束时间记录于 3.28.6
Transcribed by & Date/转录人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

	Times/次数
3.27.5 Adjust to 5±5 °C. Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 5±5min (Target:5min) at 5±5 °C (target: 5°C). **调温至 5±5 °C。控内温 5±5 °C（目标温度:5°C），维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 5±5min(目标时间：5min)，待用。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：2 °C，High limit/高限：8°C**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After dissolution, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 溶解结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: DMF dissolution total quantity is a CPP./备注：N, N-二甲基甲酰胺溶解总用量是关键工艺参数

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.27.（9）Fmoc-L-Cys(Trt)-OH dissolution /Fmoc-S-三苯甲基-L-半胱氨酸溶解

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.27.1			
Ins 3.27.4			
DMF activation total quantity / N, N-二甲基甲酰胺活化总用量			

★The operator shall confirm that DMF activation total quantity is within the Normal Operating Range (NOR) 【405.09~456.81kg】/操作人员须确认 N, N-二甲基甲酰胺活化总用量处于正常操作范围（NOR）内: 【405.09~456.81kg】	☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

3.28.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times/次数**
	1

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Transcribed by & Date/转录人及日期: _____

Before using, confirm hastelloy agitator tank oxygen content≤1.0%.

使用前确认哈氏合金搅拌罐氧气含量≤1.0%。

500 L Hastelloy agitator tank

哈氏合金搅拌罐: TJ4S-1200-AT01

Oxygen detector ID. No. /氧气检测器编号: _____

Oxygen content/氧气含量= _____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Before using, confirm the hastelloy agitator tank TJ4S-1200-AT01's calibration information is within valid date. /使用前确认哈氏合金搅拌罐 TJ4S-1200-AT01 的仪表在校准有效期内。

Checked by & Date/核实人及日期: _____

Before using, confirm that cleaning of the hastelloy agitator tank TJ4S-1200-AT01 has been completed/使用前，确认哈氏合金搅拌罐 TJ4S-1200-AT01 清洁已完成

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.28.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times/次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

CAUTION/注意：DIC is a key EHS concern chemical, highly toxic, skin sensitizing, respiratory sensitizing. /N,N'二异丙基碳二亚胺为 EHS 重点关注化学品，高毒，皮肤致敏，呼吸致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Note:备注：Ins 3.26.4~Ins3.26.8 ,Ins 3.27.1~Ins3.27.2 and Ins3.28.3~Ins3.28.4 need to be performed simultaneously.操作 3.26.4~Ins3.26.8 与操作 3.27.1~操作 3.27.2 和操作 3.28.3~操作 3.28.4 同时进行

3.28.3 Preparation of DIC/DMF Solution/配置 DIC/DMF 溶液	**Times /次数**
Add 61.75kg DMF (0.95W, 1.0Vol) into the hastelloy agitator tank TJ4S-1200-AT01. Start the agitator of the tank to 60±20 rpm (target: 60 rpm).Then add DIC(3.30eq) by peristaltic pump. **向哈氏合金搅拌罐 TJ4S-1200-AT01 中加入 61.75kg N,N-二甲基甲酰胺（0.95W, 1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg），开启搅拌转速至 60±20 rpm (目标转速: 60 rpm)。通过蠕动泵泵入_____kg N,N'-二异丙基碳二亚胺（3.30eq）（Acceptable weighing range/可接受称量范围：_____kg ~_____kg）。**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期: _____

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Weighing of DMF / N,N-二甲基甲酰胺称重:

Control Number/内部控制号: YL000336 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/ 还需加入净重目标值	Gross qty of drum that to be charged/ 待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DMF/N，N-二甲基甲酰胺	Begin/开始:	End/结束:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Start the agitator of the tank to 60±20 rpm (target: 60 rpm). /开启搅拌转速至 60±20 rpm (目标转速: 60 rpm)。

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.28.7 /结束时间记录于 3.28.7

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期: _____

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Weighing of DIC / N,N'-二异丙基碳二亚胺称重:

Control Number/内部控制号: YL003564 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

DIC addition/ 加 N,N'-二异丙基碳二亚胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Addition type 加料方式	Lot number/批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/投料前	After charging/投料后	Net weight/净重	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
☐Full drum 整桶 ☐Non-full drum / 非整桶											
☐Full drum 整桶 ☐Non-full drum / 非整桶											
Total/合计											
DIC / N,N'-二异丙基碳二亚胺	**Begin/开始:**						**End/结束:**				

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

3.28.4 Continue agitation of tank to 60±20 rpm (target: 60rpm). Maintain stirring for NLT 15min for later use. **维持搅拌罐搅拌转速至 60±20 rpm(目标转速：60rpm)。继续搅拌至少 15min，待用。**	**Times/次数**
	1

Maintain &Stir 保持搅拌	**Performed by & Date/执行人及日期:**
Maintain &Stir 保持搅拌	**Begin/开始:**
	The end time is documented in Ins3.28.7 /结束时间记录于 3.28.7

Transcribed by & Date/转录人及日期:

Prior to DIC/DMF solution transfer.Confirm the temperature is within 10~30 ℃. / DIC/DMF 溶液转移前，确认温度在 10~30℃。

Temperature/温度: _____ **□Yes/是 □No/否**

Checked by & Date/核实人及日期: _____

Solution No. /溶液编号：CPo140021-_____-01-_____-3.28.4-01

3.28.5 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数**
	1

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). / 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.28.9 /结束时间记录于 3.28.9

Transcribed by & Date/转录人及日期:

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.28.9 /结束时间记录于 3.28.9

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	376 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Transcribed by & Date/转录人及日期:

激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

3.28.6 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to15~25°C（Target:20 °C）. **合成釜 TJ4S-1213-Sy16 调温至 15~25°C（目标温度:20C）。**	**Times/次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**
Activation tank Stir /激活罐搅拌	**End/结束:**
	The start time is documented in Ins 3.27.1/开始时间记录于 3.27.1

Transcribed by & Date/转录人及日期:

3.28.7 Transfer the DIC/DMF solution (Prepared in Ins.3.28.4) into the mixture at internal temperature 15~25°C（Target: 20°C）. **控内温 15~25°C（目标温度: 20°C），转移操作 3.28.4 配制的 N,N'-二异丙基碳二亚胺/N,N-二甲基甲酰胺溶液。**	**Times/次数** **1**

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Upon completion of material transfer, perform nitrogen purge on the dedicated DIC/DMF transfer line: maintain nitrogen flow rate ≥ 6 m³/h for a continuous duration of ≥ 4 min to displace residual DIC/DMF solution in the line/物料转移完成后，对 DIC/DMF 专用输送管线进行氮气吹扫: 控制氮气流量≥6 m³/h，连续吹扫时长≥4 min。用以置换管线内残留的 DIC/DMF 溶液

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Nitrogen Purge/氮气吹扫	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Agitator tank Stir 搅拌罐搅拌	**Performed by & Date/执行人及日期:**	
Agitator tank Stir /搅拌罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.28.3/开始时间记录于 3.28.3	
Transcribed by & Date/转录人及日期:		

3.28.8 Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to 20±5ºC (target: 20ºC). **合成釜 TJ4S-1213-Sy16 内温调温至 20±5℃(目标温度：20 ℃)。**	**Times/次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

★Note:The total quantity of DMF is a CPP./备注：DMF 总用量是关键工艺参数。

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

	Times/次数
3.28.9 The mixture is allowed to react at internal temperature【20±5 ºC (target: 20 ºC)】, start timing when the temperature reaches 15 ºC). The reaction is 3.5h~8h(target: 4h) . After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black,react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【20±5ºC（目标温度 20ºC，15ºC 开始计时）】，反应时间是 3.5h~8h(目标时间：4h)，反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h(目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：17 ºC ， High limit/高限： 23ºC**	**1**
The total reaction Fmoc-L-Cys(Trt)-OH coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ Fmoc-S-三苯甲基-L-半胱氨酸偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.28.11./偶联反应结束时间即为 Ins 3.28.11 洗涤 1 的开始时间	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :15~25℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:15~25℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.
取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5～10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.28.9-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.28.9-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**End/结束:**
	The start time is documented in Ins 开始时间记录于 3.28.5 /开始时间记录于 3.28.5
Transcribed by & Date/转录人及日期:	
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**End/结束:**
	The start time is documented in Ins3.28.5 /开始时间记录于开始时间记录于 3.28.5
Transcribed by & Date/转录人及日期:	

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期	
	Checked by & Date/核实人及日期:	

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内
★Note: DMF coupling total quantity is a CPP./备注：N, N-二甲基甲酰胺偶联总用量是关键工艺参数

Ins/操作号	Weight/重量	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期
Ins 3.27.1			
Ins 3.27.4			
Ins 3.28.2			
Ins3.28.3			
DMF coupling total quantity / N, N-二甲基甲酰胺偶联总用量			

★The operator shall confirm that DMFcoupling total quantity is within the Normal Operating Range (NOR) 【521.18~587.72kg】/操作人员须确认 N, N-二甲基甲酰胺偶联总用量处于正常操作范围（NOR）内: 【521.18~587.72kg】	☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.28.10 Activation Tank Rinse/激活罐洗涤	**Times /次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 616.20kg DMF (9.48W,10.0Vol) by spray balls. Circulate and clean the activating liquid transfer pipeline1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐中加入 616.20kg N, N-二甲基甲酰胺（9.48W ,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，循环清洗激活液转移管路 1，液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Operation name/ 操作名称	**Time point/时间节点**	**Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1**	**liquid charging port cleaning pipeline 2,/ 液体加料口清洗管路 2**	**Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3**
		Wash /洗涤 1	**Wash /洗涤 2**	**Wash /洗涤 3**
DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/ 执行人及日期:**			
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/循环清洗	**Performed by & Date/ 执行人及日期:**			
Circulate and clean/循环清洗	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

3.28.11 Post-coupling wash./偶联后洗涤	**Times /次数**
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min(target:5min), start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Operation name/ 操作名称	**Time point/时间节点**	**Wash /洗涤 1**	**Wash /洗涤 2**	**Wash /洗涤 3**
Transfer/转移	**Begin/开始**			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**			
Bubble/鼓氮气	**Begin/开始**			
	End/结束			
	Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.28.（9）Fmoc-L-Cys(Trt)-OH coupling/Fmoc-S-三苯甲基-L-半胱氨酸偶联

Stir /搅拌	**Performed by & Date/ 执行人及日期:**			
Stir /搅拌	**Begin/开始**			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	**Begin/开始**			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	385 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

CAUTION/注意：Piperidine 哌啶：Strong local irritant and may cause permanent injury after short exposure to small amounts. /强烈的局部刺激，短时间少量接触可能造成永久性伤害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times /次数
3.29.1 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25±5°C (target: 25°C). Add 483.60kg 20% (V/V) piperidine/DMF (7.44W, 8.0Vol) solution into the synthesis reactor. **将合成釜 TJ4S-1213-Sy16 夹套调温至 25±5°C（目标温度：25°C）。向合成釜中加入 483.60kg 【20%（v/v）哌啶/DMF 溶液（7.44W, 8.0Vol）】（Acceptable weighing range/可接受称量范围：454.58kg~512.62kg）。**	1

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Confirm and record that the 20% Piperidine solution concentration is within the NOR range./ 确认并记录 20%哌啶溶液浓度在正常操作范围内:17.50%~22.50%

20%Piperidine solution concentration confirmation/20%哌啶溶液浓度确认：_____

□Yes/是 □No/否

Confirm and record that the 20%Piperidine solution temperature is within the NOR range./ 确认并记录 20%哌啶溶液温度在正常操作范围内：20 °C ~30 °C

20%Piperidine solution temperature confirmation/20%哌啶溶液温度确认：_____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期：_____

Checked by & Date/核实人及日期：_____

Piperidine addition/ 加哌啶溶液	**Performed by & Date/执行人及日期:**		
Piperidine addition/ 加哌啶溶液	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

	Times /次数
3.29.2 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust to 25±5℃ (target: 25℃). **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min（目标时间：5 min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，调温至 25±5℃(目标温度：25℃)。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target :5 min) 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min)

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.29.3 /结束时间记录于 3.29.3
Transcribed by & Date/转录人及日期:	
Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.29.3 /结束时间记录于 3.29.3
Transcribed by & Date/转录人及日期:	

Adjust to 25±5 ºC (target: 25ºC)./ 调温至 25±5 ℃(目标温度：25 ºC)

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

	Times/次数
3.29.3After stirring for 110~130 min (target: 2 h) at 25±5 ℃ (target: 25 ℃), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温 25±5 ℃(目标温度：25 ℃)， 搅拌 110~130 min（目标时间：2 h）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC，High limit/高限：28ºC**	**1**

Stir/搅拌	**Begin/开始:**	**End/结束:**
Stir/搅拌	**Transcribed by & Date/转录人及日期:**	
De-Fmoc/脱 Fmoc	**Begin/开始:**	**End/结束:**
	20%Piperidine solution addition completion time is defined as De-Fmoc start time, documented in Ins 3.29.1/20%哌啶溶液投料结束时间为脱保护反应开始时间记录于 3.29.1。	
	The De-Fmoc end time is defined as start of 1st post deprotection wash,documented in Ins 3.29.4/ 脱保护结束时间定义为脱保护后第一次洗涤开始时间记录于 3.29.4	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.29.2 /开始时间记录于 3.29.2	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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	QF05098	12.0	09/29/2026	388 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
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Signature &Date/签名及日期: _____

Note:备注：Ins 3.29.4~Ins3.29.8 and Ins 3.30.1~Ins3.30.3 need to be performed simultaneously.操作 3.29.4~操作 3.29.8 与操作 3.30.1~操作 3.30.3 同时进行

3.29.4 Post-deprotection wash/脱保护后洗涤	**Times /次数**
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min(目标时间: 5min)后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Add 567.45kg IPAc(8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)					
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**				
IPAC addition/ 加乙酸异丙酯	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/ 转录人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/ 转录人及日期:				
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。					
Filter/过滤	**Begin/开始**				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

3.29.5 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.29.6 Resin wash reswelling 1/洗涤后溶胀 1	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

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3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
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3.29.7 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Add 616.20kg DMF(9.48W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg）。**	**1**

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.29.8 Resin wash reswelling 2/洗涤后溶胀 2	**Times /次数**
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2 min (target: 5 min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2 min (target: 15 min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2 min (目标时间：5 min) 后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2 min（目标时间：15 min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.29.（10）Boc-L-Glu-OtBu De-Fmoc/Boc-L-谷氨酸-1-叔丁酯脱保护

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

Note:备注：Ins 3.29.4~Ins3.29.8 and Ins 3.30.1~Ins3.30.3 need to be performed simultaneously.操作 3.29.4~操作 3.29.8 与操作 3.30.1~操作 3.30.3 同时进行

3.30.1 Add solvent to the activation tank/激活罐加溶剂	**Times /次数**
Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). **将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度: 20°C）。通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg），开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。**	**1**

Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 夹套调温至 20±5°C（目标温度：20°C）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Add 369.85kg DMF (5.69W, 6.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01./ 通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 369.85kg N, N-二甲基甲酰胺（5.69W, 6.0Vol）（Acceptable weighing range/可接受称量范围: 351.36kg~388.34kg）

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	394 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.31.3/结束时间记录于 3.31.3

Transcribed by & Date/转录人及日期:

3.30.2Add Boc-L-Glu-OtBu (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. **通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg Boc-L-谷氨酸-1-叔丁酯（2.00eq）(Acceptable weighing range/可接受称量范围: _____kg~_____kg)。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge Boc-L-Glu-OtBu / **加 Boc-L-谷氨酸-1-叔丁酯**	**Performed by & Date/执行人及日期:**	
Charge Boc-L-Glu-OtBu / **加 Boc-L-谷氨酸-1-叔丁酯**	**Begin/开始:**	**End/结束:**

Transcribed by & Date /转录人及日期:

Boc-L-Glu-OtBu confirmation/Boc-L-谷氨酸-1-叔丁酯确认: Control No./内部控制号: YL008618

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Retrieved by Simon Cushen on 29 Oct 2026 at 12:35PM GMT+00:00.

Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.30.3 Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for NLT 15min. Confirm no visible solids .Adjust the activation tank TJ4S-1213-Sy16-R01 temperature to 5±5℃(target: 5℃). 维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌至少 15min，确认无明显固体，激活罐温度 TJ4S-1213-Sy16-R01 调温至 5±5℃(目标温度: 5℃)。	**Times /次数**
	1

Maintain Stir /保持搅拌	**Performed by & Date/执行人及日期:**	
Maintain Stir /保持搅拌	**Begin/开始:**	
	The end time is documented in Ins3.31.3 /结束时间记录于 3.31.3	
Transcribed by & Date/转录人及日期:		
Confirm no visible solid/确认无明显固体:	**Confirmed by& Date /确认人及日期:**	
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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ASYMCHEM	Document No./文件编号	Ver./版本	Effective Date/生效日期	Page/页码
	QF05098	12.0	09/29/2026	397 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

CAUTION/注意： PyBOP is high energy compounds, impact sensitive, skin sensitizing. /1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐是高能化合物，撞击敏感性，皮肤致敏。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.30.4Add PyBOP (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 5±5℃ (target: 5ºC). **控内温 5±5℃（目标温度:5℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐（2.00eq）（Acceptable weighing range/可接受称量范围：_____kg~ kg）。**	1

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge PyBOP /加 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	**Performed by & Date/执行人及日期:**	
Charge PyBOP /加 1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

PyBOP confirmation/1H-苯并三唑-1-基氧三吡咯烷基六氟磷酸盐确认：Control No./内部控制号：YL002855

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认		☐Conforms/相符 ☐ Does not conform/不相符		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

Internal temperature/内部温度	Highest values/最高值:	Lowest values/最低值:

Transcribed by & Date/转录人及日期:

CAUTION/注意: OxymaPure is highly toxic, high-energy compound. /2-肟氰乙酸乙酯是高毒，高能化合物。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.30.5 Add Oxyma Pure (2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01 at 5±5℃ (target: 5℃). **控内温 5±5℃（目标温度:5℃），通过投料袋向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg 2-肟氰乙酸乙酯（2.00eq）（Acceptable weighing range/可接受称量范围: _____kg ~_____kg）**	**1**

Check and record that the solid charge port is free of solid accumulation from the previous step./ 检查并记录固体投料口无来自上一步骤的固体残留。

Checked by & Date/核实人及日期: _____

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Before solid material charging, confirm/加固体物料前，确认:
➢Before performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

Prior to charging each amino acid, the operator shall perform a Raman scan to confirm the identity of the material. The Raman scan result shall be compared against the reference spectrum of the corresponding amino acid. Only materials that conform to the reference spectrum may be charged. If the Raman scan result does not conform, stop the operation, quarantine the material, and initiate an investigation per site SOP. Record the Raman scan result, operator signature, and date/time in the table below./ 每次投氨基酸前，操作人员应进行拉曼扫描以确认物料。拉曼扫描结果应与相应氨基酸的参考光谱进行比对。仅与参考光谱相符的物料方可投料。如拉曼扫描结果不相符，应停止操作，对物料进行隔离，并依据厂区 SOP 启动异常事件调查。在下表中记录拉曼扫描结果、操作人员签字及日期 / 时间。

Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Performed by & Date/执行人及日期:**	
Charge OxymaPure / 加 2-肟氰乙酸乙酯	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期: _____

OxymaPure confirmation/2-肟氰乙酸乙酯确认: Control No./内部控制号: YL005180

Check items/检查项	Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称	☐Yes/是; ☐No/否		
Project batch number/ 项目批次号	☐Yes/是; ☐No/否		
Operation number/ 操作号	☐Yes/是; ☐No/否		
Control No./内部控制号	☐Yes/是; ☐No/否		
Raman Scan Confirmation/拉曼扫描确认	☐Conforms/相符 ☐ Does not conform/不相符		

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.30.6 Rinse reactor wall with 61.75kg DMF (0.95W, 1.0Vol) by spray balls. **通过淋洗球用 61.75kg N,N-二甲基甲酰胺（0.95W，1.0Vol）（Acceptable weighing range/可接受称量范围: 58.05kg~65.46kg）淋洗釜壁。**	**Times /次数**
	1

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Retrieved by Simon Cushen on 29 Oct 2026 at 12:35PM GMT+00:00.

Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

CAUTION/注意：DIPEA/N,N-二异丙基乙胺：It can enter the body through inhalation, ingestion and skin absorption. It is strongly irritating to the eyes, skin, mucous membranes and upper respiratory tract. Inhalation of high-concentration vapors may cause inflammation, edema and spasm of the larynx/bronchi, even chemical pneumonia and pulmonary edema. Exposure may cause symptoms such as burning sensation, cough, headache, nausea and vomiting. /可通过吸入、食入、皮肤吸收侵入人体，对眼、皮肤、黏膜和上呼吸道有强刺激性，吸入高浓度蒸气会引发喉/支气管炎症水肿、痉挛，甚至化学性肺炎、肺水肿，接触后会出现烧灼感、咳嗽、头痛、恶心呕吐等症状
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

	Times/次数
3.30.7 Add DIPEA (3.00eq) through metering pump into activation tank TJ4S-1213-Sy16-R01 at a reference rate of 20~40 kg/h at 5±5°C (target: 5°C). The addition rate depends on temperature control. **控内温 5±5°C(目标温度：5°C)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N-二异丙基乙胺（3.00eq）（Acceptable weighing range/可接受称量范围：_____kg~_____kg），参考加料速度 20~40kg/h，实际以控内温为主。** **Weighing of DIPEA/N,N-二异丙基乙胺称重:** **Control Number/内部控制号：YL000555 Scale ID. No./秤编号：_____** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____**	**1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期：_____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。
Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:
1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。
Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

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3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Add DIPEA (3.00eq) through metering pump into activation tank TJ4S-1213-Sy16-R01 at a reference rate of 20~40 kg/h at 5±5ºC (target: 5ºC). The addition rate depends on temperature control.

控内温 5±5℃(目标温度：5℃)，通过计量泵向激活罐 TJ4S-1213-Sy16-R01 中加入_____kg N,N-二异丙基乙胺（3.00eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，参考加料速度 20~40 kg/h.

DIPEA Addition/ 加 N,N-二异丙基乙胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

DIPEA /N,N-二异丙基乙胺: Control Number/内部控制号: YL000555

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/ 还需加入净重目标值	Gross qty of drum that to be charged/ 待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DIPEA Addition/加 N,N-二异丙基乙胺	Begin/开始:	End/结束:
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.30.（10）Boc-L-Glu-OtBu activation/Boc-L-谷氨酸-1-叔丁酯活化

3.30.8 Adjust the temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C). **将激活罐 TJ4S-1213-Sy16-R01 调温至 20±5℃（目标温度：20℃）。**	**Times /次数**
	1

Adjust the temperature of activation tank TJ4S-1213-Sy16-R01 to 20±5°C (target: 20°C)/ 将激活罐 TJ4S-1213-Sy16-R01 调温至 20±5℃（目标温度：20℃）。

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

3.30.9 At 20±5℃ (target: 20℃). Continue agitation of activation tank to 115±5 rpm (target: 115 rpm). Maintain stirring for 20±5min (Target: 20min) after addition. **滴毕，控内温 20±5℃(目标温度：20℃)，维持激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)，继续搅拌 20±5min(目标时间：20min)。**	**Times /次数**
	1

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	**End/结束**
	Stirring start/搅拌开始时间	Transfer start in Ins 3.31.1/Ins3.31.1 的转移开始时间

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Batch size (output range): 157.69~292.80 kg

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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

3.31.1 Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. **将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。**	**Times /次数** **1**

★Note:The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./ 备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

3.31.2 Add 61.75 kg DMF (0.95W, 1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. **转移完毕，通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 61.75 kg N,N-二甲基甲酰胺（0.95W,1.0Vol）（Acceptable weighing range/可接受称量范围：58.05kg~65.46kg）淋洗釜壁。**	**Times /次数** **1**

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期: _____

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Operations Instruction
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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

3.31.3 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. **开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。**	**Times/次数** **1**

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm)./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后,开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	
	The end time is documented in Ins3.31.5 /结束时间记录于 3.31.5	
Transcribed by & Date/转录人及日期:		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:**	
	The end time is documented in Ins3.31.5 /结束时间记录于 3.31.5	
Transcribed by & Date/转录人及日期:		

Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16./激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜 TJ4S-1213-Sy16 中。

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.30.1/开始时间记录于 3.30.1	
Transcribed by & Date/转录人及日期:		

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Operations Instruction
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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

3.31.4Adjust the internal temperature in synthesis reactor TJ4S-1213-Sy16 to【 25±5ºC (target: 25ºC)】. **合成釜 TJ4S-1213-Sy16 内温调温至【25±5℃(目标温度：25 ℃)】。**	**Times /次数**
	1

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

3.31.5 The mixture is allowed to react at internal temperature【 25±5 ºC (target: 25 ºC) 】,start timing when the temperature reaches 20 ºC. The reaction is 3.5h~8h(target: 4h).After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (the resin is yellow to orange, react completely; the resin is blue to black,react incompletely) or the reaction has stirred for 3.5h~8h(target: 4h), whichever is longer. Stop the stirrer and turn off the nitrogen. Start drain the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【25±5ºC（目标温度 25ºC，20ºC 开始计时）】反应时间是 3.5h~8h(目标时间：4h)。反应 1h±15min（目标时间:1h）后取样进行溴酚蓝检测。每隔 1h±15min 取样一次，直至连续两次取样结果阴性（树脂黄色至橙色，反应完全；树脂蓝色至黑色，反应不完全），或者反应已经搅拌了 3.5h~8h (目标时间：4h)，以较长时间的情况为准。停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **Low limit/低限：22 ºC ， High limit/高限： 28ºC**	**Times /次数**
	1

The total reaction Boc-L-Glu-OtBu coupling time: NMT 600 min (Sourced from the holding time study in the PAR study report.)/ Boc-L-谷氨酸-1-叔丁酯偶联总的反应时间不超过 600min(来源于 PAR 研究报告中稳定时间研究)。

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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

React/反应	**Performed by & Date/执行人及日期:**	
	Begin/开始:	**End/结束:**
	The completion time of the coupling reaction serves as the start time of Wash 1 specified in Ins 3.31.7./偶联反应结束时间即为 Ins 3.31.7 洗涤 1 的开始时间	
Transcribed by & Date/转录人及日期:		
Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**
Transcribed by & Date/转录人及日期:		

★After coupling is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 偶联结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Note: The coupling temperature is a CPP, and the coupling IPC hold temperature is a CPP./备注：偶联温度是 CPP，偶联 IPC 维持温度是关键工艺参数。

The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) :20~30℃ /操作人员须确认反应内温处于正常操作范围（NOR）内:20~30℃

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 5–10 mL of the mixture from the third portion for testing in accordance with the Bromophenol-blue colorimetric test record. Perform IPC testing immediately after sampling and perform testing in accordance with the Bromophenol-blue colorimetric test record.

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 5 ~ 10 mL，按照溴酚蓝比色检测记录开展检测，取样完成后立即进行 IPC 检测

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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	Negative or Positive/ 阳性或阴性	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.31.5-01			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-02			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-03			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-04			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-05			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-06			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-07			☐Negative /阴性 ☐Positive /阳性		
CPo140021-_____-01-_____-3.31.5-08			☐Negative /阴性 ☐Positive /阳性		

Note:For a failing coupling test.Continue hourly sampling up to the NMT 480 min hold limit; if still positive, stop, notify PM/QA and raise a deviation; no re-coupling without an approved deviation.偶联反应检测不合格的处置规则:按每小时 1 次的频率继续取样检测，直至达到不超过（NMT）480 min的保温时限；若到达时限后检测仍为阳性，停止反应，通知项目经理（PM）/ 质量保证部门（QA）并发起偏差报告；未经偏差批准，不得进行补偶联操作。

Confirm the end point of the reaction and move to following operation. /确认反应终点后进行下一步操作。

The acceptance criterion is two consecutive negative test results./连续两次检测结果为阴性即为合格

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

Stop the stirrer and turn off the nitrogen. Start drain the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./ 停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins3.31.3 /开始时间记录于 3.31.3	
Transcribed by & Date/转录人及日期:		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.31.3 /开始时间记录于 3.31.3	
Transcribed by & Date/转录人及日期:		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.31.6 Activation Tank Rinse/激活罐洗涤	**Times /次数**
Rinse the activation tank TJ4S-1213-Sy16-R01 with 616.20kg DMF (9.48W,10.0Vol) by spray balls. Circulate and clean the activating liquid transfer pipeline1, liquid charging port cleaning pipeline 2 and solid charging port cleaning pipeline 3, clean every pipeline for 2±1min. **通过淋洗球向激活罐中加入 616.20kg N, N-二甲基甲酰胺（9.48W ,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，循环清洗激活液转移管路 1，液体加料口清洗管路 2 和固体加料口清洗管路 3，每条线路各清洗 2±1min。**	**3**

Batch size (output range): 157.69~292.80 kg

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3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

Operation name/ 操作名称	Time point/时间节点	Clean the activating liquid transfer pipeline 1/循环清洗激活液转移管路 1	liquid charging port cleaning pipeline 2,/ 液体加料口清洗管路 2	Solid charging port cleaning pipeline3 / 固体加料口清洗管路 3
		Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
DMF addition/ 加 N,N-二甲基甲酰胺	Performed by & Date/ 执行人及日期:			
DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			
Circulate and clean/ 循环清洗	Performed by & Date/ 执行人及日期:			
Circulate and clean/ 循环清洗	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			

3.31.7 Post-coupling wash./偶联后洗涤	Times/ 次数
After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to the synthesis reactor TJ4S-1213-Sy16 . Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **待合成釜排废完毕后将激活罐中体系转移至合成釜中，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间：10min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.31.（10）Boc-L-Glu-OtBu coupling/Boc-L-谷氨酸-1-叔丁酯偶联

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Transfer/转移	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h), Bubble with nitrogen for 5±2min(target:5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm).After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 10±2min（目标时间: 10min）后，停止搅拌，关闭氮气				
Bubble/鼓氮气	Performed by & Date/ 执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Stir /搅拌	Performed by & Date/ 执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/ 转录人及日期:			
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point/通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。				
Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/ 执行人及日期:			
	Checked by & Date/ 核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32.Iodine oxidation/碘氧化

Before using, confirm glass lined agitator tank oxygen content≤1.0%.

使用前确认搪玻璃搅拌罐氧气含量≤1.0%。

500 L Glass lined agitator tank

500L 搪玻璃搅拌罐： TJ4S-1213-AT09

Oxygen detector ID. No. /氧气检测器编号：_____

Oxygen content/氧气含量= _____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期：_____

Before using, confirm the glass lined agitator tank TJ4S-1213-AT09's calibration information is within valid date. /使用前确认搪玻璃搅拌罐 TJ4S-1213-AT09 的仪表在校准有效期内。

Checked by & Date/核实人及日期：_____

Before using, confirm that cleaning of the glass lined agitator tank TJ4S-1213-AT09 has been completed/使用前，确认搪玻璃搅拌罐 TJ4S-1213-AT09 清洁已完成

Checked by & Date/核实人及日期：_____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.32.1~Ins 3.32.4 and Ins 3.32.9 need to be performed simultaneously.操作 3.32.1~操作 3.32.4 与操作 3.32.9 同时进行

	Times /次数
3.32.1 Add 246.35kg DMF (3.79W,4.0Vol) into glass lined agitator tank TJ4S-1213-AT09. Start the agitator of glass lined agitator tank to 60±20rpm (target: 60 rpm). **向搪玻璃搅拌罐 TJ4S-1213-AT09 中加入 246.35kg N,N-二甲基甲酰胺（3.79W,4.0Vol）（Acceptable weighing range/可接受称量范围：231.57kg~261.13kg）。开启搪玻璃搅拌罐搅拌转速至 60±20 rpm (目标转速:60 rpm)。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Checked by & Date/核实人及日期: _____

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:

Performed by & Date/执行人及日期:

Stir/搅拌	Begin/开始:
	The end time shall be recorded upon completion of the nitrogen purge in Instruction 3.32.8./结束时间记录于 3.32.8 氮气吹扫结束后

Performed by & Date/执行人及日期:

Weighing of DMF / N,N-二甲基甲酰胺称重:

Control Number/内部控制号: YL000336 **Scale ID. No./秤编号:** _____

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Addition type 加料方式	Lot number/批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/还需加入净重目标值	Gross qty of drum that to be charged/待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/投料前	After charging/投料后	Net weight/净重	Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DMF/N，N-二甲基甲酰胺	Begin/开始:	End/结束:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

3.32.2 Adjust to -5±5°C (target: -5°C). **调温至-5±5°C（目标温度：-5°C）。**	**Times /次数** **1**

Adjust temp/调温	**Begin/开始:**	**End/结束:**

Performed by & Date/执行人及日期:

3.32.3 Add Iodine (5.00eq) with powder feeding bags into glass lined agitator tank TJ4S-1213-AT09 after dispensing in portions, with an interval of 5–10 minutes(reference) between each portion (30-40kg per portion). **通过投料袋加入_____kg 碘单质（5.00eq）(Acceptable weighing range/可接受称量范围:** **_____kg~_____kg)，分批加入，每批 30~40kg，每次间隔 5~10min(参考)。**	**Times /次数** **1**

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

- Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签
- The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Charge iodine/加碘单质	**Begin/开始:**	**End/结束:**

Performed by & Date/执行人及日期:

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Iodine confirmation/碘单质确认：Control No./内部控制号：YL003092

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
No. /编号	Net weight/ 净重	All in/全部加入	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	420 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

After solid material charging, confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.32.4 Continue agitation of glass lined agitator tank TJ4S-1213-AT09 to 60±20 rpm (target: 60rpm). Maintain stirring for NLT 15min for later use. **维持搪玻璃搅拌罐 TJ4S-1213-AT09 搅拌转速至 60±20 rpm(目标转速：60rpm)，继续搅拌至少 15min，暂存待用。**	**Times/次数** **1**

Stir/搅拌	**Begin/开始:**	
	The end time shall be recorded after the nitrogen purge.in Instruction 3.32.8 /结束时间记录于 3.32.8 氮气吹扫结束后	
Performed by & Date/执行人及日期:		
Maintain &Stir 保持搅拌	**Begin/开始:**	**End/结束:**
	The end time shall be recorded after the nitrogen purge.in Instruction 3.32.8 /结束时间记录于 3.32.8 氮气吹扫结束后	
Performed by & Date/执行人及日期:		

Solution No. /溶液编号：CPo140021-_____-01-_____-3.32.4-01

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.32.5 Add 739.05 Kg DMF (11.37W, 12.0Vol) into synthesis reactor TJ4S-1213-Sy16 by spray balls. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 739.05 Kg N,N-二甲基甲酰胺（11.37W,12.0Vol）(Acceptable weighing range/可接受称量范围: 694.71kg~783.39kg)。**	**Times/次数** **1**

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.32.6 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). **开启合成釜 TJ4S-1213-Sy16 下口鼓氮气，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。**	**Times/ 次数** **1**

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.32.8/结束时间记录于 3.32.8
Transcribed by & Date/转录人及日期:	

Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.32.8 /结束时间记录于 3.32.8
Transcribed by & Date/转录人及日期:	

3.32.7 Adjust the internal temperature to【-5±5℃ (target: -5℃)】. **内温调温至【-5±5℃（目标温度：-5℃）】。**	**Times/ 次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

	Times /次数
3.32.8 Transfer the I_2 /DMF solution into synthesis reactor TJ4S-1213-Sy16 at internal temperature【-5±5℃ (target: -5℃)】through metering pump within 【30~80min (target: 50min). After addition, stop stirring and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump within 【0~25min (target: 0min)】. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【-5±5℃（目标温度：-5℃）】，通过计量泵向合成釜中转移碘/DMF 溶液，滴加时间【30~80min（目标时间：50min）】。滴加完毕，停止搅拌，关闭氮气，通过隔膜泵【0~25min (target: 0min)】内从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **High limit/高限:-2ºC Low limit/低限：-8ºC** **Pump number/泵编号：** _____ **Material charging pipe number/物料管路编号：** _____ **Confirm the calibration of the tantalum mass flow meter has been completed./确认钽材质的质量流量计是否已完成校准** **□Yes/是 □No/否** **Confirmed by & Date/确认人及日期：** _____ **Checked by & Date/核实人及日期：** _____	**1**

★Iodine solution transfer time is a CPP/碘溶液转移时间是关键控制参数

★Oxidation reaction time is a CPP/碘氧化反应时间是关键控制参数

★Oxidation exposure time(transfer plus oxidation reaction time) is a CPP/氧化暴露时间（碘溶液转移时间+氧化反应时间）时间是关键控制参数

★Oxidation temperature is a CPP/碘氧化温度是关键控制参数

★Note:The Post-Oxidation 1st and 2nd wash temperature are CPP./ 备注：碘氧化后前两次洗涤温度是关键控制参数。

Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China) (Referred as TJ4)/凯莱英医药集团（天津）股份有限公司（中国天津市滨海新区经济技术开发区南大街 265 号）（简称 TJ4 厂区）
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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Transfer the I_2 /DMF solution into synthesis reactor TJ4S-1213-Sy16 at -5±5℃ (target: -5℃) through metering pump within 30~80min (target: 50min). /控内温-5±5℃（目标温度：-5℃），向合成釜中通过计量泵转移碘/DMF 溶液，滴加时间 30~80min（目标时间：50min）

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Upon completion of material transfer, perform nitrogen purge on the dedicated I_2/DMF transfer line: maintain nitrogen flow rate ≥ 6 m³/h for a continuous duration of ≥ 4 min to displace residual I_2/DMF solution in the line/物料转移完成后，对 I_2/DMF 专用输送管线进行氮气吹扫：控制氮气流量≥6 m³/h，连续吹扫时长≥4 min。用以置换管线内残留的 I_2/DMF 溶液

Nitrogen Purge/氮气吹扫	**Begin/开始:**	**End/结束:**

After completion of the nitrogen purge, stop the agitation of agitator tank TJ4S-1213-AT09.氮气吹扫结束后，关闭搅拌罐 TJ4S-1213-AT09 的搅拌。

Transcribed by & Date/转录人及日期:

★After Iodine solution transfer is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 碘溶液转移结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the Iodine solution transfer time is within the respective Normal Operating Ranges (NOR): 【30~80min（Target time: 50min）】/工艺技术人员确认碘溶液转移时间处于正常操作范围（NOR）范围内【30~80min（目标时间：50min）】。

- **The Iodine solution transfer start time is defined as Iodine solution transfer start time /碘溶液转移开始时间定义为碘溶液转移开始时间。**
- **The Iodine solution transfer end time is defined as end time of nitrogen purge./ 氮气吹扫结束时间定义为碘溶液转移结束时间**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Iodine solution transfer time /碘溶液转移时间 = The end time of nitrogen purge /氮气吹扫结束时间- Start of transfer time/转移开始时间

Time/时间	The end time of nitrogen purge / 氮气吹扫结束时间	Start of transfer time/转移开始时间	Iodine solution transfer time /碘溶液转移时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the Iodine solution transfer time is within the respective Normal Operating Ranges (NOR): 【30~80min（Target time: 50min）】/工艺技术人员确认碘溶液转移时间处于正常操作范围（NOR）范围内【30~80min（目标时间: 50min）】。	**☐Yes/是 ☐No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

Internal temperature/内部温度	**Highest values/最高值:**	**Lowest values/最低值:**

Transcribed by & Date/转录人及日期:

★Note:The oxidation temperature is a CPP./ 备注: 碘氧化温度是关键控制参数。

the operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) 【-5±5℃ (target: -5℃)】/操作人员须确认反应内温处于正常操作范围（NOR）内【-5±5℃ (target: -5℃)】:

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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3.32 Iodine oxidation/碘氧化

After addition, stop stirring and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.滴加完毕，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.32.6 /开始时间记录于 3.32.6	
Transcribed by & Date/转录人及日期: _____		
Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins3.32.6 /开始时间记录于 3.32.6	
Transcribed by & Date/转录人及日期: _____		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After Iodine oxidation and draining are completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 碘氧化排废结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the oxidation reaction time is within the respective Normal Operating Ranges (NOR): 【0~25min (target: 0min)】/工艺技术人员确认碘氧化时间处于正常操作范围（NOR）范围内【0~25min（目标时间: 0min）】。

- **The oxidation reaction start time is defined as Iodine solution transfer end time/碘溶液结束转移时间定义为碘氧化开始时间。**
- **The oxidation reaction end time is defined as end of draining./ 排废结束时间定义为碘氧化结束时间**

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	426 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Oxidation reaction time /碘氧化时间 = End of draining./ 排废结束时间- Iodine solution transfer end time/碘溶液转移结束时间

Time/时间	End of draining./排废结束时间	Iodine solution transfer end time/碘溶液转移结束时间	Oxidation reaction time /碘氧化时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the oxidation reaction time is within the respective Normal Operating Ranges (NOR): 【0~25min (target: 0min)】/工艺技术人员确认碘氧化时间处于正常操作范围（NOR）范围内【0~25min（目标时间：0min）】。	☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

Note:备注：Ins 3.32.1~Ins3.32.4 and Ins 3.32.9 need to be performed simultaneously.操作 3.32.1~操作 3.32.4 与操作 3.32.9 同时进行

3.32.9 Pre-cool the Washing solution 1/洗涤液预冷 1

Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Adjust the internal temperature to【-5~0℃ (target: -3℃)】.Later for use

通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。内温调温至【-5~0°C（目标温度：-3℃）】。待用

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			
Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。			
Stir/搅拌	**Performed by & Date/执行人及日期:**		
Stir/搅拌	**Begin/开始:**		
	The end time is documented in Ins3.32.10 /结束时间记录于 3.32.10		
Transcribed by & Date/转录人及日期:			
Adjust temp/调温	**Begin/开始:**	**End/结束:**	
Transcribed by & Date/转录人及日期:			

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

	Times /次数
3.32.10 Post iodine oxidation Wash 1 /碘氧化结束后洗涤 1 **Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0°C (target: -3°C). After confirmation of pre-cooled DMF temperature, immediately transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0°C (target: -3°C). After stirring for 15±2min (target: 15min) at【-5~0°C (target: -3°C)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.** **将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0°C (目标温度: -3°C), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0°C (target: -3°C)，控内温【-5~0°C (target: -3°C)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**
Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0°C (target: -3°C). After confirmation of pre-cooled DMF temperature, immediately Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16./将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0°C (目标温度: -3°C), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。	
Prior to DMF transfer. Confirm the internal temperature in activation tank is within -5~0 °C (target -3 °C). / N,N-二甲基甲酰胺溶液转移前，确认激活罐内溶液温度在-5~0 °C（目标温度：-3 °C）。 **Temperature/温度:** _____ **□Yes/是 □No/否** **Confirmed by & Date/确认人及日期:** _____ **Checked by & Date/核实人及日期:** _____	

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.32.9/开始时间记录于 3.32.9	
Transcribed by & Date/转录人及日期:		
Note/备注: After Ins 3.32.10 transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16, end of activation tank stir , start Ins 3.32.11 Pre-cool the Washing solution 2./ Ins 3.32.10 将激活罐 TJ4S-1213-Sy16-R01 内的体系转移至合成釜 TJ4S-1213-Sy16，关闭激活罐搅拌后，开始 Ins 3.32.11 洗涤液预冷 2。		
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0°C (target: -3°C). After stirring for 15±2min (target: 15min) at【-5~0°C (target: -3°C)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0°C (target: -3°C)，控内温【-5~0°C (target: -3°C)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:** The end time follows the temperature hold with stirring./结束时间在保温搅拌结束之后	
Transcribed by & Date/转录人及日期:		
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
When the internal temperature reaches -3±1 °C , record the temperature adjustment end time, set the -1 °C / -4 °C temperature alarms, and then start the 15±2 min hold./当内温达到-3±1°C时，记录调温结束时间，设置-1°C/-4°C温度报警，随后启动 15±2 分钟保温		
Transcribed by & Date/转录人及日期:		

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3.32 Iodine oxidation/碘氧化

Over- Internal temperature alarm setting after temperature conditioning/调温完成后确认内温超温报警设置

Over- Internal temperature warning value/内温超温预警设置数值:

High limit/高限: -1°C Low/ limit/低限: -4°C

Performed by & Date/执行人及日期:

Checked by & Date/核实人及日期:

Maintain & Stir/保温搅拌	**Performed by & Date/执行人及日期:**	
Maintain & Stir/保温搅拌	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

3.32.11 Pre-cool the Washing solution 2/洗涤液预冷 2

Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Adjust the internal temperature to【-5~0°C (target: -1°C)】.Later for use

通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围: 579.23kg~653.17kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。内温调温至【-5~0°C（目标温度: -1°C)】。待用

Note/备注: After Ins 3.32.10 transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16, end of activation tank stir , start Ins 3.32.11 Pre-cool the Washing solution 2./ Ins 3.32.10 将激活罐 TJ4S-1213-Sy16-R01 内的体系转移至合成釜 TJ4S-1213-Sy16，关闭激活罐搅拌后，开始 Ins 3.32.11 洗涤液预冷 2。

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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3.32 Iodine oxidation/碘氧化

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir/搅拌	**Begin/开始:**
	The end time is documented in Ins3.32.12 /结束时间记录于 3.32.12

Transcribed by & Date/转录人及日期:

Adjust temp/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

CAUTION/注意: DMF/N，N-二甲基甲酰胺: Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

3.32.12 Post iodine oxidation Wash 2 /碘氧化结束后洗涤 2	**Times/ 次数**
Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0℃ (target: -3℃).After confirmation of pre-cooled DMF temperature, immediately transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0℃ (target: -3℃). After stirring for 15±2min (target: 15min) at【-5~0℃ (target: -3℃)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0℃ (目标温度: -3℃),确认预冷 N,N-二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0℃ (target: -3℃)，控内温【-5~0℃ (target: -3℃)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0℃ (target: -3℃). After confirmation of pre-cooled DMF temperature, immediately Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16./将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0℃ (目标温度: -3℃), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。

Prior to DMF transfer. Confirm the internal temperature in activation tank is within -5~0℃ (target: -1 ℃). / N,N-二甲基甲酰胺溶液转移前，确认激活罐内溶液温度在-5~0℃ (目标温度:-1 ℃)。

Temperature/温度: _____ **□Yes/是 □No/否**

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.32.11/开始时间记录于 3.32.11	

Transcribed by & Date/转录人及日期:

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0℃ (target: -3℃). After stirring for 15±2min (target: 15min) at【-5~0℃ (target: -3℃)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.
开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0℃ (target: -3℃)，控内温【-5~0℃ (target: -3℃)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:** The end time follows the temperature hold with stirring./结束时间在保温搅拌结束之后

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Transcribed by & Date/转录人及日期:

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**

When the internal temperature reaches -3±1 °C , record the temperature adjustment end time, set the -1 °C / -4 °C temperature alarms, and then start the 15±2 min hold./当内温达到-3±1℃时，记录调温结束时间，设置-1℃/-4℃温度报警，随后启动15±2 分钟保温

Transcribed by & Date/转录人及日期:

Over- Internal temperature alarm setting after temperature conditioning/调温完成后确认内温超温报警设置

Over- Internal temperature warning value/内温超温预警设置数值:

High limit/高限：-1ºC Low/ limit/低限：-4ºC

Performed by & Date/执行人及日期:

Checked by & Date/核实人及日期:

Maintain&Stir/保温搅拌	**Performed by & Date/执行人及日期:**	
Maintain&Stir/保温搅拌	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

★After Washing are completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 洗涤结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the Oxidation exposure time is within the respective Normal Operating Ranges (NOR): 【30~120min (target: 50min)】/工艺技术人员确认碘氧化暴露时间处于正常操作范围（NOR）范围内【30~120min（目标时间：50min）】。

- **The Oxidation exposure start time is defined as Iodine solution transfer start time /碘溶液转移开始时间定义为碘氧化暴露开始时间。**
- **The Oxidation exposure end time is defined as start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间定义为碘氧化暴露结束时间**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Oxidation exposure time/碘氧化暴露时间= Start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间- Iodine solution transfer start time /碘溶液转移开始时间

Time/时间	Start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间	Iodine solution transfer start time /碘溶液转移开始时间	Oxidation exposure time/碘氧化暴露时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the Oxidation exposure time is within the respective Normal Operating Ranges (NOR): 【30~120min (target: 50min)】/工艺技术人员确认碘氧化暴露时间处于正常操作范围（NOR）范围内【30~120min（目标时间：50min）】。	☐**Yes/是** ☐**No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

The start time of the post-oxidation wash is defined as the start time of the post-oxidation wash maintain & stir. The end time of the post-oxidation wash is defined as the start of draining./ 碘氧化后洗涤的开始时间定义为保温搅拌开始的时间；碘氧化后洗涤的结束时间定义为排液操作的起始时间。

Internal temperature/内部温度	Post-Oxidation 1st Wash 碘氧化后第一次洗涤	Post-Oxidation 2nd Wash 碘氧化后第二次洗涤
Highest values/最高值:		
Lowest values/最低值:		
Transcribed by & Date/转录人及日期:		

★Note:The Post-Oxidation 1st and 2nd wash internal temperature are CPP./ 备注：碘氧化后前两次洗涤内温是关键控制参数。

★The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) 【-5~0°C (target: -3°C)】/操作人员须确认反应内温处于正常操作范围（NOR）内【-5~0°C (target: -3°C)】:

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

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	QF05098	12.0	09/29/2026	435 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

	Times /次数
3.32.13 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C). Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **将合成釜 TJ4S-1213-Sy16 夹套调温至 20±5°C（目标温度：20°C）。通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg），开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min 后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C). Add 616.20kg/将合成釜 TJ4S-1213-Sy16 夹套调温至 20±5°C（目标温度：20°C）。					
Adjust/调温	**Performed by & Date/ 执行人及日期:**				
Adjust/调温	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				
Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16 通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围: 579.23kg~653.17kg）。					

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/ 执行人及日期:**				
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/转录人及日期:				

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min 后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始**				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.32 Iodine oxidation/碘氧化

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 30–50 mL of the mixture from the third portion. Take solid after filtration to perform IPC testing

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 30～50 mL，过滤后取固体进行 IPC 检测

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/ 取样人及日期	ASYM-140850（$L3(Trt)_2$）	Result filled by &Date/ 结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.32.13-01					

Project manager confirms/项目经理确认:

☐ **ASYM-140850 ($L3(Trt)_2$)≤0.5%, move to modules 3.34.1~3.35.1, Move to Modules 3.34~3.35/ASYM-140850（$L3(Trt)_2$）≤0.5%，则进行模块 3.34.1 ~3.35.1；调用模块：3.34~3.35/**

☐**ASYM-140850 ($L3(Trt)_2$)＞0.5%, repeat modules 3.33.1-3.33.13, Move to Modules 3.33/ASYM-140850（$L3(Trt)_2$）＞0.5%，则进行模块3.33.1-3.33.13；调用模块：3.33;**

Confirmed by & Date/确认人及日期： _____

Checked by & Date/核实人及日期： _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Before using, confirm glass lined agitator tank oxygen content≤1.0%.

使用前确认搪玻璃搅拌罐氧气含量≤1.0%。

500 L Glass lined agitator tank

500L 搪玻璃搅拌罐： TJ4S-1213-AT09

Oxygen detector ID. No. /氧气检测器编号：_____

Oxygen content/氧气含量= _____

设备检查表文件编号/Equipment Checklist Document Number: _____

Checked by & Date/核实人及日期：_____

Before using, confirm the glass lined agitator tank TJ4S-1213-AT09's calibration information is within valid date. /使用前确认搪玻璃搅拌罐 TJ4S-1213-AT09 的仪表在校准有效期内。

Checked by & Date/核实人及日期：_____

Before using, confirm that cleaning of the glass lined agitator tank TJ4S-1213-AT09 has been completed/使用前，确认搪玻璃搅拌罐 TJ4S-1213-AT09 清洁已完成

Checked by & Date/核实人及日期：_____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期：_____

Note:备注：Ins 3.33.1 ~Ins 3.33.4 and Ins 3.33.9 need to be performed simultaneously.操作 3.33.1~操作 3.33.4 与操作 3.33.9 同时进行

	Times/次数
3.33.1 Add 246.35kg DMF (3.79W,4.0Vol) into glass lined agitator tank TJ4S-1213-AT09. Start the agitator of glass lined to 60±20rpm (target: 60 rpm). **向搪玻璃搅拌罐 TJ4S-1213-AT09 中加入 246.35kg N,N-二甲基甲酰胺（3.79W,4.0Vol）(Acceptable weighing range/可接受称量范围：231.57kg~261.13kg)。开启搪玻璃搅拌罐搅拌转速至 60±20 rpm (目标转速: 60 rpm)。**	1

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。
Checked by & Date/核实人及日期：_____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Before drummed liquid material charging, visual check there is no other extraneous matter in material. /加桶装液体物料前，应目视确认桶内无异物。

Visual Examination of Foreign Matters for Drum Solvent/桶装溶剂异物目视检查:

1. Place the solvent drum in a well-lit area (or use a portable light source) to ensure adequate visibility of the liquid contents./ 将溶剂桶置于光线充足区域（或使用便携式光源），确保液体内容物可见性良好
2. Visually inspect the solvent through the drum bung opening or transparent sight glass for any visible foreign matters, including but not limited to particulates, fibers, sediments, or unusual turbidity./ 通过桶口或透明视镜目视检查溶剂中是否有可见异物，包括但不限于颗粒、纤维、沉淀物或异常浑浊。
3. Confirm that the solvent is clear and free of visible foreign matters before charging. If any foreign matter is observed, do not use the drum and follow the applicable handling procedure for subsequent disposition./ 投料前确认溶剂澄清且无可见异物。如发现任何异物，不得使用该桶，并依据处理流程进行后续处置。

Checked by & Date/核实人及日期: _____

Confirm liquid charging pipeline is released. /确认加料管路经过放行。

Checked by & Date/核实人及日期:

Charging of drummed solvent: verify that the pipeline connections remain consistent before and after material feeding, then record the weighing reading 桶装溶剂溶液称量时，确认投料前后管路连接一致，读取称量示数。

Performed by & Date/执行人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

DMF addition/ 加 N,N-二甲基甲酰胺	Begin/开始:	End/结束:
Performed by & Date/执行人及日期:		
Stir/搅拌	**Begin/开始:**	
	The end time shall be recorded upon completion of the nitrogen purge in Instruction 3.33.8./结束时间记录于 3.33.8 氮气吹扫结束后	
Performed by & Date/执行人及日期:		

Weighing of DMF / N,N-二甲基甲酰胺称重:

Control Number/内部控制号: YL000336 **Scale ID. No./秤编号: _____**

Pump number/泵编号: _____

Material charging pipe number/物料管路编号: _____

If the target net weight≥the planned net weight, perform full drum charge; otherwise, perform partial (Non-full drum) charge. /当目标加料量≥计划加入净重，勾选进行整桶加料；否则勾选进行零头加料。

Note: Target net weight to be charged = Total target net weight – Total net weight charged.

Expected tare weight of drum after charge = Gross weight of drum that to be charged - Target net weight to be charged.

还需加入净重目标值= 加料步骤目标总净重-已加入净重总量

加料完成后预计桶皮重=待加料桶毛重-还需加入净重目标值

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Addition type 加料方式	Lot number/ 批号	Pre-charging qty calculation/投料前投料量计算					Actual Charging Qty/实际投料量				
		Total qty charged/已加入净重总量	Target qty to be charged/ 还需加入净重目标值	Gross qty of drum that to be charged/ 待加料桶毛重	Expected net weight of drum after charge /加料完成后预计桶净重	Expected tare weight of drum after charge/加料完成后预计桶皮重	Before charging/ 投料前	After charging/ 投料后	Net weight/ 净重	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
☐Full drum 整桶 ☐ Non-full drum /非整桶											
☐Full drum 整桶 ☐ Non-full drum /非整桶											
Total/合计											

DMF/N，N-二甲基甲酰胺	Begin/开始:	End/结束:

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

3.33.2 Adjust to -5±5°C (target: -5°C). **调温至-5±5°C（目标温度：-5°C）。**	**Times /次数**
	1

Adjust temp/调温	**Begin/开始:**	**End/结束:**

Performed by & Date/执行人及日期:

3.33.3 Add Iodine (5.00eq) with powder feeding bags into glass lined agitator tank TJ4S-1213-AT09 after dispensing in portions, with an interval of 5–10 minutes(reference) between each portion (30-40kg per portion). **通过投料袋向搪玻璃搅拌罐 TJ4S-1213-AT09 中加入_____kg 碘单质（5.00eq）(Acceptable weighing range/可接受称量范围：_____kg~_____kg)，分批加入，每批 30~40kg，每次间隔 5~10min(参考)。**	**Times /次数**
	1

Before material charging, confirm the material is within the retest date/ the expiry date,Confirm whether the weight is adequate for use in the current charging. /加物料前，确认物料在复检日期/有效期之内，确认重量是否满足本次投料使用。

Before solid material charging, confirm/加固体物料前，确认:

➢Before operation performed, confirm there is no non-related material / equipment / instrument / tools / pipeline / document / record / label in the intended area /操作前，确认目标区域无不相关物料/设备/仪器/工具/管路/文件/记录/标签

➢The outer packages are removed. If the outer package cannot be removed, but un-package material without damaging the package bag, and before charging, check no package fragment and other extraneous matter in material. /外包装移除；如果外包装不能移除，但在不破坏包装的情况下打开包装袋，投料前以及过程中同时检查确认物料中无包装碎片以及异物

Checked by & Date/核实人及日期: _____

Charge iodine/加碘单质	**Begin/开始:**	**End/结束:**

Performed by & Date/执行人及日期:

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Iodine confirmation/碘单质确认：Control No./内部控制号：YL003092

Check items/检查项		Is it correct?/是否正确	Performed by & Date/ 执行人及日期	Checked by & Date/ 核实人及日期
Material name/物料名称		☐Yes/是; ☐No/否		
Project batch number/ 项目批次号		☐Yes/是; ☐No/否		
Operation number/ 操作号		☐Yes/是; ☐No/否		
Control No./内部控制号		☐Yes/是; ☐No/否		
No. /编号	**Net weight/ 净重**	**All in/全部加入**	**Performed by & Date/ 执行人及日期**	**Checked by & Date/ 核实人及日期**
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
____/____		☐Yes/是; ☐No/否		
____/_____		☐Yes/是; ☐No/否		
Total/合计		Confirm the net weight is within the input range/确认合计净重在投料范围内	Recorded by & Date/记录人及日期	Checked by & Date/ 核实人及日期
		☐Yes/是; ☐No/否		

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

After solid material charging confirm/固体物料投料之后，确认:
➢The corresponding leftover material has been properly sealed and transferred to the staging area/相应的剩余物料已经恰当密封，转移到了暂存区域
➢The waste packaging material has been sealed and transferred to the corresponding waste staging area. /产生的包材废物已经密封并转移到了相应的废物暂存区域
➢The operation area/equipment surface has been cleaned/操作区域以及设备表面已经清洁

Checked by & Date/核实人及日期: _____

3.33.4 Continue agitation of glass lined agitator tank TJ4S-1213-AT09to 60±20 rpm (target: 60rpm). Maintain stirring for NLT 15min for later use. **维持搪玻璃搅拌罐 TJ4S-1213-AT09 搅拌转速至 60±20 rpm(目标转速：60rpm)，继续搅拌至少 15min，暂存待用**	**Times /次数** **1**

Stir/搅拌	**Begin/开始:**	
	The end time shall be recorded after the nitrogen purge.in Instruction 3.33.8/结束时间记录于 3.33.8 氮气吹扫结束后	
Performed by & Date/执行人及日期:		
Maintain &Stir 保持搅拌	**Begin/开始:**	**End/结束:**
	The end time shall be recorded after the nitrogen purge.in Instruction 3.33.8 /结束时间记录于 3.33.8 氮气吹扫结束后	
Performed by & Date/执行人及日期:		

Solution No. /溶液编号：CPo140021-_____-01-_____-3.33.4-01

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期: _____

3.33.5 Add 739.05 Kg DMF (11.37W, 12.0Vol) into synthesis reactor TJ4S-1213-Sy16 by spray balls. **通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 739.05 Kg N,N-二甲基甲酰胺（11.37W,12.0Vol）(Acceptable weighing range/可接受称量范围： 694.71kg~783.39kg)。**	**Times /次数** **1**

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Performed by & Date/执行人及日期:**		
Rinse with DMF/ N,N-二甲基甲酰胺淋洗	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			

3.33.6 Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). **开启合成釜 TJ4S-1213-Sy16 下口鼓氮气，鼓氮气 5±2min（目标时间：5min）后。开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)。**	**Times /次数** **1**

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**
Bubble/鼓氮气	**Begin/开始:**
	The end time is documented in Ins3.33.8 /结束时间记录于 3.33.8
Transcribed by & Date/转录人及日期:	
Stir /搅拌	**Performed by & Date/执行人及日期:**
Stir /搅拌	**Begin/开始:**
	The end time is documented in Ins3.33.8 /结束时间记录于 3.33.8
Transcribed by & Date/转录人及日期:	

3.33.7 Adjust to -5±5℃ (target: -5℃). **调温至-5±5℃（目标温度：-5℃）。**	**Times /次数** **1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

□3.33 Iodine oxidation/碘氧化

	Times /次数
3.33.8 Transfer the I_2 /DMF solution into synthesis reactor TJ4S-1213-Sy16 at internal temperature【-5±5℃ (target: -5℃)】through metering pump within【30~80min (target: 50min)】. After addition, stop stirring and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump within【0~25min (target: 0min)】. Drain until there is no drip, the operator visually confirms the draining end point. **控内温【-5±5℃（目标温度：-5℃）】，通过计量泵向合成釜中转移碘/DMF 溶液，滴加时间【30~80min（目标时间：50min）】。滴加完毕，停止搅拌，关闭氮气，通过隔膜泵【0~25min (target: 0min)】内从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。** **Over- internal temperature warning value/内温超温预警设置数值:** **High limit/高限:-2ºC Low/ limit/低限：-8ºC** **Pump number/泵编号：_____** **Material charging pipe number/物料管路编号：_____** **Confirm the calibration of the tantalum mass flow meter has been completed./ 确认钽材质的质量流量计是否已完成校准** **□Yes/是 □No/否** **Confirmed by & Date/确认人及日期：_____** **Checked by & Date/核实人及日期：_____**	**1**

★Iodine solution transfer time is a CPP/碘溶液转移时间是关键控制参数

★Oxidation reaction time is a CPP/碘氧化反应时间是关键控制参数

★Oxidation exposure time(transfer plus oxidation reaction time) is a CPP/氧化暴露时间（碘溶液转移时间+氧化反应时间）时间是关键控制参数

★Oxidation temperature is a CPP/碘氧化温度是关键控制参数

★Note:The Post-Oxidation 1st and 2nd wash temperature are CPP./ 备注：碘氧化后前两次洗涤温度是关键控制参数。

Transfer the I_2 /DMF solution into synthesis reactor TJ4S-1213-Sy16 at -5±5℃ (target: -5℃) through metering pump within 30~80min (target: 50min). /控内温-5±5℃（目标温度：-5℃），向合成釜中通过计量泵转移碘/DMF 溶液，滴加时间 30~80min（目标时间：50min）

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Upon completion of material transfer, perform nitrogen purge on the dedicated I_2/DMF transfer line: maintain nitrogen flow rate ≥ 6 m³/h for a continuous duration of ≥ 4 min to displace residual I_2/DMF solution in the line/物料转移完成后，对 I_2/DMF 专用输送管线进行氮气吹扫：控制氮气流量≥6 m³/h，连续吹扫时长≥4 min。用以置换管线内残留的 I_2/DMF 溶液

Nitrogen Purge/氮气吹扫	Begin/开始:	End/结束:

After completion of the nitrogen purge, stop the agitation of agitator tank TJ4S-1213-AT09.氮气吹扫结束后，关闭搅拌罐 TJ4S-1213-AT09 的搅拌。

Transcribed by & Date/转录人及日期:

★After Iodine solution transfer is completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 碘溶液转移结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the Iodine solution transfer time is within the respective Normal Operating Ranges (NOR): 【30~80min（Target time: 50min）】/工艺技术人员确认碘溶液转移时间处于正常操作范围（NOR）范围内【30~80min（目标时间：50min）】。

- **The Iodine solution transfer start time is defined as Iodine solution transfer start time /碘溶液转移开始时间定义为碘溶液转移开始时间。**
- **The Iodine solution transfer end time is defined as end time of nitrogen purge./ 氮气吹扫结束时间定义为碘溶液转移结束时间**

Iodine solution transfer time /碘溶液转移时间 = The end time of nitrogen purge /氮气吹扫结束时间- Start of transfer time/转移开始时间

Time/时间	The end time of nitrogen purge / 氮气吹扫结束时间	Start of transfer time/转移开始时间	Iodine solution transfer time /碘溶液转移时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the Iodine solution transfer time is within the respective Normal Operating Ranges (NOR): 【30~80min（Target time: 50min）】/工艺技术人员确认碘溶液转移时间处于正常操作范围（NOR）范围内【30~80min（目标时间：50min）】。	☐Yes/是 ☐No/否

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

□3.33 Iodine oxidation/碘氧化

Internal temperature/内部温度	Highest values/最高值:	Lowest values/最低值:

Transcribed by & Date/转录人及日期:

★Note: The oxidation temperature is a CPP./ 备注：碘氧化温度是关键控制参数。

the operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) 【-5±5℃ (target: -5℃)】/操作人员须确认反应内温处于正常操作范围（NOR）内【-5±5℃ (target: -5℃)】:

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

After addition, stop stirring and turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.滴加完毕，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**End/结束:**	
	The start time is documented in Ins3.33.6 /开始时间记录于 3.33.6	

Transcribed by & Date/转录人及日期:

Stir /搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**End/结束:**	
	The start time is documented in Ins3.33.6 /开始时间记录于 3.33.6	

Transcribed by & Date/转录人及日期:

Filter/过滤	**Performed by & Date/执行人及日期:**	
Filter/过滤	**Checked by & Date/核实人及日期:**	
Filter/过滤	**Begin/开始:**	**End/结束:**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

★After Iodine oxidation and draining are completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 碘氧化排废结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the oxidation reaction time is within the respective Normal Operating Ranges (NOR): 【0~25min (target: 0min)】/工艺技术人员确认碘氧化时间处于正常操作范围（NOR）范围内【0~25min（目标时间：0min）】。

- **The oxidation reaction start time is defined as Iodine solution transfer end time/碘溶液结束转移时间定义为碘氧化开始时间。**
- **The oxidation reaction end time is defined as end of draining./ 排废结束时间定义为碘氧化结束时间**

Oxidation reaction time /碘氧化时间 = End of draining./ 排废结束时间- Iodine solution transfer end time/碘溶液转移结束时间

Time/时间	End of draining./ 排废结束时间	Iodine solution transfer end time/碘溶液转移结束时间	Oxidation reaction time /碘氧化时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the oxidation reaction time is within the respective Normal Operating Ranges (NOR): 【0~25min (target: 0min)】/工艺技术人员确认碘氧化时间处于正常操作范围（NOR）范围内【0~25min（目标时间：0min）】。	**☐Yes/是** **☐No/否**

Confirmed by & Date/确认人及日期:

Checked by & Date/核实人及日期:

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。

Confirm that the above content has been acknowledged./确认已知悉上述内容。

Signature &Date/签名及日期:

Note:备注：Ins 3.33.1 ~Ins3.33.4 and Ins 3.33.9 need to be performed simultaneously.操作 3.33.1~操作 3.33.4 与操作 3.33.9 同时进行

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

3.33.9 Pre-cool the Washing solution 1/洗涤液预冷 1

Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Adjust the internal temperature to-5~0℃ (target: -3℃).Later for use
通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围：579.23kg~653.17kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。内温调温至-5~0℃（目标温度：-3℃)。待用

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**
Transcribed by & Date/转录人及日期:			
Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。			
Stir/搅拌	**Performed by & Date/执行人及日期:**		
Stir/搅拌	**Begin/开始:**		
	The end time is documented in Ins3.33.10 /结束时间记录于 3.33.10		
Transcribed by & Date/转录人及日期:			
Adjust temp/调温	**Begin/开始:**	**End/结束:**	
Transcribed by & Date/转录人及日期:			

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

□3.33 Iodine oxidation/碘氧化

	Times /次数
3.33.10 Post iodine oxidation Wash 1 /碘氧化结束后洗涤 1 **Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0°C (target: -3°C). After confirmation of pre-cooled DMF temperature, immediately .Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0°C (target: -3°C). After stirring for 15±2min (target: 15min) at【-5~0°C (target: -3°C)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.** **将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0°C (目标温度: -3°C), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0°C (target: -3°C)，控内温【-5~0°C (target: -3°C)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0°C (target: -3°C). After confirmation of pre-cooled DMF temperature, immediately Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16./
将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0°C (目标温度: -3°C), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。

Prior to DMF transfer. Confirm the internal temperature in activation tank is within -5~0°C (target: -3°C). / N,N-二甲基甲酰胺溶液转移前，确认激活罐内溶液温度在-5~0°C(目标温度：-3°C)。

Temperature/温度: _____

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.33.9/开始时间记录于 3.33.9	
Transcribed by & Date/转录人及日期:		
Note/备注: After Ins 3.33.10 transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16, end of activation tank stir , start Ins 3.33.11 Pre-cool the Washing solution 2./ Ins 3.33.10 将激活罐 TJ4S-1213-Sy16-R01 内的体系转移至合成釜 TJ4S-1213-Sy16，关闭激活罐搅拌后，开始 Ins 3.33.11 洗涤液预冷 2。		
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0°C (target: -3°C). After stirring for 15±2min (target: 15min) at【-5~0°C (target: -3°C)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0°C (target: -3°C)，控内温【-5~0°C (target: -3°C)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	
Stir /搅拌	**Begin/开始:** The end time follows the temperature hold with stirring./结束时间在保温搅拌结束之后	
Transcribed by & Date/转录人及日期:		
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
When the internal temperature reaches -3±1 °C, record the temperature adjustment end time, set the -1 °C / -4 °C temperature alarms, and then start the 15±2 min hold../ 当内温达到-3±1°C时，记录调温结束时间，设置-1°C/-4°C温度报警，随后启动 15±2 分钟保温搅拌		
Transcribed by & Date/转录人及日期:		

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Over- Internal temperature alarm setting after temperature conditioning/调温完成后确认内温超温报警设置

Over- Internal temperature warning value/内温超温预警设置数值:

High limit/高限: -1°C Low/ limit/低限: -4°C

Performed by & Date/执行人及日期:

Checked by & Date/核实人及日期:

Maintain&Stir/保温搅拌	**Performed by & Date/执行人及日期:**	
Maintain&Stir/保温搅拌	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

3.33.11 Pre-cool the Washing solution 2/洗涤液预冷 2

Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator of activation tank to 115±5 rpm (target: 115 rpm). Adjust the internal temperature to-5~0°C (target: -1°C).Later for use
通过淋洗球向激活罐 TJ4S-1213-Sy16-R01 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围: 579.23kg~653.17kg)，开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。内温调温至-5~0°C（目标温度: -1°C）。待用

Note/备注: After Ins 3.33.10 transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16, end of activation tank stir , start Ins 3.33.11 Pre-cool the Washing solution 2./ Ins 3.33.10 将激活罐 TJ4S-1213-Sy16-R01 内的体系转移至合成釜 TJ4S-1213-Sy16，关闭激活罐搅拌后，开始 Ins 3.33.11 洗涤液预冷 2。

DMF addition/ 加 N，N 二甲基甲酰胺	**Performed by & Date/执行人及日期:**		
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始:**	**End/结束:**	**Qty/重量:**

Transcribed by & Date/转录人及日期:

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Start the agitator of activation tank to 115±5 rpm (target: 115 rpm)./ 开启激活罐搅拌转速至 115±5 rpm (目标转速: 115 rpm)。

Stir/搅拌	**Performed by & Date/执行人及日期:**
Stir/搅拌	**Begin/开始:**
	The end time is documented in Ins3.33.12 /结束时间记录于 3.33.12

Transcribed by & Date/转录人及日期:

Adjust temp/调温	**Begin/开始:**	**End/结束:**

Transcribed by & Date/转录人及日期:

CAUTION/注意: DMF/N，N-二甲基甲酰胺: Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

3.33.12 Post iodine oxidation Wash 2 /碘氧化结束后洗涤 2	**Times /次数**
Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0℃ (target: -3℃). After confirmation of pre-cooled DMF temperature, immediately .Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16.Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0℃ (target: -3℃). After stirring for 15±2min (target: 15min) at【-5~0℃ (target: -3℃)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0℃ (目标温度: -3℃),确认预冷 N,N-二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0℃ (target: -3℃)，控内温【-5~0℃ (target: -3℃)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**1**

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
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☐3.33 Iodine oxidation/碘氧化

Set the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to -5~0℃ (target: -3℃). After confirmation of pre-cooled DMF temperature, immediately Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16./将合成釜中 TJ4S-1213-Sy16 夹套温度设定为-5~0℃ (目标温度: -3℃), 确认预冷 N,N - 二甲基甲酰胺温度合格后。立即将激活罐 TJ4S-1213-Sy16-R01 中体系转移至合成釜中 TJ4S-1213-Sy16。

Prior to DMF transfer. Confirm the internal temperature in activation tank is within -5~0℃ (target: -1℃). / N,N-二甲基甲酰胺溶液转移前，确认激活罐内溶液温度在-5~0℃（目标温度：-1℃）。

Temperature/温度: _____ **□Yes/是 □No/否**

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Transfer/转移	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	
Activation tank Stir /激活罐搅拌	**Performed by & Date/执行人及日期:**	
Activation tank Stir /激活罐搅拌	**End/结束:**	
	The start time is documented in Ins 3.33.11/开始时间记录于 3.33.11	

Transcribed by & Date/转录人及日期:

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). Adjust internal temperature to -5~0℃ (target: -3℃). After stirring for 15±2min (target: 15min) at【-5~0℃ (target: -3℃)】, stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.
开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间: 5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，内温调温至-5~0℃ (target: -3℃)，控内温【-5~0℃ (target: -3℃)】搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Bubble/鼓氮气	**Performed by & Date/执行人及日期:**	
Bubble/鼓氮气	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stir/搅拌	**Performed by & Date/执行人及日期:**	

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Operations Instruction
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☐3.33 Iodine oxidation/碘氧化

Stir /搅拌	**Begin/开始:** The end time follows the temperature hold with stirring./结束时间在保温搅拌结束之后	
Transcribed by & Date/转录人及日期:		
Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
When the internal temperature reaches-3±1 ℃ , record the temperature adjustment end time, set the -1 ℃ / -4 ℃ temperature alarms, and then start the 15±2 min hold../当内温达到-3±1℃时，记录调温结束时间，设置-1℃/-4℃温度报警，随后启动15±2 分钟保温		
Transcribed by & Date/转录人及日期:		
Over- Internal temperature alarm setting after temperature conditioning/调温完成后确认内温超温报警设置 **Over- Internal temperature warning value/内温超温预警设置数值:** **High limit/高限: -1ºC Low/ limit/低限: -4ºC**		
Performed by & Date/执行人及日期:		
Checked by & Date/核实人及日期:		
Maintain&Stir/保温搅拌	**Performed by & Date/执行人及日期:**	
Maintain&Stir/保温搅拌	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point.通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。		
Filter/过滤	**Begin/开始:**	**End/结束:**
	Performed by & Date/执行人及日期:	
	Checked by & Date/核实人及日期:	

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Operations Instruction
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☐3.33 Iodine oxidation/碘氧化

★After Washing are completed, the operator shall confirm that all Critical Process Parameters (CPP) are within their respective Normal Operating Ranges (NOR)/ 洗涤结束后，操作人员须确认所有关键工艺参数（CPP）均处于各自的正常操作范围（NOR）内

★Process chemist confirms the Oxidation exposure time is within the respective Normal Operating Ranges (NOR): 【30~120min (target: 50min)】/工艺技术人员确认碘氧化暴露时间处于正常操作范围（NOR）范围内【30~120min（目标时间：50min）】。

- **The Oxidation exposure start time is defined as Iodine solution transfer start time /碘溶液转移开始时间定义为碘氧化暴露开始时间。**
- **The Oxidation exposure end time is defined as Start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间定义为碘氧化暴露结束时间**

Oxidation exposure time/碘氧化暴露时间= Start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间- Iodine solution transfer start time /碘溶液转移开始时间

Time/时间	Start of transferring 1st post-oxidation DMF wash./ 氧化后第一次 DMF 洗涤液转移开始时间	Iodine solution transfer start time /碘溶液转移开始时间	Oxidation exposure time/碘氧化暴露时间	Calculated by & Date/计算人及日期	Checked by & Date/核实人及日期

★Process chemist confirms the Oxidation exposure time is within the respective Normal Operating Ranges (NOR): 【30~120min (target: 50min)】/工艺技术人员确认碘氧化暴露时间处于正常操作范围（NOR）范围内【30~120min（目标时间：50min）】。	☐**Yes/是** ☐**No/否**
Confirmed by & Date/确认人及日期:	
Checked by & Date/核实人及日期:	

The start time of the post-oxidation wash is defined as the start time of the post-oxidation wash maintain & stir. The end time of the post-oxidation wash is defined as the start of draining./ 碘氧化后洗涤的开始时间定义为保温搅拌开始的时间；碘氧化后洗涤的结束时间定义为排液操作的起始时间。

Internal temperature/内部温度	Post-Oxidation 1st Wash 碘氧化后第一次洗涤	Post-Oxidation 2nd Wash 碘氧化后第二次洗涤
Highest values/最高值:		
Lowest values/最低值:		
Transcribed by & Date/转录人及日期:		

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

□3.33 Iodine oxidation/碘氧化

★Note:The Post-Oxidation 1st and 2nd wash internal temperature are CPP./ 备注：碘氧化后前两次洗涤内温是关键控制参数。

★The operator shall confirm that the reaction internal temperature is within the Normal Operating Range (NOR) 【-5~0℃ (target: -3℃)】/操作人员须确认反应内温处于正常操作范围（NOR）内【-5~0℃ (target: -3℃)】:

□Yes/是 □No/否

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

CAUTION/注意：DMF/N，N-二甲基甲酰胺：Irritation of eyes, skin and nose. May cause nausea. /刺激眼睛、皮肤和鼻子。可能引起恶心。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

	Times /次数
3.33.13 Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C). Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **将合成釜 TJ4S-1213-Sy16 夹套调温至 20±5℃（目标温度：20℃）。通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）（Acceptable weighing range/可接受称量范围：579.23kg~653.17kg），开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min 后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**4**

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C)/ 将合成釜 TJ4S-1213-Sy16 夹套调温至 20±5℃（目标温度：20℃）					

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Adjust/调温	**Performed by & Date/ 执行人及日期:**				
Adjust/调温	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				
Add 616.20kg DMF (9.48W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16/通过淋洗球向合成釜 TJ4S-1213-Sy16 中加入 616.20kg N,N-二甲基甲酰胺（9.48W,10.0Vol）(Acceptable weighing range/可接受称量范围: 579.23kg~653.17kg）					
DMF addition/ 加 N,N-二甲基甲酰胺	**Performed by & Date/ 执行人及日期:**				
DMF addition/ 加 N,N-二甲基甲酰胺	**Begin/开始**				
	End/结束				
	Qty/重量				
	Transcribed by & Date/转录人及日期:				
Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min), Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后，开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15min±2min（目标时间:15min）后，停止搅拌，关闭氮气					
Bubble/鼓氮气	**Performed by & Date/ 执行人及日期:**				
Bubble/鼓氮气	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				
Stir /搅拌	**Performed by & Date/ 执行人及日期:**				
Stir /搅拌	**Begin/开始**				
	End/结束				
	Transcribed by & Date/转录人及日期:				
Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./ 通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。					

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	460 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

☐3.33 Iodine oxidation/碘氧化

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4
Filter/过滤	Begin/开始				
	End/结束				
	Performed by & Date/ 执行人及日期:				
	Checked by & Date/ 核实人及日期:				

Standard sampling method/标准取样方法

Wear nitrile gloves and a 3M 6000 series full-face respirator fitted with 6006CN cartridges during sampling. Collect samples via an inline sampler. Discard the first two sample portions, then withdraw 30–50 mL of the mixture from the third portion. Take solid after filtration to perform IPC testing

取样过程中佩戴丁腈手套及配备 6006CN 滤毒盒的 3M 6000 系列全面罩呼吸器。采用在线取样器取样，弃去前两份样品，取第三份混合液 30 ~ 50 mL，过滤后取固体进行 IPC 检测

Sampling number/ 取样编号	Sampling time/取样时间	Sampled by & Date/取样人及日期	ASYM-140850 （$L3(Trt)_2$）	Result filled by &Date/结果填写人及日期	Checked by & Date/核实人及日期
CPo140021-_____-01-_____-3.33.13-01					

Project manager confirms/项目经理确认:

☐**ASYM-140850 ($L3(Trt)_2$)≤0.5%, move to Ins. 3.34.1~3.35.1, and move to Module 3.34~3.35 /ASYM-140850（$L3(Trt)_2$）≤0.5%，则进行操作3.34.1 ~操作3.35.1；并调用模块：3.34~3.35**

☐ **If ASYM-140850 ($L3(Trt)_2$) > 0.5%, immediately notify the Project Manager (PM) and Quality Assurance (QA) department and initiate an investigation./若ASYM-140850 ($L3(Trt)_2$) > 0.5%，通知项目经理（PM）及质量保证部门（QA），并启动调查**

Investigation Document No./调查文件编号: _____

As an emergency action during the investigation, repeat the Ins. 3.33.1 to 3.33.13, and move to Module 3.33.作为调查的紧急措施，可重复氧化操作3.33.1-3.33.13,并调用模块：3.33

（From/上接：_____；To/下转：_____）;

Confirmed by & Date/确认人及日期: _____

Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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	QF05098	12.0	09/29/2026	461 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.33.De-swell/收缩

CAUTION/注意：IPAC/乙酸异丙酯： Vapors irritate eyes and respiratory tract; high concentrations can be anesthetic. Liquid irritates eyes but causes no serious injury; may cause dermatitis; no serious effects if swallowed. /蒸汽会刺激眼睛和呼吸道；高浓度的蒸汽可能是麻醉剂。液体刺激眼睛，但不会造成严重伤害；可能会引起皮炎；吞食不会造成严重影响。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期：_____

	Times /次数
3.34.1 Add 567.45kg IPAc (8.73W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加入 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min(目标时间: 5min)后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min（目标时间：15min）后，停止搅拌关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	**3**

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Add 567.45kg IPAc (8.73W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加入 567.45kg 乙酸异丙酯（8.73W,10.0Vol）(Acceptable weighing range/可接受称量范围：533.40kg~601.50kg)				
IPAC addition/ 加乙酸异丙酯	**Performed by & Date/ 执行人及日期:**			
IPAC addition/ 加乙酸异丙酯	**Begin/开始**			
	End/结束			
	Qty/重量			
	Transcribed by & Date/ 转录人及日期:			

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.34 De-swell/收缩

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h) .Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./ 开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min（目标时间：15min）后，停止搅拌关闭氮气

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3
Bubble/鼓氮气	Performed by & Date/执行人及日期:			
Bubble/鼓氮气	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			
Stir /搅拌	Performed by & Date/执行人及日期:			
Stir /搅拌	Begin/开始			
	End/结束			
	Transcribed by & Date/转录人及日期:			

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始			
	End/结束			
	Performed by & Date/执行人及日期:			
	Checked by & Date/核实人及日期:			

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.34 De-swell/收缩

CAUTION/注意：MTBE/甲基叔丁基醚：Inhalation: May cause dizziness or suffocation. Contact may irritate or burn eyes or skin. May be harmful if swallowed. /吸入:可引起头晕或窒息。接触可能刺激或灼伤眼睛或皮肤。吞食可能有害。
Confirm that the above content has been acknowledged./确认已知悉上述内容。
Signature &Date/签名及日期: _____

	Times /次数
3.34.2 Add 481.65 kg MTBE (7.41W, 10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point. **向合成釜 TJ4S-1213-Sy16 中加入 481.65 kg 甲基叔丁基醚（7.41W,10.0Vol）(Acceptable weighing range/可接受称量范围: 452.75kg~510.55kg)，开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min(目标时间: 5min)后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min（目标时间：15min）后，停止搅拌，关闭氮气，通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。**	5

Operation name/ 操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4	Wash /洗涤 5
Add 481.65 kg MTBE(7.41W,10.0Vol) by spray balls into the synthesis reactor TJ4S-1213-Sy16./ 向合成釜 TJ4S-1213-Sy16 中加入 481.65 kg 甲基叔丁基醚（7.41W,10.0Vol）(Acceptable weighing range/可接受称量范围: 452.75kg~510.55kg)						
MTBE addition/ 加甲基叔丁基醚	**Performed by & Date/ 执行人及日期:**					
MTBE addition/ 加甲基叔丁基醚	**Begin/开始**					
	End/结束					
	Qty/重量					
	Transcribed by & Date/ 转录人及日期:					

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.34 De-swell/收缩

Turn on the nitrogen of synthesis reactor to 8.5±4m³/h (target: 8.5 m³/h). Bubble with nitrogen for 5±2min (target: 5min). Start the agitator of synthesis reactor to 20±10rpm (target: 20 rpm). After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen./开启合成釜下口鼓氮气，氮气流速开至 8.5±4m³/h (目标流速: 8.5 m³/h)，鼓氮气 5±2min（目标时间：5min）后开启固相合成釜搅拌转速至 20±10 rpm (目标转速: 20 rpm)，搅拌 15±2min（目标时间：15min）后，停止搅拌，关闭氮气

Operation name/操作名称	Time point/时间节点	Wash /洗涤 1	Wash /洗涤 2	Wash /洗涤 3	Wash /洗涤 4	Wash /洗涤 5
Bubble/鼓氮气	Performed by & Date/执行人及日期:					
Bubble/鼓氮气	Begin/开始					
	End/结束					
	Transcribed by & Date/转录人及日期:					
Stir /搅拌	Performed by & Date/执行人及日期:					
Stir /搅拌	Begin/开始					
	End/结束					
	Transcribed by & Date/转录人及日期:					

Start draining the solution from synthesis reactor with diaphragm pump. Drain until there is no drip, the operator visually confirms the draining end point./通过隔膜泵从合成釜中排出溶液。排液至无液体滴出，由操作人员目视确认排液终点。

Filter/过滤	Begin/开始					
	End/结束					
	Performed by & Date/执行人及日期:					
	Checked by & Date/核实人及日期:					

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.34.Dry/烘料

	Times /次数
3.35.1 Adjust T_{jacket} to 25±5°C(target: 25°C). Sweep with nitrogen above, nitrogen flow is 10m³/h±4m³/h (target: 10 m³/h), and vacuum underneath at T_{jacket}=25±5°C(target: 25°C). After 12h, sample the solid for LOD analysis every 12±4h until LOD ≤8%. **夹套温度调温至 25±5°C（目标温度：25°C），夹套控温 25±5°C（目标温度 25°C）烘料，上口鼓氮气，氮气流量 10m³/h±4m³/h（目标流量：10 m³/h）下口抽真空。12h 后开始取样检测 LOD，之后每隔 12±4h 取样一次直至 LOD≤8%。** **Note 1: After drying 6h, begin to stir the solid for 2~5 min at a speed of 10±5 rpm (target: 10rpm) every 2±1h. /备注：干燥 6h 后，开始搅拌，每 2±1h，搅拌 2~5min，参考搅拌转速 10±5rpm（目标转速:10rpm）。** **Note 2: After sample the solid for LOD analysis. And the sample send to QC lab for testing. /LOD 检测样品取完后，样品送至 QC 实验室进行检测。** **Over jacket temperature Alarm value/夹套超温报警设置数值：Low limit/低限：22 °C，High limit/高限：28°C**	**1**

Adjust/调温	**Performed by & Date/执行人及日期:**	
Adjust/调温	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Dry/烘料	**Performed by & Date/执行人及日期:**	
Dry/烘料	**Begin/开始:**	**End/结束:**
Transcribed by & Date/转录人及日期:		
Stirring record/搅拌记录:		

No./次数	Begin/开始	End/结束	Agitator speed/转速	Performed by & Date/执行人及日期
1				
2				
3				
4				
5				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.35 Dry/烘料

No./次数	Begin/开始	End/结束	Agitator speed/转速	Performed by & Date/执行人及日期
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

3.35 Dry/烘料

No./次数	Begin/开始	End/结束	Agitator speed/转速	Performed by & Date/执行人及日期
28				
29				
30				
31				
32				
33				
34				
35				

Standard sampling method/标准取样方法

Take 0.5~5g resins from sampling point in the glove box and submit for analysis testing. During sampling, operators should wear nitrile gloves and 3M 6000 series Full-face piece respirator, 6006CN cartridges.
从手套箱内取样口取树脂 0.5~5g，直接送样检测。取样过程中，应该戴丁腈手套、3M 6000 系列全面罩, 6006CN 滤毒盒。

Batch size (output range): 157.69~292.80 kg

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MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

Operations Instruction
操作指令

3.35 Dry/烘料

Sampling number/ 取样编号	Sampling time/ 取样时间	Sampled by & Date/取样人及日期	LOD/干燥失重	Result meet acceptance criteria or not/结果是否满足接受限度 LOD≤8%	Result filled by &Date/结果填写人及日期	Checked by & Date/ 核实人及日期
CPo140021-____-01-__-3.35.1-01				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-02				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-03				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-04				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-05				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-06				☐Pass/通过; ☐Fail/不通过		
CPo140021-____-01-__-3.35.1-07				☐Pass/通过; ☐Fail/不通过		

Process chemist confirms the end point of the drying and move to following operation.

工艺技术人员确认烘料终点后进行下一步操作;

Process chemist & Date/ 工艺技术人员签字及日期: _____

Checked by & Date/核实人签字及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

Wear nitrile gloves and 3M 6000 series Full-face piece respirator, 6006CN cartridges, Particulate filter, White disposable chemical protective clothing. 戴丁腈手套、3M 6000 系列全面罩, 6006CN 滤毒盒, 颗粒物滤棉, 白色防化服。
ASYM-140851: OEB3; OEL (10~100ug/m³). Exposure control level ≤10ug/m3./暴露控制水平≤10ug/m³。

After drying, the intermediate is collected via the material receiving port. All material collection tools are placed inside the glove box, and material collection is performed by operating the tools through gloves. Operators shall not have direct contact with materials throughout the entire process. The intermediate is directly charged into double-layer low-density polyethylene bags. Upon completion of weighing, labels shall be affixed to the outer polyethylene bag. The packaged materials are then placed into high-density polyethylene drums and temporarily stored in the warehouse./烘干后的中间体通过收料口进行收料，收料工具均在手套箱内，通过手套使用工具进行收料，全程操作人员不直接接触物料。中间体直接收入双层低密度聚乙烯袋。称重完毕后贴标签至外层聚乙烯袋上后，放置于高密度聚乙烯桶中，暂存至库房。

Put Boc-L3-SS peptidyl resin (ASYM-140851) into an inner pharmaceutical-grade LDPE powder charging bag, then enclose the inner bag in an outer pharmaceutical-grade LDPE bag. Purge the interior of the package with nitrogen, and place desiccant between the two layers (when net weight ≤ 1 kg, desiccant amount shall be NLT 10% of the net weight; when net weight > 1 kg, desiccant amount NLT 1% of net weight + 100 g).The inner bag is sealed with a sealing clip, and the outer bag is sealed separately with a cable tie.After labeling and weighing, place the material into a lidded HDPE packaging drum for temporary storage, then transfer to cold storage at 2–8 °C under sealed and light-protected conditions.

将烘干后的 Boc-L3-SS 肽树脂（ASYM-140851）装入内层药用级低密度聚乙烯（LDPE）粉体投料袋，外套一层药用级 LDPE 袋；采用氮气吹扫包装内部，在两层袋之间放置干燥剂（当净重≤1kg 时，干燥剂添加量不得低于净重的 10%；当净重＞1kg 时，干燥剂添加量不少于净重 ×1%＋100g）。内层袋采用密封卡箍密封，外层袋采用扎带单独密封。 完成贴签与称重后，将物料装入带盖高密度聚乙烯（HDPE）包装桶中暂存，随后转移至 2–8℃冷库，在密封、避光条件下贮存。

Intermediate lot number/中间体编号: CPo140021-_____-01-_____

Lot number of inner grade LDPE powder charging bag/内层低密度聚乙烯粉末投料袋批号: _____

Lot number of outer grade LDPE bag/外层低密度聚乙烯袋批号: _____

Lot number of desiccant/干燥剂批号: _____

Note: The solid in each bag≤25Kg. If the project has special requirements, need to do risk assessment and get approval.
备注: 每袋物料≤25Kg，若项目有特殊要求，需经过风险评估和审批。_

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

包装层级 Packaging Level	规格参数 Specification	包装类型 Package Type	密封方式 Seal Type	Confirmed by & Date/确认人及日期	Checked by & Date/核实人及日期
一级包装 Primary package	容积：30 L 或 50 L Volume: 30 L or 50 L	药用级粉体投料袋 Pharmaceutical-grade powder feeding bag	每袋采用密封卡箍密封 Each bag sealed with a sealing clip		
二级包装 Secondary package	尺寸：长 1100 mm × 宽 700 mm × 厚 0.09 mm Dimensions: 1100 L × 700 W × 0.09 T (mm)	药用级低密度聚乙烯（LDPE）袋 Pharmaceutical-grade LDPE bag	每袋采用扎带密封 Each bag sealed with a cable tie		
外包装 Outer packaging	容积：50 L Volume: 50 L	带盖存储用高密度聚乙烯（HDPE）桶 HDPE drum with lid for storage	每桶采用卡箍密封 Each drum sealed with clips		
干燥剂 Desiccant	尺寸：宽 78±4 mm、长 150±7 mm 重量：100 g Dimensions: W:78±4 mm, L:150±7 mm Weight: 100 g	硅胶干燥剂 Silica gel desiccant	放置于一级包装与二级包装之间 Placed between primary and secondary packages		

Key raw materials, in-process products and products for PPQ and commercial production should be weighed after the weighing implements are checked using the standard weights. For the balance/scale with the maximum capacity of greater than 40kg, check with the standard weight of 20kg; for the balance/scale with maximum capacity is not more than 40kg, check with the standard weight of 1/2 maximum capacity. /用于工艺性能确认和商业化生产的主原料、过程产物以及产品的称量前，应使用标准砝码对称量器具进行检查，量程大于 40kg 的电子秤，采用 20kg 的标准砝码进行检查；量程不大于 40kg 的电子秤，采用最大量程 1/2 的标准砝码进行检查。

Checking record for standard weights/标准砝码检查记录:

Weight No. 砝码编号	Weight 砝码重量（kg）	Reading 示重（kg）	Less than half of verification scale interval. / 小于检定分度值的一半	Performed by & Date/执行人及日期	Checked by & Date/ 核实人及日期
			☐Y ☐N		

Collect/收料（Time/时间）	Begin/开始:	End/结束:

Weighing of in-process intermediate/中间体称重:

Scale ID. No./秤编号: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

Peptidyl resin number/肽树脂编号	Drum No. /桶编号	Intermediate Qty/ 中间体质量			Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Tare weight/ 皮重	Total weight/ 总重	Net weight/ 净重		
CPo140021-_____-01-_____						

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

Peptidyl resin number/肽树脂编号	Drum No. /桶编号	Intermediate Qty/中间体质量			Performed by & Date/执行人及日期	Checked by & Date/核实人及日期
		Tare weight/皮重	Total weight/总重	Net weight/净重		
CPo140021-_____-01-_____						
Total/合计						

Acceptance criterion: The dried material is light yellow to dark yellow solid./ 合格判定标准：干燥物料为浅黄色至深黄色固体

Appearance/外观: _____

Record the inspection result as Conforms or Does not conform.

☐**Yes/是**; ☐**No/否**

Inspected by and date/检查人以及日期: _____

Checked by QA & Date/QA 核实人及日期: _____

Non-conformance action: In the event of non-conformance, notify QA immediately and initiate a formal deviation process./ 不合格处置：若出现不合格情况，立即通知质量保证部门（QA）并启动正式偏差流程。

Label type (quality form number)/标签类型（表格编号）: ☐QF02092 ☐Other/其他: _____

Label filling and pasting performed by & Date/标签填写及粘贴人及日期: _____

Label filling and pasting checked by &Date/标签填写及粘贴核实人及日期: _____

Synthesis completed, print the corresponding operation log and attach it to this MOR/合成结束，打印出相应的操作记录并作为此文件附件。

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

1	3000L Peptide Solid Phase Synthesizer /多肽固相合成仪 TJ4S-1213-Sy16 Summary Report of Operation Records/操作记录汇总报表附件编号: CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期: _____
2	3000L Peptide Solid Phase Synthesizer /多肽固相合成仪 TJ4S-1213-Sy16 Temperature log/温度记录附件编号: CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期: _____
3	500L Glass lined agitator tank2/500L 搪玻璃搅拌罐 2TJ4S-1213-AT09 Temperature log/温度记录附件编号: CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期: _____
4	3000L Stainless steel agitator tank3/3000L 不锈钢搅拌罐 3 TJ4S-1213-SPS01-AT01 的 20%V/V Piperidine solution preparation log number/20%哌啶/N,N-二甲基甲酰胺溶液配制记录附件编号: CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期: _____
5	3000L Stainless steel agitator tank3/3000L 不锈钢搅拌罐 3 TJ4S-1213-SPS01-AT02 的 20%V/V Piperidine solution preparation log number/20%哌啶/N,N-二甲基甲酰胺溶液配制记录附件编号: CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期: _____

Batch size (output range): 157.69~292.80 kg

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Operations Instruction
操作指令

Ins.4/操作 4: Collect/收料

6	3000L Peptide Solid Phase Synthesizer /多肽固相合成仪 TJ4S-1213-Sy16 Raw Material Summary Table/物料使用汇总表附件编号：CPo140021-_____-01-_____ Print and confirm that the parameters in the attachment meet the process requirements. / 打印并确认附件中各参数满足工艺要求。 Checked by & Date/核实人及日期：_____

When materials are transferred between different buildings in the same plant or between different areas within the same building, it should be ensured that they are sealed during the transfer process. The inner packaging of the material should not be opened before transferring to the designated operation area to avoid contamination caused by exposure of the material. /物料在同一厂区不同建筑物之间或相同建筑物内不同区域之间进行转移时，应确保在转移过程中处于密闭状态。在转移到指定的操作区域前，物料的内包装不得被打开，避免物料暴露造成污染。

Transfer the peptide resin to warehouse for temporary storage. During transfer, use an insulated box when transferring. Transfer time not exceed 1h

将肽树脂从_____转移至_____暂存。转移时使用保温箱，转移时间不超过 1h。

Storage conditions: seal and avoid light, storage at 2~8ºC. /贮藏条件：密封，避光 2~8ºC 保存。

Package Qty./包装数：_____

Confirmation of package/包装方式确认：HDPE drum/高密度聚乙烯桶

Checked by & date/核实人及日期：_____

Transfer begin Time/ 转移开始时间：_____	Transfer end Time/ 转移结束时间：_____	Transfer time/ 转移用时：_____

Performed by & Date/填写人及日期：_____

Confirm transfer time not exceed 1h/确认转移时间不超过 1h

Checked by & date/核实人及日期：_____

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Appendix/附录

1. Supplemental Record/附加页

If the tables provided in the Operations Instructions are not enough to record all information, you can record them in the following table. If there is still no enough record, project personnel should request <OR supplemental page> (QF05037) from QA Dept.

若操作指令部分的表格不够用，可以把信息记录到如下表格中。如仍有填写不尽的应到 QA 部门领取《OR 附加页》（QF05037）进行记录。

Note/注:

1. Be consistent with the recording requirements of the operating part of the OR text. For example, if there is weighing operation, need to record the Performed by& Date and Checked by & Date; if there is temperature, need to record the Entered by and Date (record in the Performed by& Date column). /和 OR 正文中操作部分的记录要求一致。例如有称重操作的，需要记录执行人及日期和核实人及日期；有温度的，需要记录执行人及日期(记录在执行人及日期栏)。
2. IPC Sample/material number rule: Batch No. + Instruction No. + Sequent number, sequent number should take OR and other corresponding OR supplemental page into consideration. For example, the first sample/material number in instruction 12 of batch 5, step 3 of project CPo122345-01 is CPo122345-01-03-05-12-05. /IPC 样品/物料编号原则：生产批号+操作指令号+顺序号。顺序号应考虑该操作指令内 OR 及其他相关 OR 附加页中已有的顺序号。例如：项目 CPo122345-01 第三步第五批第 12 个操作指令中产生的第 5 个样品/物料编号，其编号为：CPo122345-01-03-05-12-05。
3. All Chinese test added to the following table or OR text pages must be translated to English before Asymchem QA's review. /在 Asymchem QA 审核前，所有添加到下面表格和 OR 正文的中文都必须翻译成英文。

Batch size (output range): 157.69~292.80 kg

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Title: 140021 Step: 1 Preparation of: 140851

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Operation Name/ 操作名称	Operation Description /操作内容 Include: Process parameters (e.g. Temp., vacuum, pressure, pH) 包含：工艺参数（温度、真空度、压力、pH）等	Time/ 时间	Performed by& Date/ 执行人/日期	Checked by & Date/ 核实人/日期	Note/ 备注

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	QF05098	12.0	09/29/2026	485 / 494

MOR #& Ver: MOR-TJ414002101.04

Title: 140021 Step: 1 Preparation of: 140851

From/上接: _____（Ins. No. & Page#/指令编号和页码）

Operation Name/ 操作名称	Operation Description /操作内容 Include: Process parameters (e.g. Temp., vacuum, pressure, pH) 包含：工艺参数（温度、真空度、压力、pH）等	Time/ 时间	Performed by& Date/ 执行人/日期	Checked by & Date/ 核实人/日期	Note/ 备注

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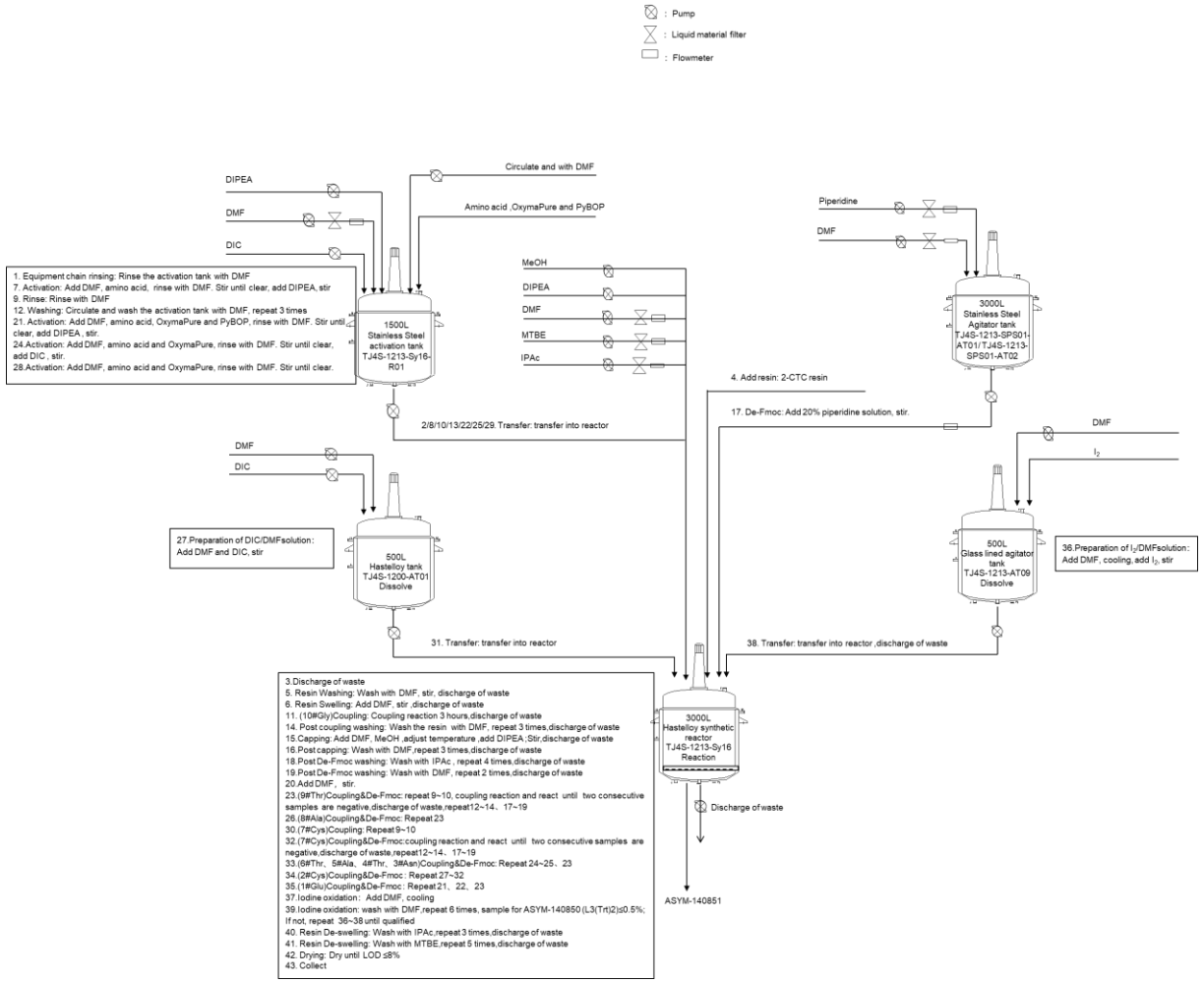
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2. Process flow chart /工艺流程图

English/英文:



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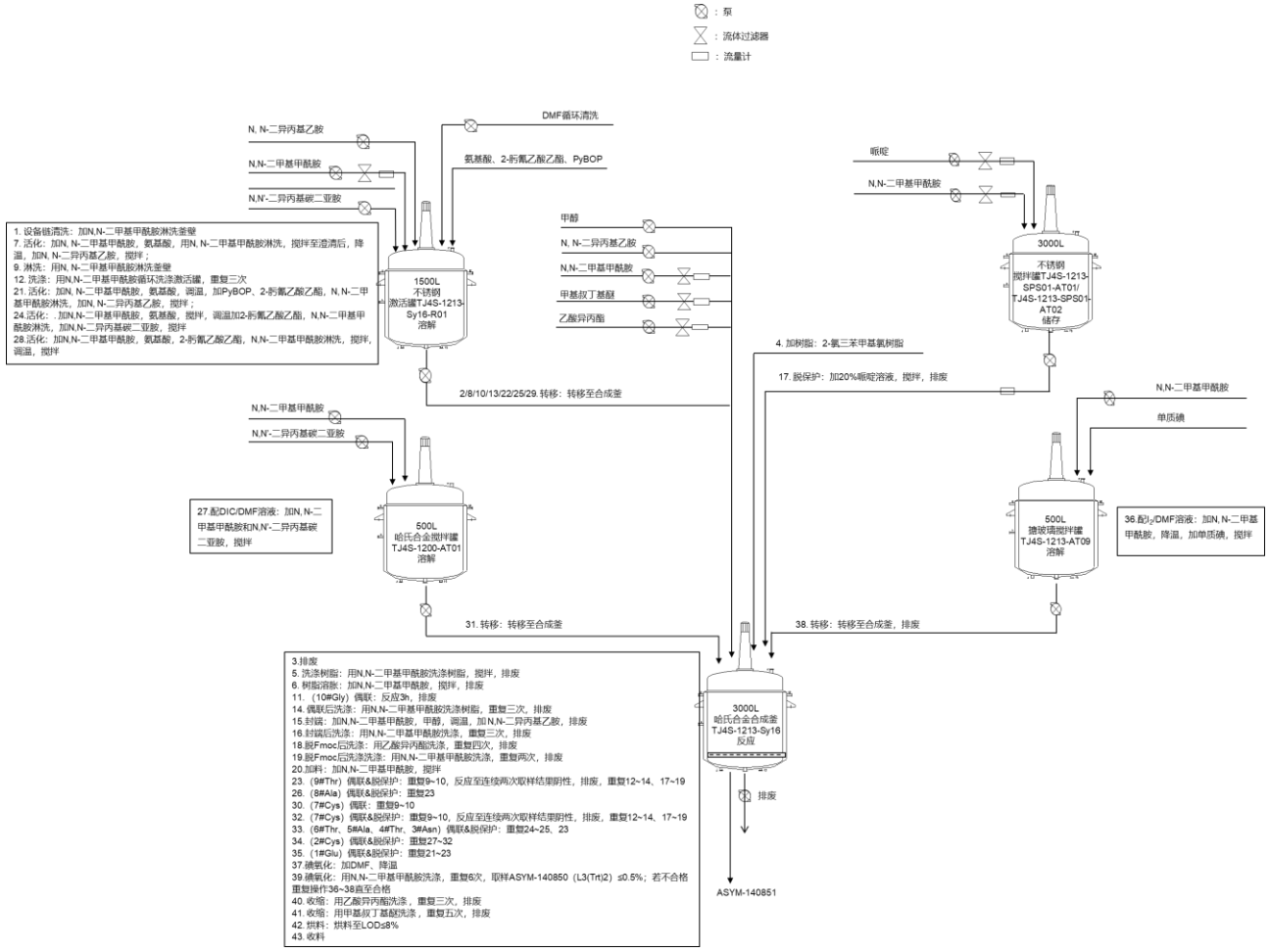
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Chinese/中文:



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3. English abbreviations /英文缩写表

English Abbreviation 英文缩写	English Full Name 英文全名	Chinese Name 中文名
IPC	In Process Control	过程控制
SDS	Safety Data Sheet	安全数据表
Qty	Quantity	质量
Ins.	Instruction	指令
MPa	Mega Pascal	兆帕斯卡
KF	Karl-Fischer	卡尔费休
LOD	Loss on Drying	干燥失重
MP	Melting Point	熔点
BP	Boiling Point	沸点
NMT (NGT)	No More Than (No greater Than)	不大于，即小于等于
NLT	No Less Than	不小于，即大于等于

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Post-batch Approvals/操作完毕确认

I have reviewed the above operations record and confirmed that it accurately represents how the intermediate noted on the record was produced.
我本人已经阅读了以上的操作记录，并确信此记录准确地记述了记录上中间体的生产过程。

Production Director & Date/生产主管及日期: _____

Project Director & Date/项目主管及日期: _____

QA project manager & Date/QA 项目经理及日期: _____

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Version Instructions/版本说明

1. The MOR is written with the Standardization language (SOP DS-QA-001 <Batch operation record management procedure (DS)>. MOR 是按照标准化语言撰写（SOP DS-QA-001《批操作记录管理规程(DS)》）。

2. This is fourth version of MOR-TJ414002101, so the contrast with the last:

此版本为 MOR-TJ414002101 第四版，与上一版的对比说明如下:

Brief description of changes /更改内容简要描述:

1. MOR documentation optimization/MOR 文件优化

   Upgraded batch record framework, traceability mechanisms and dual-person verification rules for critical operations to fully align with commercial GMP expectations and ALCOA+ data integrity principles. 升级批记录框架、追溯机制与关键操作双人复核规则，全面对齐商业化 GMP 要求与 ALCOA＋数据完整性原则。

2. Process control refinement within validated PAR ranges/验证范围内工艺管控精细化

   Standardized CPP time definitions, optimized operational endpoints and parameter tolerances, and refined key temperature ranges based on PAR study findings, improving process robustness and operability without changing the core process design./基于 PAR 研究规范 CPP 时间定义，优化操作终点与参数容差，细化关键温度范围，在不改变核心工艺设计的前提下提升工艺稳健性与可操作性。

3. Analytical method optimization/分析方法优化

   Tightened the LOD acceptance specification, updated core IPC test methods (coupling completion, KF moisture), and adjusted appearance inspection governance mode to improve detection reliability and intermediate batch-to-batch consistency/收紧 LOD 接收标准，更新偶联反应完成确认、水分等核心 IPC 检测方法，调整外观检测管控模式，提升检测可靠性与中间体批次一致性。

Details in Change file, Change No. Change-CPo140021-030106.01/以上更改详细内容见变更文件，编号: Change-CPo140021-030106.01

3. MOR numbering method: MOR-site code+product code + step No. + version#(2digits), as MOR-TJ3111234A01.01. /MOR 编号方式为 MOR-厂区代码+产品代码+步骤号+2 位版本号，如 MOR-TJ3111234A01.01

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Document Approval Signatures

Approved Date: 29 Oct 2026

Approval Verdict: Approve	Yiwen Zhang (L084212@lilly.com) Approver 29-Oct-2026 11:59:24 GMT+0000
Approval Verdict: Approve	Novie Atakiwe (L033124@lilly.com) Quality Approver 29-Oct-2026 12:01:31 GMT+0000
Approval Verdict: Approve	Michelle Cronin-Carroll (C203669@lilly.com) Approver 29-Oct-2026 12:35:12 GMT+0000
Reviewer / Approver Additional Details	Note: Michelle Cronin-Carroll is signing on behalf of Yu-Lin Zhao.

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